

Accelerate AI Adoption within Abu Dhabi Government

Use Cases

MAY 2024

Abu Dhabi’s ambition is to become an AI Native Government achieving Whole-Of-Emirate Impact



Citizen Experience

Improve the quality and speed of citizen & resident interaction with the government



Operational Efficiency

Increase process efficiency through automation to become more cost efficient



Productivity Enhancement

Improve worker output with new AI-powered applications and tools



Achievement of Government Goals

Enable achievement of horizontal and vertical Government goals through AI



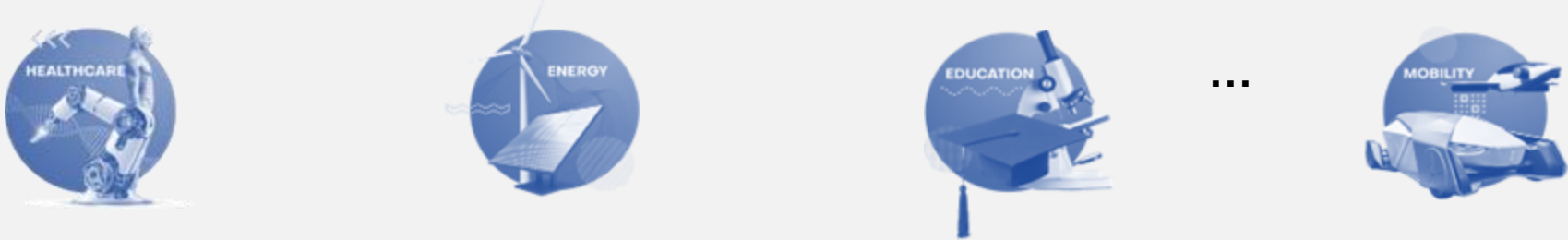
Uplift Output Quality

Uplift Government output quality by improving accuracy and citizen engagement

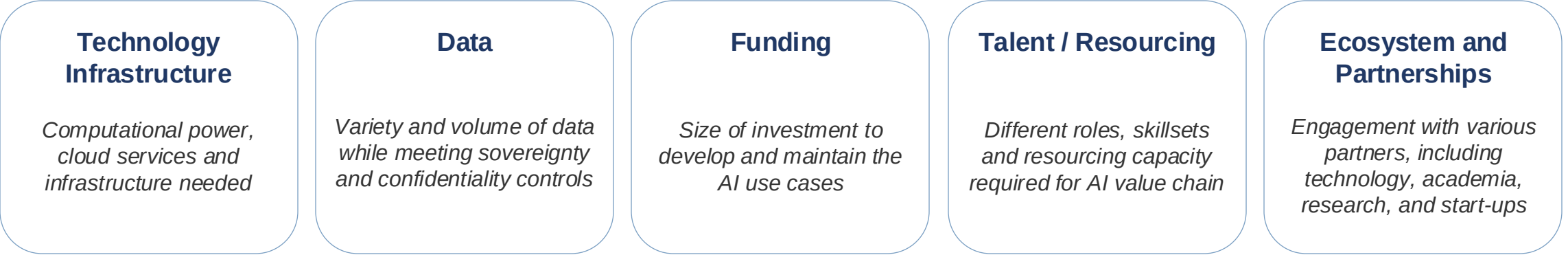
At the heart of this ambition lies an understanding of the AI use cases across the Abu Dhabi Government and underlying capabilities required to deliver them

To achieve this Ambition, driving AI adoption across Government ...

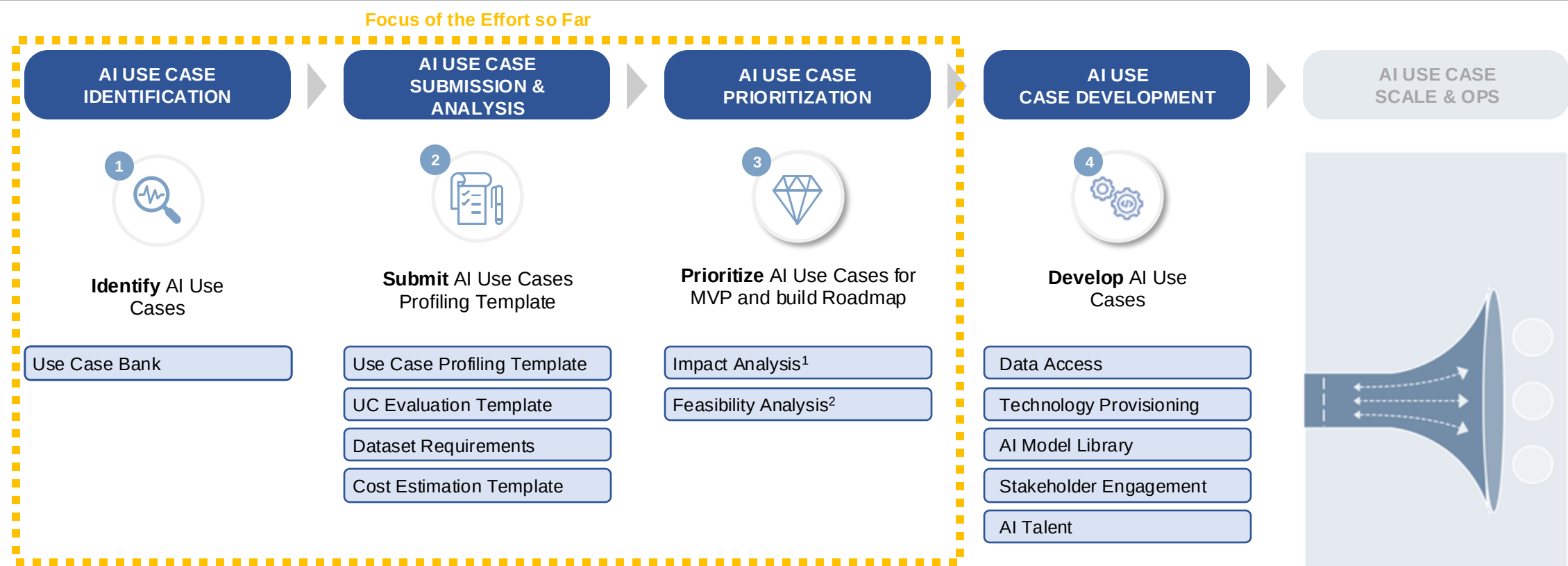
... starts with definition of AI use cases across sectors...



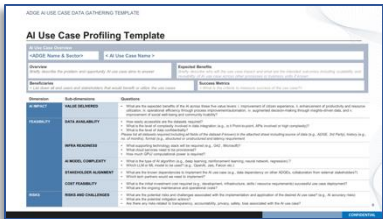
... which enables understanding of underlying requirements across



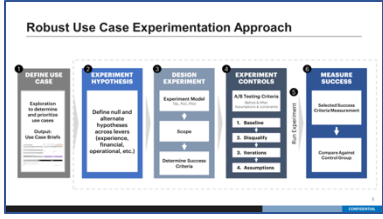
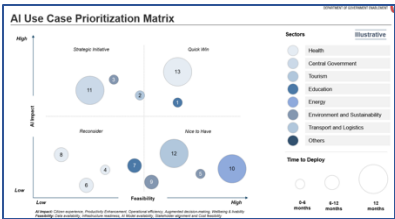
Accordingly, we follow a standardized AI use case submission and prioritization process



Supporting Tools and Templates

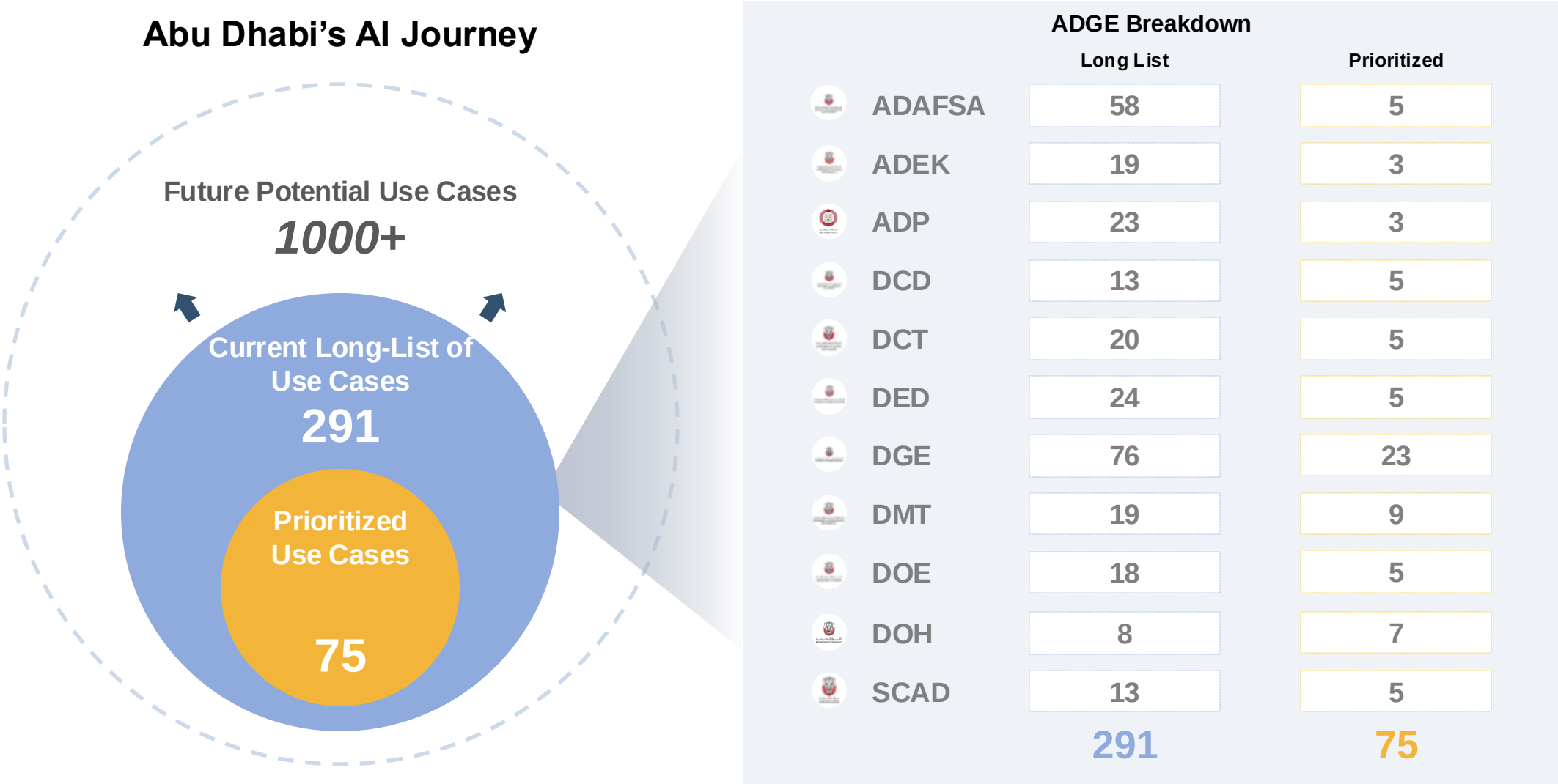


Category	Sub-category	Strategic Importance	Impact Potential	Feasibility	Stakeholder Alignment	Cost Feasibility
Technology	Healthcare	High	High	Medium	High	Medium
Technology	Education	Medium	Medium	High	Medium	High
Technology	Finance	High	High	High	High	Medium
Technology	Manufacturing	Medium	Medium	Medium	Medium	High
Technology	Transportation	Medium	Medium	Medium	Medium	Medium
Technology	Energy	High	High	Medium	Medium	Medium
Technology	Environment	Medium	Medium	Medium	Medium	Medium
Technology	Government	High	High	Medium	High	Medium
Technology	Other	Medium	Medium	Medium	Medium	Medium
People	Healthcare	High	High	Medium	High	Medium
People	Education	Medium	Medium	High	Medium	High
People	Finance	High	High	High	High	Medium
People	Manufacturing	Medium	Medium	Medium	Medium	High
People	Transportation	Medium	Medium	Medium	Medium	Medium
People	Energy	High	High	Medium	Medium	Medium
People	Environment	Medium	Medium	Medium	Medium	Medium
People	Government	High	High	Medium	High	Medium
People	Other	Medium	Medium	Medium	Medium	Medium



1. **Expected Impact:** Citizen Experience, Productivity Enhancement, Operational Efficiency, Achievement of Gov. Goals and Output Quality
2. **Feasibility:** Data availability, Infrastructure readiness, AI Model availability, Stakeholder alignment and Cost feasibility

This has resulted so far with 291 use cases collected and 75 prioritized



Note (*): Assessment is based on prioritized use cases only



ADAFSA – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility

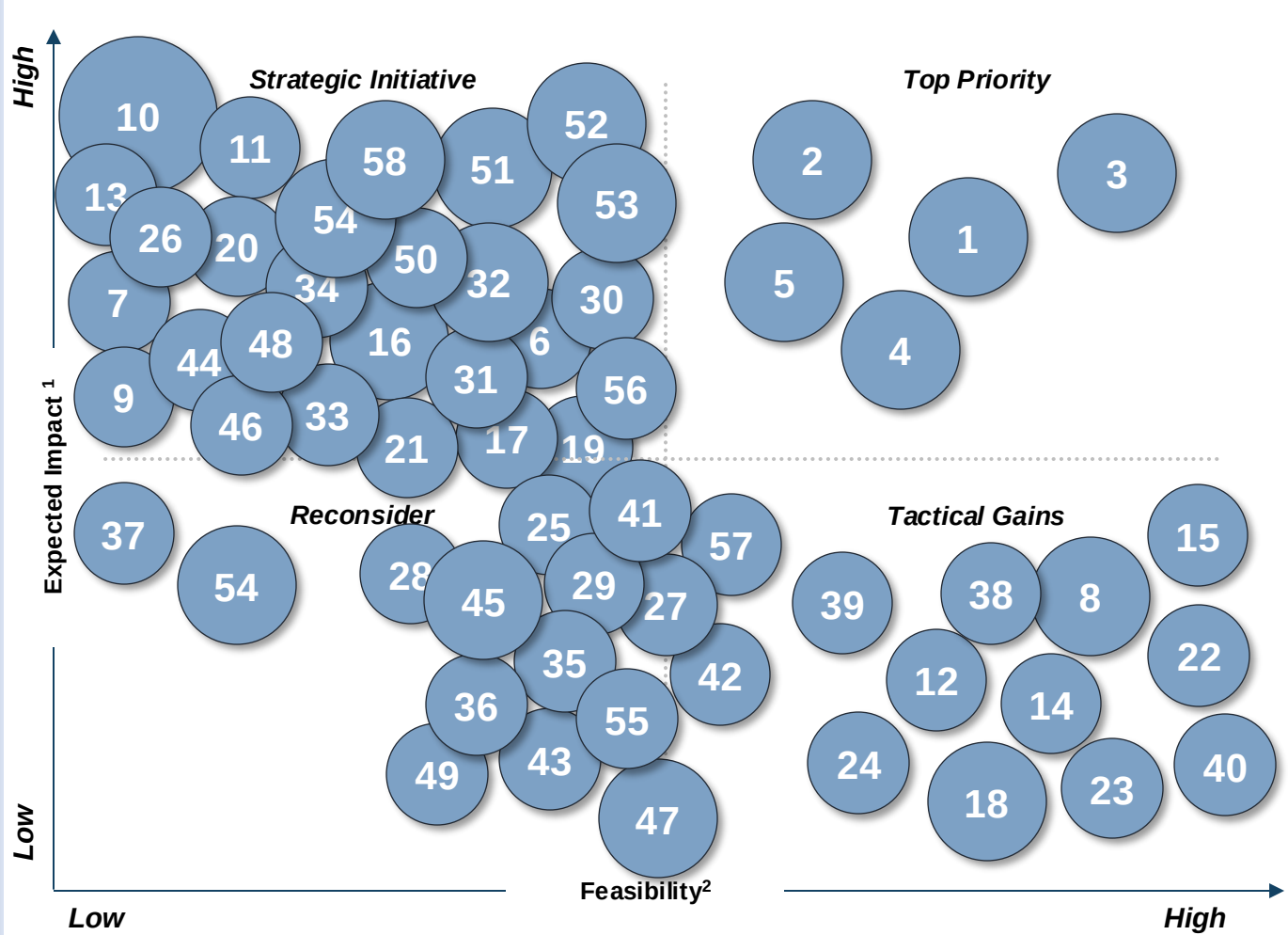
ADAFSA



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ABU DHABI AGRICULTURE AND FOOD
SAFETY AUTHORITY

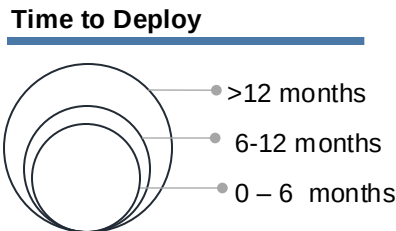
58

Identified AI Use Cases



Identified Use Cases ³

1	AI Agricultural inspection
2	Abu Dhabi Agriculture & Food Safety Data Platform Phase 2
3	AI Food Safety inspection parameters detected (Expansion Phase)
4	AI Veterinary
5	AI Import Diversification Engine
6	Predictive Disease Management System
7	AI-enhanced Pest Monitoring
8	Robotic Pest Mgmt.
9	AI-Based Fraud Detection
10	Blockchain-Enabled Traceability
11	Image Recognition for Packaging Authenticity
12	AI-Driven Soil Management
13	Remote Sensing for Soil Management
14	Precision Farming
15	Chatbot to Help and Assist Farmers
16	Weather Prediction to Assist Farmers
17	AI-Driven Market Predictions
18	Risk Management with AI Solutions
19	Dynamic Pricing Strategies with AI
20	Forecasting consumer price index(CPI) to anticipate rising food prices



1. **Expected Impact:** Citizen Experience, Productivity Enhancement, Operational Efficiency, Achievement of Gov. Goals and Output Quality
2. **Feasibility:** Data availability, Infrastructure readiness, AI Model availability, Stakeholder alignment and Cost feasibility

Summary and prioritization of identified use cases, across impact and feasibility

ADAFSA



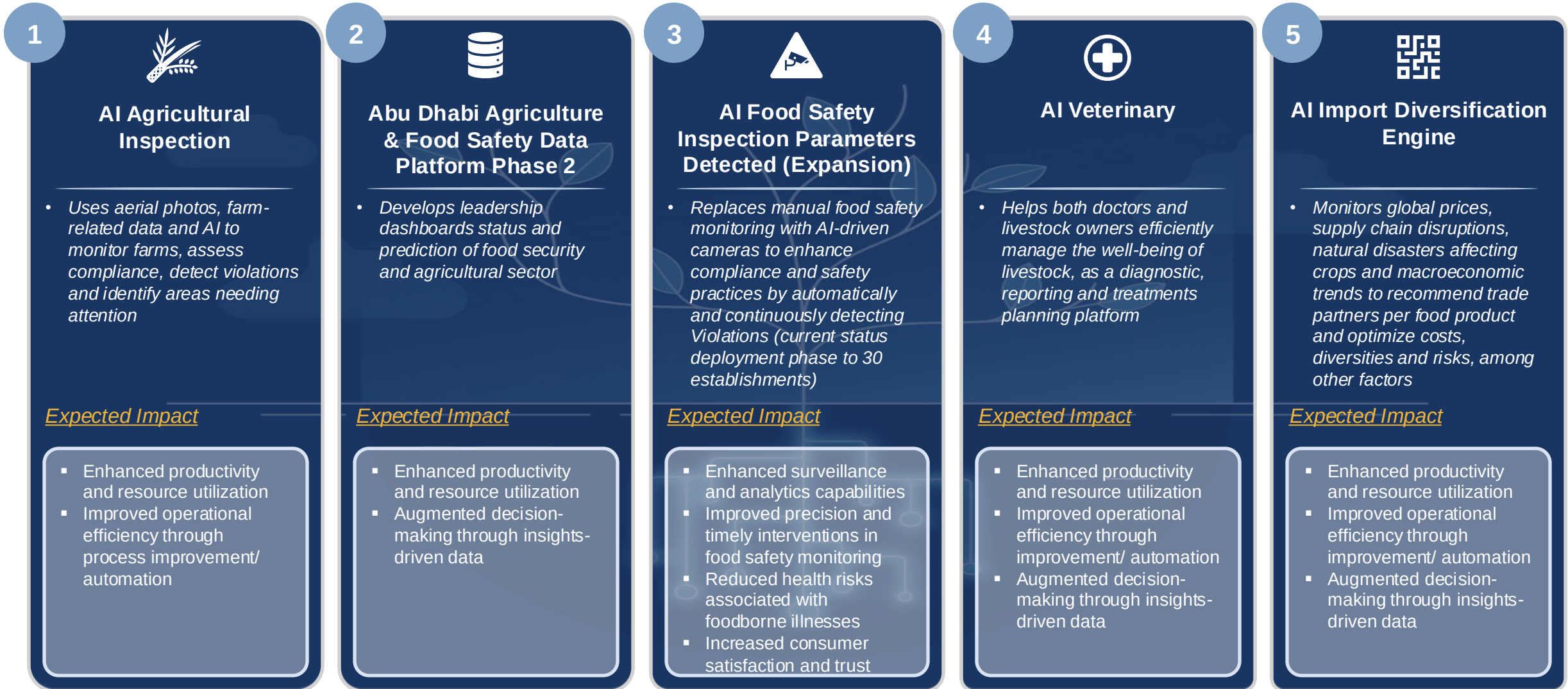
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Identified AI Use Cases

ADAFSA Long List of Non-prioritized Use Cases

21	Data-driven recommendations on the pricing of the products	34	Proactive maintenance & monitoring of equipment and machinery	47	Estimating green house gases(GHG) emissions in agriculture value chain
22	Internal knowledge mining based chatbot	35	Disease detection and health monitoring of livestock	48	Slaughterhouse production estimation
23	Assessment of farmer creditworthiness	36	Optimizing crop yield	49	Monitoring heat stress and temperature on livestock
24	Analyzing "automated milking systems" data to forecast yield	37	Predicting honey harvests	50	Predicting appropriate time for pruning and thinning date palm trees
25	Land use and land cover classification	38	Creating optimal irrigation scheduling and distribution	51	Monitoring and detecting regenerative agricultural practices
26	Predicting evolving food trends in the market	39	Automated human capacity planning using Data Analytics and AI for call center	52	Predicting AMR in livestock
27	Date palm tree identification and classification per tree	40	AI enabled monitoring and control systems for poultry farms / animals	53	Fish Disease and health management
28	Bird identification and automated weighing systems	41	Generate alerts / early warnings for scenarios leading to a potential breach of controls	54	Monitoring drinking and feeding behavior of the livestock
29	Augmented Risk estimation for sampling	42	Automate repetitive and continuous audit processes	55	Livestock vocalization analysis
30	Farm area segmentation and Representative sample selection	43	Forecasting Tick-Borne Diseases	56	Designing optimal feeding schedule for livestock
31	Pest outbreak prediction & hotspot detection in crops	44	Early determination of fertile poultry eggs	57	Food contamination detection system in the warehouse
32	Identify opportunities and priorities for investments in Abu Dhabi	45	Track unauthorized food being sold at online marketplaces and platforms	58	Monitoring pollination and alerting farmers to take appropriate measures in timely manner
33	Analysis of the products of the livestock to determine the quality	46	Reproduction monitoring and detecting optimal breeding time		

ADAFSA has prioritized 5 use cases from their identified long list





Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~10.1M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	Medium	High	High	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	High	Medium	Medium

1. Business Overview

Description

- The purpose of this use case is to utilize aerial photos, farm-related data, and AI to monitor farms, assess compliance, detect violations, and identify areas needing attention, thereby optimizing inspections and monitoring activities in the agricultural control and extension sectors.
- Key beneficiaries of this use case include ADAFSA, which will leverage the insights provided to enhance compliance with agricultural standards, improve resource utilization, and increase the efficiency of field workers.

1.2

Pain Points

- Manual and Time-consuming Inspections: Field visits for inspections are time-consuming, taking about 30 minutes per visit.
- Resource Constraints: The current system not effectively optimize resource utilization and productivity, resulting in less efficient operations and higher operational costs.

1.3

Expected Impact

- Optimization of inspections and monitoring activities in the agricultural control and agricultural extension sectors

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Ensure access to high-quality satellite images and GIS data,
- Secure the necessary computing power and technology stack

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Satellite Aerial Images	Medium				Medium	Confidential
GIS Data	Medium				Medium	Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

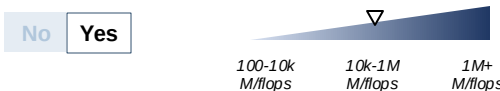
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient CV and ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Data Integration and Infrastructure Readiness: Ensure accessible, high-quality datasets



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					~20M AED		6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Medium	High	High	High	Medium				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
Medium	Low	Medium	Medium	Medium	

1. Business Overview

1.1

Description

- The purpose of this use case is to provide leadership dashboards and predictive insights for food security and the agricultural sector, integrating various sources of agricultural data. This platform supports enhanced decision-making by providing up-to-date information on global price indices, supply chain threats, food safety data, and more.
- Key beneficiaries of this use case include decision makers, governmental stakeholders, and the public benefit from published data, allowing them to make informed decisions about food security and agricultural management in Abu Dhabi.

1.2

Pain Points

- Dependencies: Dependencies on multiple stakeholders, including government entities and private sectors, can complicate coordination and implementation efforts.

1.3

Expected Impact

- Access to agriculture sector data and its indicators from various sources (such as data on local production, trade, investment, alerts)
- Global, price indices, supply chain threats, food safety data, traceability and security threats biological, environmental and other data that support decision-making

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Data owners' sharing policies limitations

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
ADGE Databank						Public
Private Sector Data						Confidential

2.2

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No

Yes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Infrastructure and Cost Management: Ensure scalable and robust technology platforms
- Data Integration and Confidentiality: Seamless data integration while maintaining confidentiality

AI Food Safety inspection parameters detected (Expansion Phase)



ADAFSA

Status:

Started

Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					0.01M AED		0-6 Months	6 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
High	Medium	High	High	Medium				
2. Feasibility								
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility				
High	High	Medium	Medium	Medium				

1.1

1. Business Overview

Description

- The purpose of this use case is to replace manual food safety monitoring with AI-driven cameras, enhancing compliance and safety practices by automatically detecting violations and continuously monitoring food establishments. This project aims to improve food safety standards, reduce human error, and ensure timely interventions to prevent foodborne illnesses.
- Key beneficiaries of this use case include Food establishment owners, ADAFSA, the Abu Dhabi Government, and the general public benefit from improved food safety and reduced health risks associated with foodborne illnesses.

1.2

Pain Points

- Limited Surveillance and Analytics Capabilities
- Resource Constraints and Scalability Issues
- Manual and Time-consuming Inspections

1.3

Expected Impact

- Improvement of food safety standards, reduction in human error and inconsistency.
- enhanced efficiency in monitoring by (24/7) violation detection, and potentially lower health risks associated with foodborne illnesses.

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- AI Violation Accuracy
- Integration & Operational challenges from Food Establishment

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Live Video Feeds	High	Large		Food Establishments	High	Confidential

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
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AI Technology

- Computer Vision
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Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

NoYes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Computer Vision and ML build/operational infrastructure
- Data Management and Security: Ensure real-time processing with high privacy standards
- Infrastructure and Stakeholder Coordination: Leverage robust AI and video analytics infrastructure



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	High	High	Low

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Low	High	AI Model Complexity Medium	Medium	Medium

1. Business Overview

1.1

Description

- This use case aims to support both doctors and livestock owners in managing the well-being of livestock through an AI-powered diagnostic, reporting, and treatment planning platform. It facilitates visual preliminary diagnoses, test result notifications, outbreak alerts, and remote veterinary consultations.
- Key beneficiaries of this use case include livestock owners benefit from reduced mortality rates, improved treatment outcomes, and increased engagement with veterinary services, resulting in better overall livestock health.

1.2

Pain Points

- High Volume of Requests with Limited Resources
- Manual and Inefficient Processes

1.3

Expected Impact

- Visual preliminary diagnosis. Tests' results access and notifications. Outbreaks notifications.
- Remote veterinarian consultancies.

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Timely availability of data
- Data Quality
- Data Integration
- Model Accuracy
- Data Privacy

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Historical Biomedical Data	Difficult	Medium		ADAFSA	Restricted	High
Imagery Data	Difficult	Large		ADAFSA	Restricted	High

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

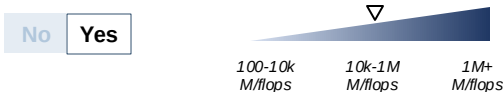
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
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GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Adequate data for solutions
- Right NLP and ML build/operational infrastructure
- Infrastructure and Stakeholder Engagement: Utilize robust cloud infrastructure and foster collaborative relationships
- Data Accessibility and Integration: Ensure comprehensive data integration and easy access while maintaining confidentiality



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	High	Medium	Medium

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	Medium	Medium	Medium

1. Business Overview

1.1

Description

- The purpose of this use case is to monitor global prices, supply chain disruptions, natural disasters, and macroeconomic trends to recommend optimal trade partners for food products. The system aims to optimize costs, diversify suppliers, and manage risks by providing real-time data and predictive insights.
- Key beneficiaries of this use case include Federal and local governments, policymakers, and local businesses benefit from increased trade partner diversity, improved preparedness for global supply chain disruptions, and optimized trade strategies.

1.2

Pain Points

- Difficult Access: Obtaining necessary data (UAE Customs Data, consumer prices, global events) is challenging due to restrictions and high complexity.

1.3

Expected Impact

- Up-to-date monitoring of global supply chain disruptions, Recommendation of new trade strategies/partners, Increased preparedness for future expected events

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Timely availability of data
- Data Quality
- Data Integration
- Model Accuracy

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Customs Data	Difficult	Large		Federal Government	Restricted	High
Consumer Prices	Difficult	Large		Various Data Providers	Restricted	High

2.2

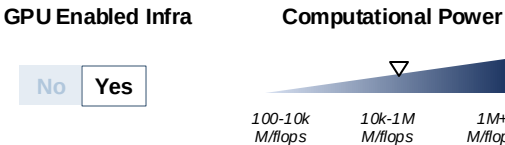
AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
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Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage



2.4

Key Considerations

- Adequate data for solutions
- Sufficient ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Data Complexity and Integration: Ensure seamless data integration from diverse sources
- Infrastructure and Model Sophistication: Utilize robust cloud infrastructure for handling complex AI models

Furthermore, an additional 53 use cases have been defined by ADAFSA

ADAFSA Long List of Non-prioritized Use Cases					
	Title	Description	AI Impact	Feasibility	Time to Deploy
6	Predictive Disease Management System	AI predicts disease outbreaks using environmental data, enabling proactive measures for crop and livestock protection, reducing losses and ensuring food security	●	●	0-6 Months
7	AI-enhanced Pest Monitoring	AI analyzes sensor data to detect pests, triggering targeted interventions for reduced crop damage and optimized pesticide use	●	●	0-6 Months
8	Robotic Pest Mgmt.	AI-guided robots identify and eliminate pests, reducing chemical pesticide reliance and minimizing environmental impact while enhancing crop health	●	●	6-12 Months
9	AI-Based Fraud Detection	AI algorithms detect anomalies in food production processes, flagging potential fraud instances and ensuring product authenticity, safety, and compliance	●	●	0-6 Months
10	Blockchain-Enabled Traceability	AI-powered blockchain ensures food provenance, enhancing transparency and trust by tracking products from farm to table, reducing food fraud risks	●	●	>12 Months
11	Image Recognition for Packaging Authenticity	AI image recognition technology verifies packaging authenticity, preventing counterfeit products from entering the market, safeguarding consumer trust and brand reputation	●	●	0-6 Months
12	AI-Driven Soil Management	AI interprets soil data for precise nutrient recommendations, enhancing soil fertility and sustainability, improving crop yields and long-term land productivity	●	●	0-6 Months
13	Remote Sensing for Soil Management	AI-analyze satellite imagery for soil health assessment, guiding targeted interventions like erosion control, preserving soil quality, and optimizing agricultural practices	●	●	0-6 Months
14	Precision Farming	AI-powered precision farming optimizes soil use, water resources, and crop rotation, promoting healthy soil ecosystems, sustainable agriculture, and environmental conservation	●	●	0-6 Months
15	Chatbot to Help and Assist Farmers	Chat bot to provide farming and livestock information to provide easy interactions and provide farmers with knowledge to optimize their production	●	●	0-6 Months
16	Weather Prediction to Assist Farmers	Predict weather forecast using AI and ML, to assist farmers in planning	●	●	6-12 Months
17	AI-Driven Market Predictions	AI analyzes market trends, forecasts price fluctuations, and provides insights for strategic decision-making, empowering farmers to navigate market volatility and optimize profitability	●	●	0-6 Months
18	Risk Management with AI Solutions	AI models assess market risks, enabling farmers to implement risk mitigation strategies, hedge against price volatility, and adapt to changing market dynamics effectively	●	●	6-12 Months
19	Dynamic Pricing Strategies with AI	AI recommends dynamic pricing strategies based on real-time market data, optimizing product pricing, maximizing revenue, and enhancing competitiveness in volatile markets	●	●	0-6 Months

● Low

● Medium

● High

ADAFSA



58

Identified AI Use Cases

Furthermore, an additional 53 use cases have been defined by ADAFSA

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Identified AI Use Cases

ADAFSA Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
20	Forecasting consumer price index(CPI) to anticipate rising food prices	- To generate early warnings about potential increases in food prices - Analysis of trends and patterns in CPI and commodity data would help identify indicators that can signal upcoming price rises	●	●	0-6 months
21	Data-driven recommendations on the pricing of the products	- To predict the price of agricultural commodities which will be used to optimize pricing and marketing strategies - To achieve competitive advantage by understanding price dynamics in the market and in turn attracting more customers	●	●	0-6 months
22	Internal knowledge mining based chatbot	- The internal organization chatbot would help employees with a fast and seamless resolution of their queries and data/statistics needs - The internal chatbot, through a common platform for query resolution, would provide consistency in information dissemination and improve communication	●	●	0-6 months
23	Assessment of farmer creditworthiness	To develop a system that utilizes AI/ML technologies to assess the creditworthiness of farmers effectively and extend the most appropriate line of credit	●	●	0-6 months
24	Analyzing “automated milking systems” data to forecast yield	The use case utilizes Artificial Intelligence (AI) for analyzing and forecasting milk production yield using automated milking systems by processing the vast amount of data collected making accurate predictions about milk production. Thus, helping in monitoring milk production traits, or milk yield	●	●	0-6 months
25	Land use and land cover classification	- To understand what is covering the land surface and how the land is being used - To classify the land into various broad categories e.g., industrial zones, residential areas, water bodies, agricultural land, forest, etc. It's possible to further classify the vegetation areas such as cropland and specific crops, pasture, orchards, groves, vineyards, nurseries, etc.	●	●	0-6 months
26	Predicting evolving food trends in the market	To develop a system that can provide insights into evolving food preferences of consumers by analyzing vast amounts of data and leveraging advanced analytics and AI thereby assisting businesses to proactively adapt and cater to consumer expectations and demands	●	●	0-6 months
27	Date palm tree identification and classification per tree	To develop a system that will use AI/ML to accurately identify and categorize different species, subspecies, and cultivars based on their morphological, physiological, and genetic characteristics	●	●	0-6 months

● Low

● Medium

● High

Furthermore, an additional 53 use cases have been defined by ADAFSA

ADAFSA



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Identified AI Use Cases

ADAFSA Long List of Non-prioritized Use Cases				
Title	Description	AI Impact	Feasibility	Time to Deploy
28 Bird identification and automated weighing systems	- To propose an automated method for quickly and accurately counting the number of chickens in a poultry farm and monitoring chicken populations by studying the morphology of the chickens and observing their physiological behaviors - To develop a real-time, automated, and non-invasive system to measure the weights of chickens	●	●	6-12 months
29 Augmented Risk estimation for sampling	Testing of food samples is required to assess food quality and evaluate whether proper safety controls are in place. Risk-based sampling ensures that statistically significant data with proper sample size and sampling frequency are being adopted in testing various food safety-related parameters such as the presence of contaminations, pathogens, nutrient content, etc.	●	●	6-12 months
30 Farm area segmentation and Representative sample selection	To understand different characteristics of agricultural farms (such as size, topography, type of crop grown, farming techniques used, geographic location, etc.) and divide farm areas into different segments based on these characteristics	●	●	0-6 months
31 Pest outbreak prediction & hotspot detection in crops	- To predict the possibility of pest outbreaks and determine hotspots for pest infestation in AD - To minimize the damage induced by pests in crops during harvest and postharvest processing, as well as to maximize productivity and ensure agricultural sustainability, advanced pest hotspot detection and prevention mechanism are highly important for ADAFSA	●	●	0-6 months
32 Identify opportunities and priorities for investments in Abu Dhabi	- Identify Venture Capitalists focus areas of investment and identify trends in investment in agriculture and recommend segments/products to focus on and invest in - Attract global companies to establish advanced AgriTech ecosystem in Abu Dhabi and benefit from globally competitive tax rates, advanced infrastructure, ability to handle large volume commodities and economic business hub geographical location	●	●	6-12 months
33 Analysis of the products of the livestock to determine the quality	- To determine the quality of products accurately and efficiently. Products can include meat, milk, etc. - To reduce human errors and bias in the analysis process and evaluate the quality more objectively and with greater consistency	●	●	0-6 months
34 Proactive maintenance & monitoring of equipment and machinery	- To anticipate possible failures of equipment and machinery well before the actual shutdown - To implement a data-driven system that constantly analyzes the streaming data collected through IOT sensors regarding the conditions of equipment and machinery to identify any abnormal behavior	●	●	0-6 months

Furthermore, an additional 53 use cases have been defined by ADAFSA

ADAFSA



هيئة أبوظبي للزراعة والسلامة الغذائية
ABU DHABI AGRICULTURE AND FOOD
SAFETY AUTHORITY

58

Identified AI Use Cases

ADAFSA Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
35	Disease detection and health monitoring of livestock	- To identify and address health issues as early as possible in order to prevent disease outbreaks and improve animal welfare and productivity - To help veterinarians take appropriate actions to prevent the spread of diseases including reducing mortality rates	●	●	0-6 months
36	Optimizing crop yield	- To create an optimal irrigation schedule for watering the farm-land to effectively manage water resources for agriculture and balance water availability and plant requirements - To maximize crop productivity and minimize water wastage	●	●	0-6 months
37	Predicting honey harvests	- To create a machine learning model that predicts harvest potential per apiary - To detect the drivers for honey production and design a precision apiculture system (PAS) platform for determining the influence of environmental parameters inside & outside the hive	●	●	0-6 months
38	Creating optimal irrigation scheduling and distribution	- To create an optimal irrigation schedule for watering the farm-land to effectively manage water resources for agriculture and balance water availability and plant requirements - To maximize crop productivity and minimize water waste	●	●	0-6 months
39	Automated human capacity planning using Data Analytics and AI for call center	By applying AI and data analytics techniques, such as predictive modeling and machine learning algorithms, the use case can forecast future call volumes, predict staffing needs, and optimize scheduling	●	●	0-6 months
40	AI enabled monitoring and control systems for poultry farms / animals	The primary objective of control systems in poultry/animal farms is to ensure the health and welfare of poultry and animals. By utilizing cameras and sensors, vital parameters such as temperature, humidity, CO2, etc. can be continuously monitored to detect any breach in the acceptable range of these parameters enabling the prevention of health issues or adverse environmental conditions.	●	●	0-6 months
41	Generate alerts / early warnings for scenarios leading to a potential breach of controls	- Early identification of potential risks and anomalies across the sectors and offices - Alert the internal audit team to start focusing on identified high-risk areas	●	●	0-6 months
42	Automate repetitive and continuous audit processes	- Identification of potential risks and anomalies across the sectors and offices - Move from on-demand process-based risk assurance to continuous risk reviews - Assist in preparation of audit programs, identify supporting documents, preparation of testing sheets, and suggest entry in the risk registers - Allow the internal audit team to focus on high-risk areas	●	●	0-6 months
43	Forecasting Tick-Borne Diseases	To develop a system that enables enhancing public health preparedness and response by forecasting and mitigating the impact of tick-borne diseases in livestock	●	●	0-6 months

● Low

● Medium

● High

Furthermore, an additional 53 use cases have been defined by ADAFSA

ADAFSA Long List of Non-prioritized Use Cases					
	Title	Description	AI Impact	Feasibility	Time to Deploy
44	Early determination of fertile poultry eggs	To develop an automated system that can help in the early identification of fertile poultry eggs thereby enhancing the efficiency of poultry breeding practices by accurately determining the viability of eggs prior to incubation	●	●	0-6 months
45	Track unauthorized food being sold at online marketplaces and platforms	To develop an AI-based system that can monitor the selling activity of food products in Abu Dhabi and identify instances of unauthorized food products being sold and distributed	●	●	6-12 months
46	Reproduction monitoring and detecting optimal breeding time	- To optimize the time of insemination and make sure that the livestock are inseminated in a timely manner - To improve the efficiency and success of animal breeding programs, breeders can ensure they are maximizing the chances of successful reproduction and producing healthy offspring	●	●	0-6 months
47	Estimating green house gases(GHG) emissions in agriculture value chain	To estimate greenhouse gases (GHG) in the agriculture value chain. The below sources would be analyzed to estimate the GHG emission - Pre-harvest – Land preparation, Harvest – Cultivation, Fertilizer application, Livestock presence, Irrigation , Post-harvest- storage, transport and processing	●	●	6-12 months
48	Slaughterhouse production estimation	To develop an automated system for estimating meat production and respective grades by assessing the weights and body measurements of live animals	●	●	0-6 months
49	Monitoring heat stress and temperature on livestock	- To monitor heat stress in livestock and improve animal health and welfare - Due to the high concentration of livestock in a relatively limited space, farm animals tend to be exposed to high temperatures. This may have a devastating effect on both their physical and mental well-being in the long term	●	●	0-6 months
50	Predicting appropriate time for pruning and thinning date palm trees	- To exercise pruning and thinning in a precise and timely manner, as improper pruning and thinning can often weaken or ruin the plant - To prescribe the farmers and farm owners appropriate time for pruning and thinning based on image analysis and inputs from SMEs	●	●	0-6 months
51	Monitoring and detecting regenerative agricultural practices	- Regenerative agriculture is a form of farming that aims to rehabilitate and restore the quality of the soil used for agricultural purposes - Five practices that are widely associated with regenerative farming are: 1) abandoning tillage, 2) eliminating bare soil, 3) fostering plant diversity, 4) encouraging water percolation into the soil, and 5) integrating livestock and cropping operations	●	●	6-12 months

ADAFSA



58

Identified AI Use Cases

Furthermore, an additional 53 use cases have been defined by ADAFSA

ADAFSA Long List of Non-prioritized Use Cases				
Title	Description	AI Impact	Feasibility	Time to Deploy
52 Predicting AMR in livestock	- To detect and prevent the emergence and spread of AMR in the livestock population at an early stage - To plan appropriate interventions such as targeted antibiotic use, improved farm management practices, and biosecurity measures	●	●	6-12 months
53 Fish Disease and health management	To detect disease in fish, which is a significant issue affecting the quality of aquaculture	●	●	6-12 months
54 Monitoring drinking and feeding behavior of the livestock	To enhance the overall management and well-being of farm animals, this system aims to provide real-time monitoring and analysis of livestock behavior, allowing farmers to make informed decisions regarding the health and nutrition of their livestock	●	●	6-12 months
55 Livestock vocalization analysis	- Vocal behavior of livestock is a useful indicator of their physiological and psychological functioning, as it provides information about age, gender, breeding status, and any extraordinary conditions, such as pain, hunger, thirst, estrus, separation, etc. - To extract and analyze the features of animals' vocal sounds and identify the animal species	●	●	0-6 months
56 Designing optimal feeding schedule for livestock	To design an optimal feeding schedule for livestock i.e., suggesting the quantity, nutrients, and frequency at which the animals need to be fed	●	●	0-6 months
57 Food contamination detection system in the warehouse	- To detect food contamination at an early stage within the warehouse environment, enabling timely intervention and preventing contaminated food from reaching consumers - Food contamination and spoilage/deterioration of food quality during storage are major challenges for the food industry. A system capable of detecting food contamination and spoilage during storage will enable early detection and ensure food safety.	●	●	0-6 months
58 Monitoring pollination and alerting farmers to take appropriate measures in timely manner	- To monitor pollination activity either naturally or through manual intervention by farmers and generate alerts in case of abnormalities - To enhance pollination by analyzing bee activity, weather conditions, and flower availability - To enhance pollination by analyzing manual hand pollination processes such as brushing etc.	●	●	6-12 months

ADAFSA



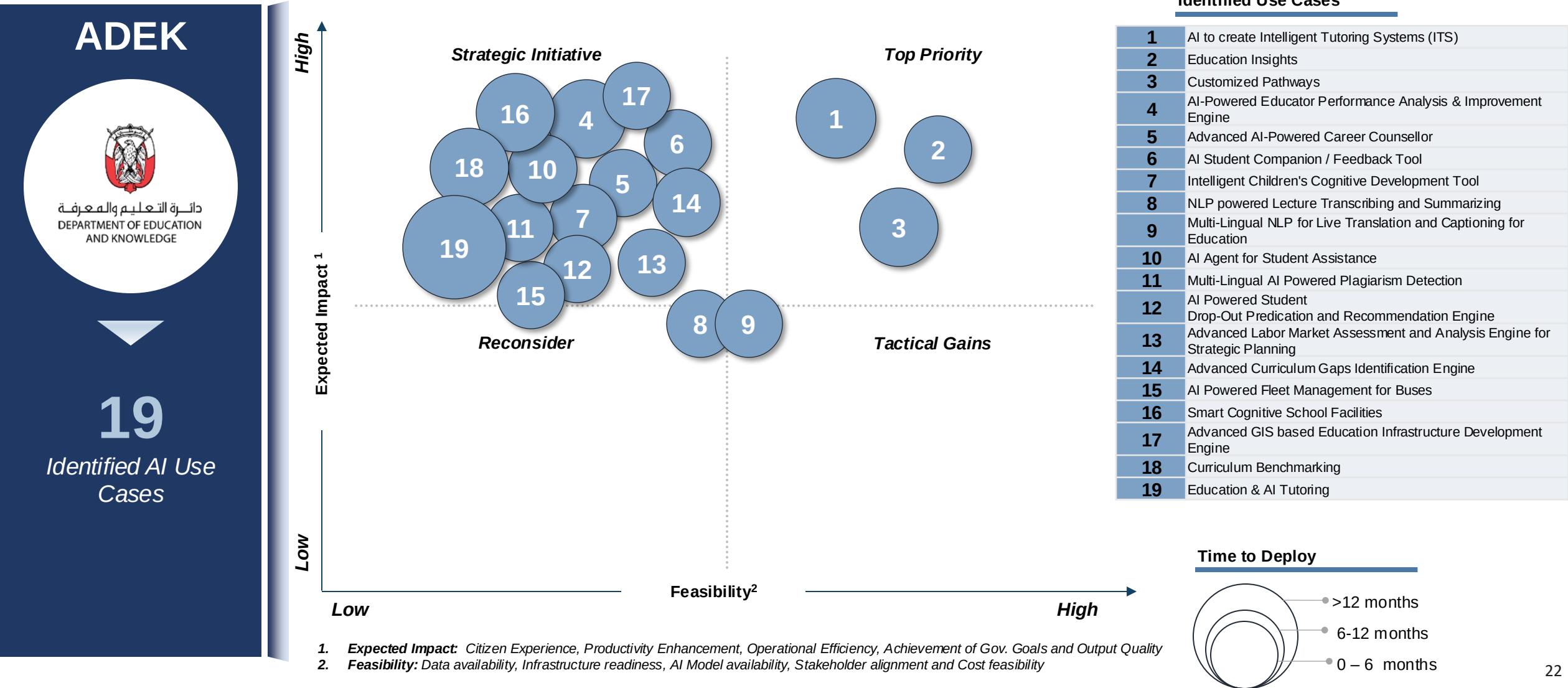
58

Identified AI Use Cases



ADEK – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



ADEK has prioritized 3 use cases from their identified long list

1



AI to create Intelligent Tutoring Systems (ITS)

- The challenge is providing individualized attention to each student given the student-teacher ratio. The opportunity is to use AI to create Intelligent Tutoring Systems (ITS) that provide personalized instruction and feedback to students.

Expected Impact

- This will primarily impact students and teachers
- The intended outcome is improved learning outcomes and efficiency in teaching
- The ITS could be scalable and reusable across different subjects and grade levels

2



Education Insights

- ADEK is enhancing its education system by leveraging AI to analyze data from PISA and other international assessments. This aims to deepen insights, elevate knowledge levels, and expand data analysis capabilities.

Expected Impact

- PISA and other international assessments are in the Emirate long term agenda, focusing on improving educational system
- ADEK is increasing knowledge about information available and today already start getting important insights
- This project can elevate the knowledge levels and expand the reach of the data analysis

3



Customized Pathways

- When creating customized pathways for academics, physical activities, or artistic talents, schools need to consider resource allocation and flexibility. Understanding a school's capacity for offering customized pathways is crucial.

Expected Impact

- AI in the field of nurturing talents can provide customized learning experience, identify students strengths and weakness, facilitating access to educational resources aligned with learners needs and preference



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Medium	Medium	Medium	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Low	High	High	Low

1. Business Overview

Description

- The purpose of this use case is to use AI to create Intelligent Tutoring Systems (ITS) that provides personalized instruction and feedback for students, thereby providing individualized attention to each student and overcoming the challenge of the student-teacher ratio. This would enhance the student experience, alleviate teachers' workload, and keep parents well informed of their children's progress
- Key beneficiaries of this use case are the students, teachers, school administrators, and parents

1.2

Pain Points

- Personalized learning
- Administrative efficiency
- Predictive analytics
- Student engagement
- Accessibility
- Research

1.3

Expected Impact

- Significantly enhance learning outcomes through highly personalized educational experiences tailored to students
- Optimizes the efficiency of teaching by reducing the time educators spend on routine tasks

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Data privacy concerns including confidentiality and security of student data
- Accuracy of the AI models to avoid misguided decisions

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Student performance data	TBD	TBD	TBD	TBD	Easy	High

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation Augmentation - Advisory - Interaction	- Dashboard Product/ app. - Embedded Model - Other:	- Descriptive Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition ML (supervised, unsupervised, reinforcement learning) Generative AI

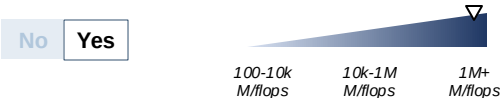
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient ML build/operational infrastructure.
- Adequate data for solutions.
- Partnership with AI firms or EdTech companies



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	High	Medium	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Low	Low	High	High

1. Business Overview

Description

- The purpose of this use case is to enhance the educational system managed by leveraging AI to analyze data from PISA and other international assessments (being in the Emirate's long-term agenda). This aims to deepen insights, elevate knowledge levels, and expand data analysis capabilities
- Key beneficiaries of this use case are ADEK, the students, teachers, school administrators and parents

1.2 Pain Points

- Personalized learning
- Administrative efficiency
- Predictive analytics
- Student engagement
- Accessibility
- Research

1.3 Expected Impact

- Increasing knowledge of existing information and deepen insight instantly
- Elevate knowledge levels across educational system and expand capacity for advanced data analysis driving improvement in educational strategies and outcomes

1.4 Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5 Risks

- Data privacy concerns including confidentiality and security of student data
- Accuracy of the AI models to avoid misguided decisions

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
International Assessment Data	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

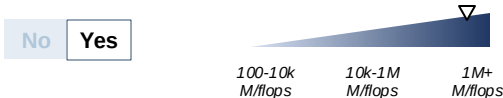
AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation Advisory - Interaction	- Dashboard - Product/ app. Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition ML (supervised, unsupervised, reinforcement learning) Generative AI

2.3 Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4 Key Considerations

- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- Adequate data for solutions
- Partnership with AI consulting firms or EdTech companies



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Medium	Medium	Medium	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Low	High	High	Medium

1. Business Overview

Description

- The purpose of this use case is to enhance the educational offerings by providing customized pathways for academics, physical activities, or artistic talents. These will allow students to pursue individual interests and strengths, leading to improved engagement and outcomes. However, implementing these pathways requires careful consideration of resource allocation and flexibility. Understanding a school's capacity to offer these personalized options is crucial for effective planning and execution
- Key beneficiaries of this use case are the students, teachers, school administrators and parents

1.2

Pain Points

- Personalized learning
- Administrative efficiency
- Predictive analytics
- Student engagement
- Accessibility
- Research

1.3

Expected Impact

- Enhanced student engagement
- Improved academic performance and student satisfaction
- Greater educational flexibility
- Enhanced teacher effectiveness
- Optimized resource use

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Data privacy concerns including confidentiality and security of student data
- Accuracy of the AI models to avoid misguided decisions

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Information on Talents	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

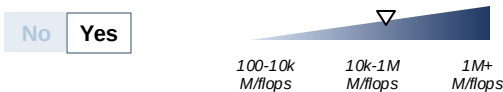
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Adequate data for solutions
- Secure personal data handling
- Partnership with AI consulting firms or EdTech companies

Furthermore, an additional of 16 use cases have been defined by ADEK

ADEK Long List of Non-prioritized Use Cases					
	Title	Description	AI Impact	Feasibility	Time to Deploy
4	AI-Powered Educator Performance Analysis & Improvement Engine	Supporting educators with course material analysis, teaching technique enhancements based on student learning styles using NLP and ML and identifying underperforming educators for improvement recommendations.	●	●	6-12 Months
5	Advanced AI-Powered Career Counsellor	Using supervised learning models to gather student data (including personality, academic performance, extracurricular activities) and guide academic and career decisions by predicting suitable courses,, and professions of interest based on their profiles	●	●	0-6 Months
6	AI Student Companion / Feedback Tool	Using an AI Teaching Assistant (chatbot) to provide personalized feedback, support students' learning journeys, and identify underperforming students for targeted improvement in Arabic language.	●	●	0-6 Months
7	Intelligent Children's Cognitive Development Tool	Improving a child early cognitive development by monitoring the child's interactions and speech using NLP and improving it based on the child's needs and build the child's unique developmental profile and personalized activity plan for each milestone as well as create reports to the parents	●	●	0-6 Months
8	NLP powered Lecture Transcribing and Summarizing	Using advanced speech recognition technology, transcribe and summarize lectures accurately by capturing key points mentioned by educators in real-time.	●	●	0-6 Months
9	Multi-Lingual NLP for Live Translation and Captioning for Education	Real-time lecture translation using deep learning and speech recognition, providing subtitles for educators' speech to aid absent students and accommodate diverse learning needs, with a focus on Arabic and other languages.	●	●	0-6 Months
10	AI Agent for Student Assistance	Using an AI-powered enrollment virtual assistant with NLP capabilities to support education applicants by answering questions, sending reminders, and aiding throughout the education journey	●	●	0-6 Months
11	Multi-Lingual AI Powered Plagiarism Detection	Detecting Arabic plagiarism using an AI-powered engine that can verify authorship quickly, effectively, and with substantial evidence.	●	●	0-6 Months
12	AI Powered Student Drop-Out Predication and Recommendation Engine	Predicting the possibility of a student's dropout using a predictive model based on the student's academic underperformance and suggest preventive measures.	●	●	0-6 Months

ADEK



19

Identified AI Use Cases

Furthermore, an additional of 16 use cases have been defined by ADEK

ADEK



19

Identified AI Use Cases

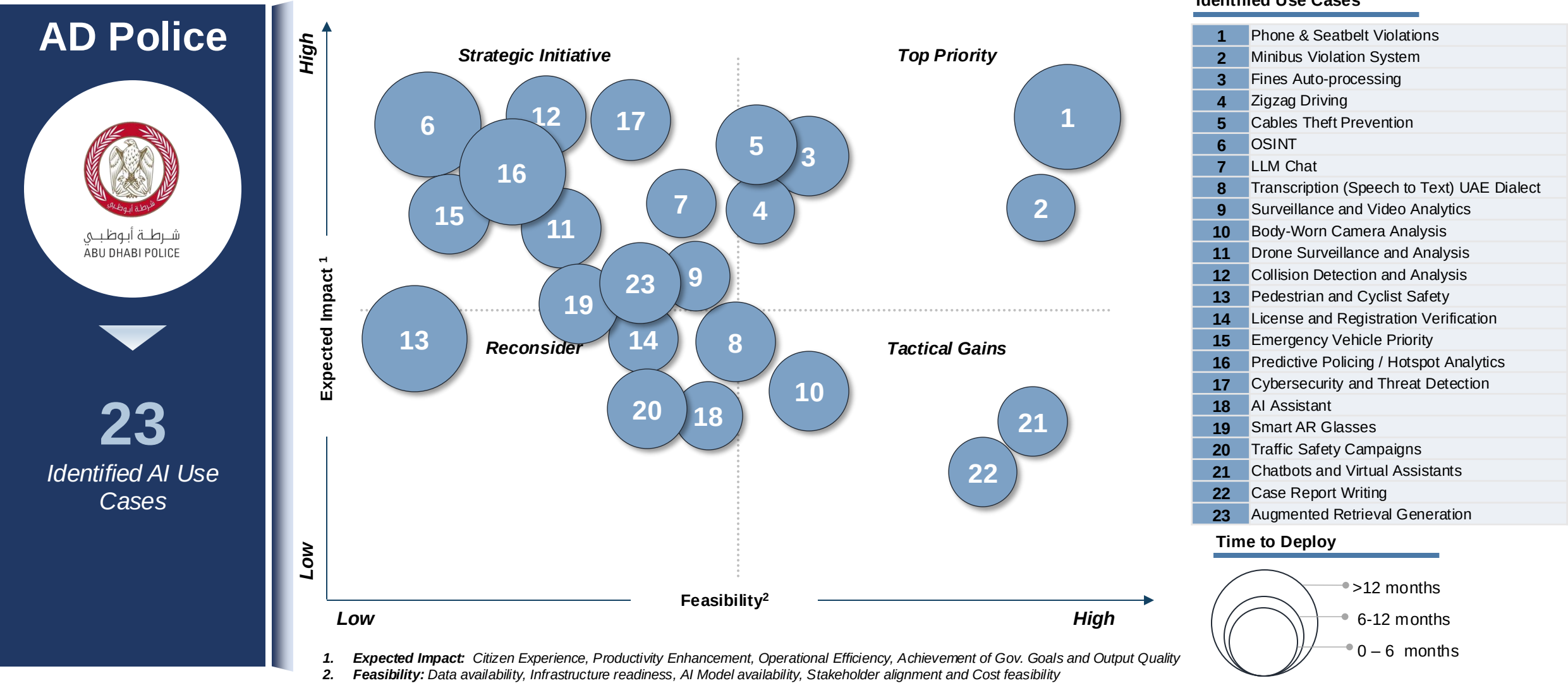
ADEK Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
13	Advanced Labor Market Assessment and Analysis Engine for Strategic Planning	Forecasting labor market demand and necessary skills across disciplines and regions by leveraging education trends, current programs, and capacity to predict the future workforce for the education sector.	●	●	0-6 Months
14	Advanced Curriculum Gaps Identification Engine	Analyzing academic entities' course curriculum using AI to give insights on possible curriculum gaps to ultimately improve the competitive offerings of universities.	●	●	0-6 Months
15	AI Powered Fleet Management for Buses	Enhancing the bus overall route and schedule to fit the students' needs as well as its safety procedures, leveraging multiple data sets (i.e. courses timing, students' schedules, locations) and different technologies such as IoT.	●	●	0-6 Months
16	Smart Cognitive School Facilities	Enhancing facility management with AI for activities like energy and inventory management and optimizing classroom design and resource allocation based on student numbers to improve efficiency and reduce costs.	●	●	6-12 Months
17	Advanced GIS based Education Infrastructure Development Engine	Using AI to analyze education infrastructure in regions/cities and identify optimal locations for schools based on capacity management, population growth, and other data models	●	●	0-6 Months
18	Curriculum Benchmarking	Enhance Academic Standards with Generative AI Insights	●	●	0-6 Months
19	Education & AI Tutoring	Benchmark and update educational curricula using AI insights and provide personalized tutoring with AI.	●	●	>12 Months

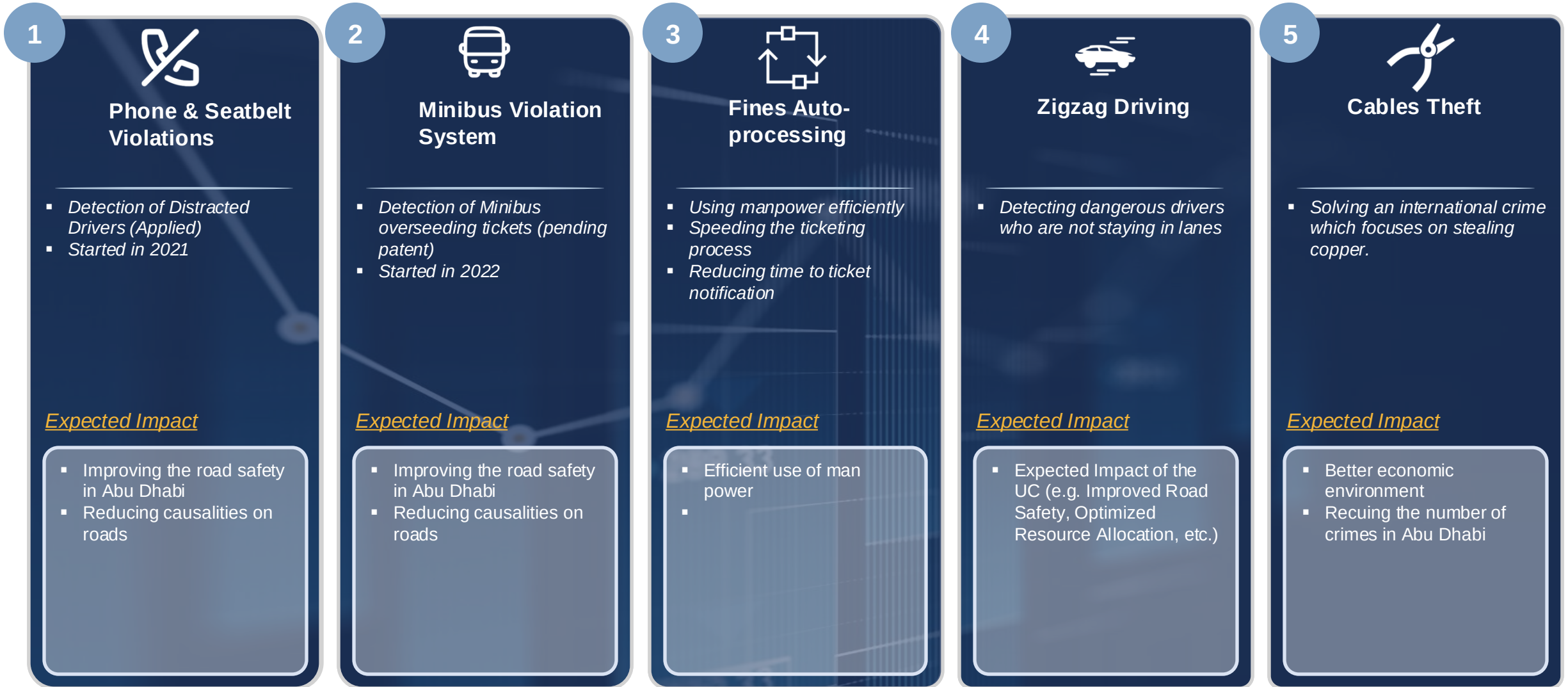


ADP – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



AD Police has prioritized 3 use cases from their identified long list





Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD AED	14 Months	15 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	Medium	High	Medium

Data Availability	Infrastructure Readiness	2. Feasibility	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Medium	High	High	Low	

1. Business Overview

Description

- One of the main challenges we face in road's are road accidents that are usually caused by the driver being distracted by focusing on his phone. Even with improvement in road infostructure and educational Campaigns to enlighten drivers, still most of accidents are caused from human error. Utilizing Computer vision Models, we could utilize it to detect drivers who are using their phones, and divers with passengers who not wearing their seatbelt to issue violations in order to lower the accidents in the road.

1.2

Pain Points

- Driver distraction from the road due to smart phones
- Not following regulations of using Seatbelt for both the driver and the passenger next to him

1.3

Expected Impact

- Using violations to enforce behavior changes from the drivers in order to lower number of extreme road accidents.
- In the cause critical accidents occurring, we expect the enforcement of seatbelt violation make passengers wear seatbelt and lower the severity of the injuries.

1.4

Expected Squad Required

- AI engineers
- Backend developers
- Data Engineer
- Data labelers
- Solution Architect
- Network engineer
- IOT engineer
- maintenance Team

1.5

Risks

- Extreme weather changes, which affect the model accuracy and may damage the edge devices).
- Model generalization to unexpected data.
- Edge device Availability
- Lack of labeled Dataset.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Phone and Seatbelt dataset	No available Data. Data is collected and labeled	Large Dataset	Images Collected from Cameras	ADP cameras	N/A	Highly Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

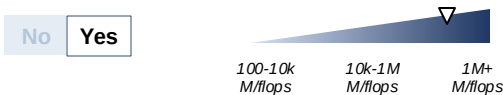
2.3

Cloud Services and Providers

- Technology Stack: Pytorch, RabbitMQ, FastAPI, DVC
- Cloud Services: No Cloud services were utilized

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Accurate data labeling pipeline to minimize False Positives
- Continues Active MLOPS process to improve the model generalizability
- Collaboration on maintenance and installation on new sites



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					TBD AED		3 Months	4 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Medium	High	High	High	High				

2. Feasibility				
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	High	Medium	High	TBD

1. Business Overview

1.1

Description

- Minibuses are an anomaly on current r installed radar systems, due to being categorize as light vehicle. These buses are utilized to transport passengers (Capacity of 14 max), however due to previously motioned challenge the drivers usually violated buses speed limited which endanger the lives of these passengers. Utilizing Computer vision Models allows to violate minibuses that are going over the speed limit

1.2

Pain Points

- The inability of Radar systems to distinguish between minibus and minivans.

1.3

Expected Impact

- Lower the potential of deadly and catastrophic accidents by enforce fines on Minibuses>

1.4

Expected Squad Required

- AI Engineer
- Data Labelers
- Soulution Architect

1.5

Risks

- Potential of False Negative being filter out by model prediction.
- Lack of model Generalizability due to sample size of the dataset.

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Minibuses and minivans (classification)	Yes	Small to medium size	Fine database	ADP	e.g. Requires Integration	Confidential

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: Pytorch, Rabbitmq, FastAPI
- Cloud Services: None

2.4

Key Considerations

- Accurate data labeling pipeline to minimize False Positives
- Continues Active MLOPS process to improve the model generalizability

GPU Enabled Infra

Computational Power

No

Yes

100-10k M/flops

10k-1M M/flops

1M+ M/flops



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD AED	8 Months	4 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	Medium	High

Data Availability	Infrastructure Readiness	2. Feasibility AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	High	High	High	Low

1. Business Overview

Description

- Before a fine is issued a human operator validates and fill the vehicle details before issuing it and fining the vehicle. The challenge is the huge queue of fines that are waiting to be procced by the human operator. This inherently increase the duration between the fine registering from the radar to be issued out. With the advancements in computer vision and ANPR technology an ensemble models of ANPR models can Automate the fine issuing stage with significantly lower error rates with human operators, and slashes the duration of fines issuing significantly

1.2

Pain Points

- Huge Amount of fines in queue that take long time to be procced.

1.3

Expected Impact

- Lower the operational cost of human operators by decreasing their numbers.
- Lower the KPI of fine trip through the system

1.4

Expected Squad Required

- AI engineers
- Backend developers
- Data Engineer
- Data labelers
- Solution Architect

1.5

Risks

- The inability of the models to identify plates of countries not trained on
- The diversity of Plates in UAE and the continues update of new plates that require to update and model.
- The models inaccurate Reading due to low-light images, weather-condition, or plate image being lower than the reequipment dimension

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Object Detection and Classification of plates	Yes	2 M Images	Fine system	ADP	Yes	Confidential

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: Pytorch, Fast API
- Cloud Services: None

GPU Enabled Infra

No Yes

Computational Power

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4

Key Considerations

- Accurate data labeling pipeline to improve accuracy
- Continues Active MLOPS process to improve the model generalizability with active learning integration to adapt to continues new plates that require identifying
- Collaboration with operators to identify potential weakness and improve the models overtime.
- Requirement of diverse CV models to minimize error rate.



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD AED	0-6 Months	4 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	Medium	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	High	High	High	Low

1. Business Overview

Description

- Zigzag Driving resemble a traffic problem in the UAE, yet not system can detect the misbehaviors on roads. Therefore, ADP is developing a customized system to detect the violator.

1.2

Pain Points

- Dangerous driving behaviors that cannot be detected

1.3

Expected Impact

- Improving the safety on Abu Dhabi Roads

1.4

Expected Squad Required

- AI engineers
- Backend developers
- Data Engineer
- Data labelers
- Solution Architect

1.5

Risks

- The inability of the models to accurately identify the violation

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Object Detection and Classification of plates	Yes	2 M Images	Fine system	ADP	Yes	Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- Technology Stack: Pytorch, Fast API
- Cloud Services: None

GPU Enabled Infra

Computational Power

No Yes

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4

Key Considerations

- Accurate data labeling pipeline to improve accuracy
- Continues Active MLOPS process to improve the model generalizability with active learning integration to adapt to continues new plates that require identifying
- Collaboration with operators to identify potential weakness and improve the models overtime.
- Requirement of diverse CV models to minimize error rate.



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD AED	6-12 Months	4 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	Medium	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	High	High	High	Low

1. Business Overview

Description

- Stealing copper cables represents a global crisis in the construction realm. Tons of copper are stolen proportionally with the international copper prices.

1.2

Pain Points

- Crimes cannot be identified

1.3

Expected Impact

- Improving the economic eco system in Abu Dhabi

1.4

Expected Squad Required

- AI engineers
- Backend developers
- Data Engineer
- Data labelers
- Solution Architect

1.5

Risks

- NA

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Object Detection and Classification of plates	Yes	2 M Images	Fine system	ADP	Yes	Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- Technology Stack: Pytorch, Fast API
- Cloud Services: None

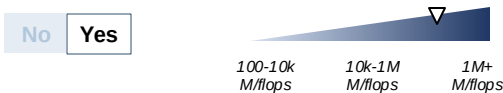
2.4

Key Considerations

- Collaboration with utility companies and DEDs.
- Changing the laws and rules which ae governing used copper exchange.

GPU Enabled Infra

Computational Power



Furthermore, an additional 20 use cases have been defined by AD Police

AD Police



23

Identified AI Use Cases

AD Police Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
6	OSINT	Monitoring the open source platform including Darkweb	High	Low	>12 months
7	LLM Chat	Developing a specialized chatbot that takes the confidentiality of ADP in consideration	High	Medium	4-6 months
8	Transcription (Speech to Text) UAE Dialect	Transcribing the Emirati dialect to speed to shorten response time	Medium	Medium	6-12 months
9	Surveillance and Video Analytics	Utilize AI to analyze surveillance footage in real-time for suspicious behavior, crowd anomalies, or unattended objects	High	Medium	0-6 Months
10	Body-Worn Camera Analysis	AI can be used to analyze footage from body-worn cameras, dashboard cameras, and in-vehicle (360-degree view) cameras to detect critical incidents and identify behavioral patterns requiring attention	Medium	Medium	6-12 Months
11	Drone Surveillance and Analysis	Use AI-enabled drones for aerial surveillance during public events, protests, or emergencies	High	Medium	6-12 Months
12	Collision Detection and Analysis	Implement AI to detect collisions in real-time, alerting emergency services and assisting with post-incident analysis to determine causes	High	Medium	6-12 Months
13	Pedestrian and Cyclist Safety	Use AI-based sensors to detect pedestrians and cyclists at intersections, adjusting traffic signals to enhance safety	Medium	Low	>12 Months
14	License and Registration Verification	Utilize AI to automatically verify vehicle registration and driver licenses during traffic stops or roadside checks	Medium	Medium	0-6 Months
15	Emergency Vehicle Priority	Use AI to create adaptive traffic signals that prioritize emergency vehicles, allowing them to move through traffic more efficiently	High	Low	6-12 Months
16	Predictive Policing / Hotspot Analytics	Predict future potential crimes and accidents by identifying crime hotspots and times based on historical data (e.g., criminals, location, accident types) and additional triggers such as weather, events, holidays allowing for more efficient police resource allocation in terms of location, resources, etc.	High	Low	>12 Months
17	Cybersecurity and Threat Detection	Implement AI-based cybersecurity tools to detect and respond to cyber threats against police networks and databases	High	High	6-12 Months
18	AI Assistant	Support the security operations team with alert monitoring, incident triage, incident documentation, escalation and routine reporting	Medium	Medium	0-6 Months
19	Smart AR Glasses	Leverage Augmented Reality glasses combined with Artificial Intelligence and facial recognition software to identify suspects on streets	Medium	Medium	6-12 Months

Furthermore, an additional 20 use cases have been defined by AD Police

AD Police



23

Identified AI Use Cases

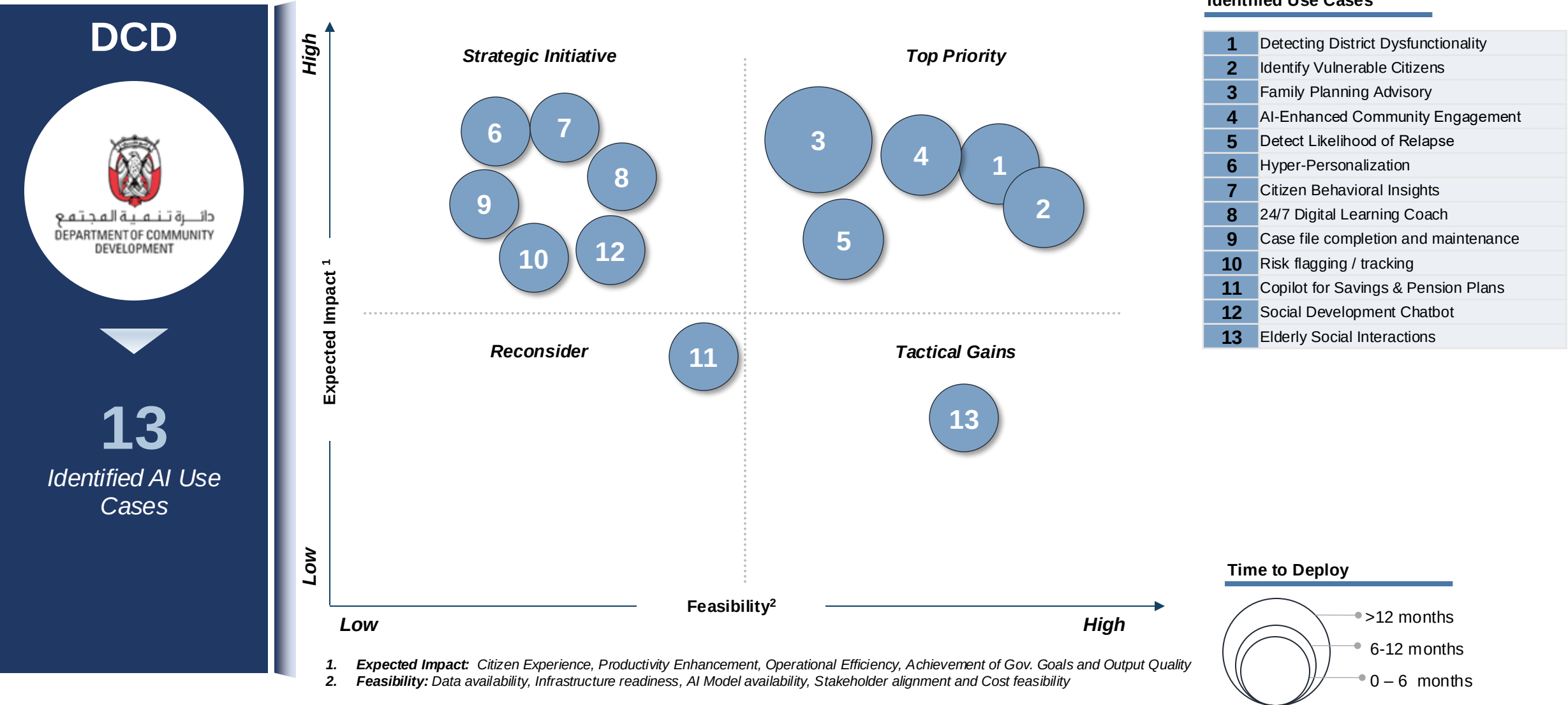
AD Police Long List of Non-prioritized Use Cases

Title	Description	AI Impact	Feasibility	Time to Deploy
20 Traffic Safety Campaigns	Implement AI to analyze traffic data and determine the most effective messaging for safety campaigns and educational programs	●	●	6-12 Months
21 Chatbots and Virtual Assistants	Implement AI-powered chatbots for non-emergency inquiries, helping to reduce the workload on human dispatchers	●	●	0-6 Months
22 Case Report Writing	Minimize errors (e.g., spelling), maintain consistent terminology, and ensure accurate data capture on critical information (e.g., license plate number) in police reports	●	●	0-6 Months
23 Augmented Retrieval Generation	Implement AI-based systems to access historical documents and enable officers to retrieve relevant information quickly, answer questions, or generate summaries, saving time in report preparation and legal processes	●	●	6-12 Months



DCD – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



DCD has prioritized 5 use cases from their identified long list

1



Detect District Dysfunctionality

- Detects signs of dysfunctionality within districts in the Emirate of Abu Dhabi

Expected Impact

- Better allocation of resources
- Improved living conditions and safety
- Implementation of preventive measures
- Fostering community engagement
- Developed collective responsibility

2



Identify Vulnerable Citizens

- Identifies at-risk individuals, who are highly vulnerable across multiple dimensions

Expected Impact

- Improved mitigation of social vulnerabilities and prevention for individuals to transition into dysfunctional or adverse social conditions

3



Family Planning Advisory

- Helps families in financial, educational and parenting management

Expected Impact

- Personalized financial planning
- Optimized educational planning, childcare and parenting support, and retirement planning
- Coordinated advisory services tailored to individual needs

4



AI-Enhanced Community Engagement

- Enhances community engagement for both donations and volunteering

Expected Impact

- Enhanced community engagement
- Improved matching of donors and volunteers with relevant projects
- Fostered community bonds

5



Detect Likelihood of Relapse

- Assesses the likelihood of relapse in substance abusers

Expected Impact

- Improved identification of triggers and risk factors associated with relapse
- Personalized patient care
- Improved timely intervention
- Reduced rates of relapse



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~5M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Low	Medium	High	High

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Medium	High	High

1. Business Overview

Description

- The purpose of the use case is to detect signs of dysfunctionality within districts of the Emirate of Abu Dhabi. The system aims to identify potential areas of concerns within the districts in real-time, allowing timely intervention and targeted allocation of resources to address emerging issues
- Key beneficiaries of this use case cover the whole of society within different districts in the Emirate of Abu Dhabi

1.2 Pain Points

- Presence of crimes, poverty, infrastructure decay, and social unrest
- Escalation of some issues before mitigation
- Delay in actions taken

1.3 Expected Impact

- Better allocation of resources
- Improved living conditions and enhanced safety
- Implementation of preventive measures
- Fostering community empowerment and engagement
- Developed collective responsibility between organizations

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Variance in human behavioral aspects which may cause fluctuation in model's result
- Transparency based on the elected algorithm, this should be mitigated through optimization

2. Data, AI and Infrastructure Overview

Data Requirements

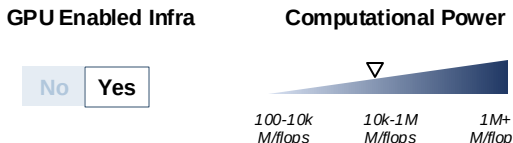
Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Socio-Economic, Demographic, Environmental, Crime, Infrastructure and Livability	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage



2.4 Key Considerations

- Sufficient ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Stakeholder Alignment: Ensure collaborative integration
- Ethical Considerations: Uphold strict ethical standards



Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility					~5M AED	6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility			
High	Low	Medium	High	High	Medium	Medium	Medium	High	High			

1.1

1. Business Overview

Description

- The purpose of the use case is to identify and assist vulnerable individuals within a community. The use case aims to identify individuals who may be at risk due to factors such as poverty, health issues, disabilities, or social isolation.
- Key beneficiaries of this use case cover the whole of society within different districts in the Emirate of Abu Dhabi

1.2

Pain Points

- Difficulty in identifying vulnerable individuals
- Limited resources for providing support to vulnerable populations
- Lack of targeted automatic identification of those in need

1.3

Expected Impact

- Early identification avoiding citizen dissatisfactory vulnerable state
- Reduction of waste and increased efficacy of social programs
- Personalized support tailored interventions
- Preventative measures with more cost-effectiveness

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Variance in human behavioral aspects which may cause fluctuation in model's result
- Transparency based on the elected algorithm, this should be mitigated through optimization
- Privacy and data security
- Challenges in integrating and standardizing data from disparate sources

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Socioeconomic, Demographic, Employment, Household, Complaint Records, Criminal History Information, Automated Survey and Complaints Data	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No

Yes

Computational Power

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Stakeholder Alignment: Collaborative engagement essential
- Ethical Considerations: Strict adherence to ethical standards



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					5-20M AED		>12 Months	11 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
High	Low	Medium	High	High				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
Medium	Medium	Medium	High	Medium	

1. Business Overview

1.1

Description

- The purpose of the use case is to provide personalized and accessible guidance and support to individuals and families to help individuals make informed decisions about family planning and achieve their desired reproductive goals
- Key beneficiaries of this use case cover the whole of society within different districts in the Emirate of Abu Dhabi

1.2

Pain Points

- Limited access to reliable information and resources
- Lack of personalized guidance and support tailored to individual needs and preferences

1.3

Expected Impact

- Personalized financial planning
- Optimized educational planning, childcare and parenting support, and retirement planning
- Coordinated advisory services tailored to individual needs

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineers
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect

- 2 AI Testers

1.5

Risks

- Variance in human behavioral aspects which may cause fluctuation in model's result
- Transparency based on the elected algorithm, this should be mitigated through optimization
- Privacy and data security
- Resistance from certain communities regarding family planning initiatives

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Income, Expense, Asset, Liability, Financial Goals, Personal Demographics	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No

Yes

Computational Power

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Ease of integration
- Dataset Collaboration: Necessary data collection partnership



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					~5M AED		6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
High	Low	Medium	High	High				
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility				
Low	Medium	Medium	High	High				

1. Business Overview

1.1

Description

- The Purpose of the use case is to enhance community engagement for both donations and volunteering
- Key beneficiaries of this use case include whole of society within different districts in the Emirate of Abu Dhabi

1.2

Pain Points

- Limited information on sense of community engagement as well as ideas to enhance it

1.3

Expected Impact

- Increased community engagement
- Enhanced community impact
- Strengthened Community Bonds
- Optimized resource allocation

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineering
- 0.5 UX/UI Designer
- 0.5 Solution Architect

- 1 Tester (AI Engineering tester and Data Science tester)

1.5

Risks

- Variance in human behavioral aspects which may cause fluctuation in model's result

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Donor and Volunteer Profile Data, Donation Opportunities & records, Forum Posts, Organization Details, Past Donation and Volunteer Activities	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No

Yes

Computational Power

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Gen AI build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Ease of Integration
- Data Dependency and Social Expert Input: Essential for accurate modeling
- Proactive Data Integration: Ensure future scalability and flexibility



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~5M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Low	Medium	High	Medium

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Medium	Medium	High	High

1. Business Overview

Description

- The Purpose of the use case is to assess the likelihood of relapse in substance abusers
- Key beneficiaries of this use case include whole of society within different districts in the Emirate of Abu Dhabi

1.2

Pain Points

- Limited information on individual likelihood to relapse back to use of substance which does not allow proactive measures to be taken

1.3

Expected Impact

- Improved identification of triggers and risk factors associated with relapse
- Personalized patient care
- Improved timely intervention
- Reduced rates of relapse

1.4

Expected Squad Required

- | | |
|--|---|
| <ul style="list-style-type: none">1 Scrum Master1 Business Analyst2 AI Scientists2 Data Engineering0.5 UX/UI Designer0.5 Solution Architect1 Tester (AI) | Engineering tester and Data Science tester) |
|--|---|

1.5

Risks

- Variance in human behavioral aspects which may cause fluctuation in model's result
- Transparency based on the elected algorithm, this should be mitigated through optimization

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Treatment History, Biometric Data, Behavioral Data, Psychosocial Indicators, Mental Health Records, Substance Craving Reports, Genetic data	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

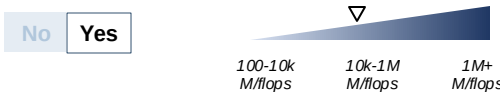
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Data Dependency and Social Expert Input: Essential for model accuracy
- Proactive Data Integration: Prepare for scalable data handling

Furthermore, an additional 8 use cases have been defined by DCD

DCD



13

Identified AI Use Cases

DCD Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
6	Hyper-Personalization	Allowing generated content to be contextualized by region/persona to support cohort specific and place-based learning and policy development	High	Medium	0-6 Months
7	Citizen Behavioral Insights	Aggregate large volumes of data and information to identify trends and patterns of behaviors across beneficiary cohort groups	High	Medium	0-6 Months
8	24/7 Digital Learning Coach	Equipped with domain-specific knowledge available to new and existing staff on-demand	High	Medium	0-6 Months
9	Case file completion and maintenance	Screening and workflow capabilities that can enhance information integrity and flag missing information credentials	High	Medium	0-6 Months
10	Risk flagging / tracking	Ensure that vulnerable beneficiaries including children and young people remain visible within the system	High	Medium	0-6 Months
11	Copilot for Savings & Pension Plans	Gen AI enabled navigation across complex savings & pension documents	Medium	Medium	0-6 Months
12	Social Development Chatbot	Leverage GenAI to enhance citizens' research efficiency for better legal outcomes, streamlining the reconsideration and appeal process for quick resolution	High	Medium	0-6 Months
13	Elderly Social Interactions	Create immersive social interactions and companionship for the elderly through AI platforms	Medium	High	0-6 Months



Low



Medium

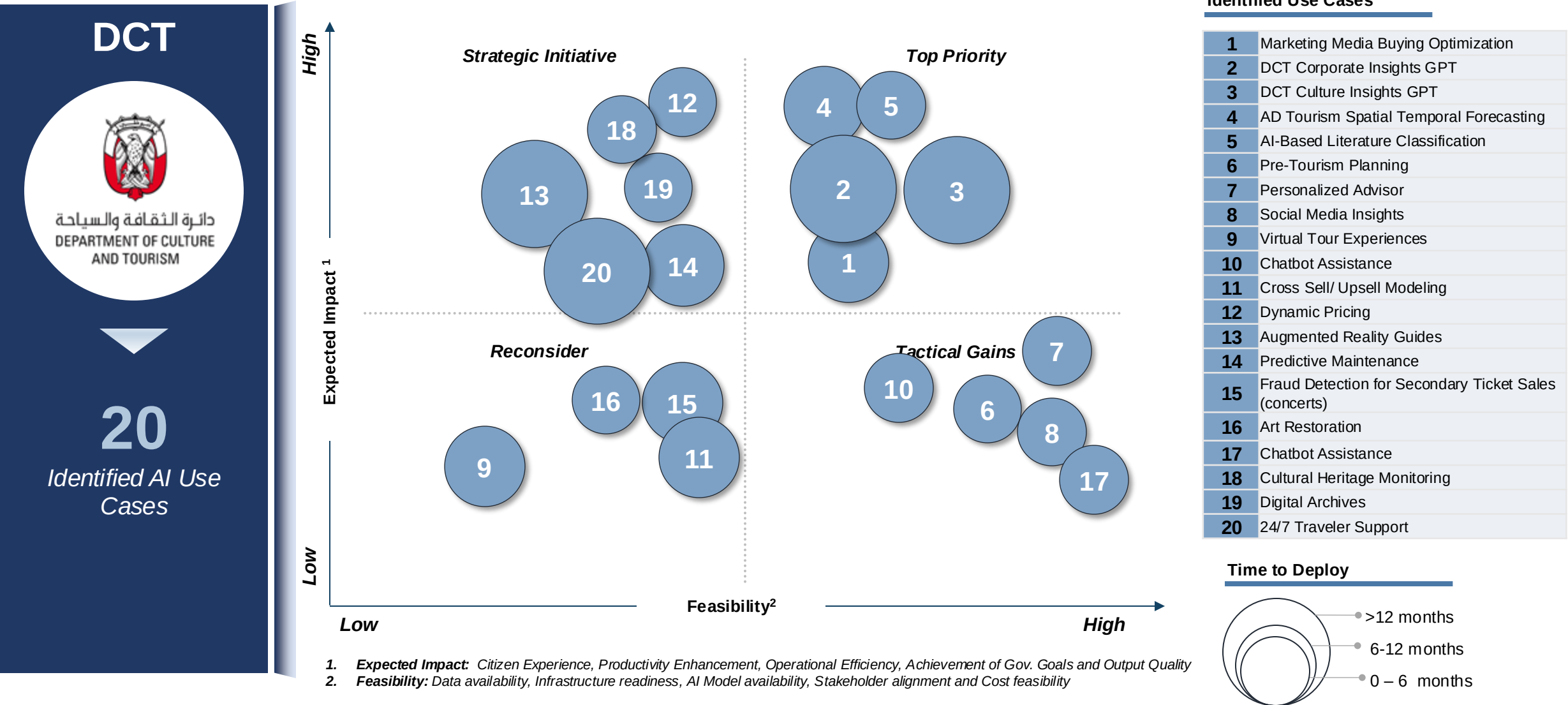


High

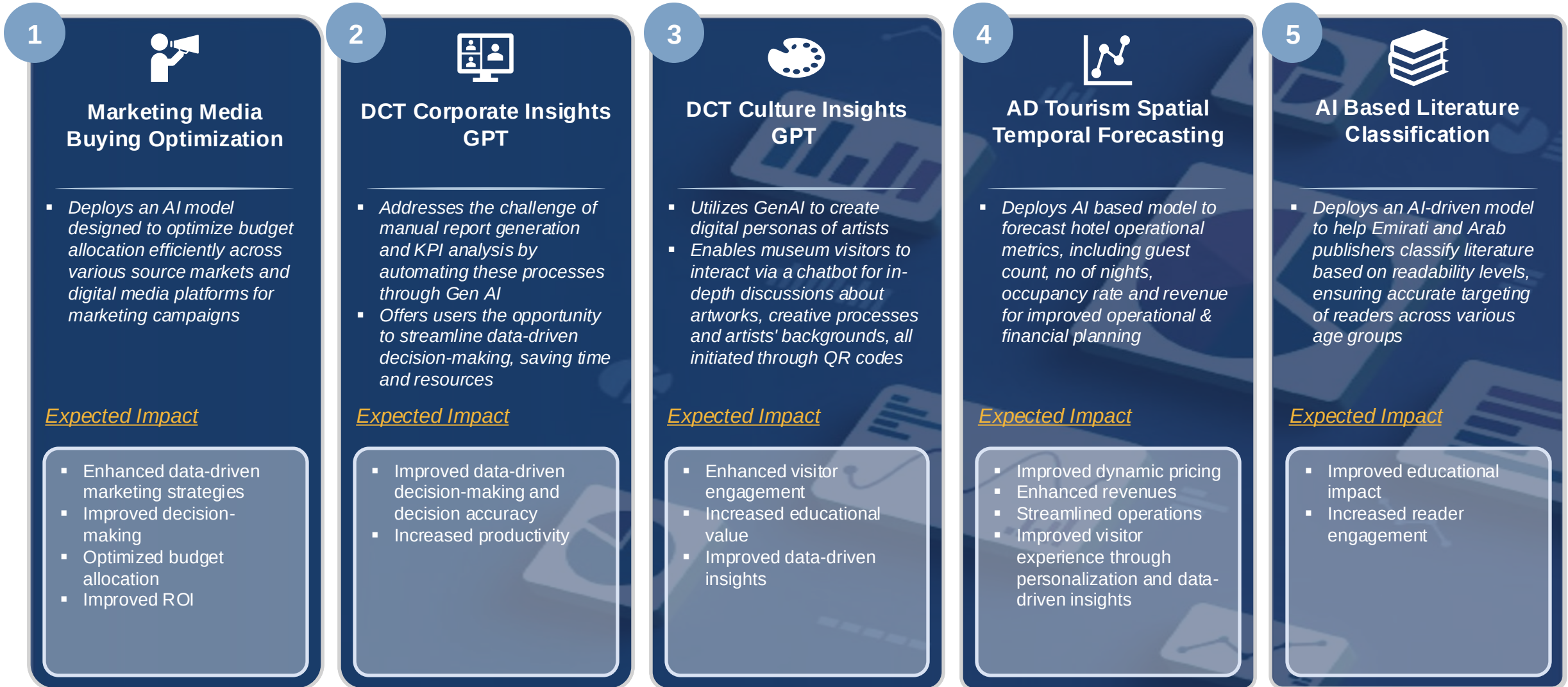


DCT – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



DCT has prioritized 5 use cases from their identified long list





Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	Medium	High	High	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	High	High	Medium	Medium

1. Business Overview

Description

- The purpose of the use case is to deploy an AI model designed to optimize budget allocation efficiently across various source markets and digital media platforms for marketing campaigns
- Key beneficiaries of this use case include the marketing and strategic affairs within the Department of Culture and Tourism

1.2

Pain Points

- Challenges in manual budget allocation and decision making
- Identifying and targeting the right audience

1.3

Expected Impact

- Improved Campaign ROI
- Improved Brand Awareness
- Increased Bookings & Revenue
- Optimal Budget Allocation

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineers
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Inaccurate or biased AI-generated insights from flawed data input or poorly designed models, leading to suboptimal media buying decisions.
- Data quality and availability ensuring that the insights generated are reliable.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Historical Marketing, Visitation & Revenue	36 months	TBD	TBD	TBD	TBD	No PPI

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

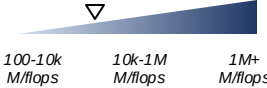
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Local and Global Data Needs: Manage diverse marketing data
- Comparative Information: Crucial performance comparisons
- Visualization tools and its connectivity with DB and analytics system



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	High	High	Low

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	High	Medium	Medium

1. Business Overview

Description

- The purpose of this use case is to address the challenge of manual report generation and KPI analysis by automating these processes through Gen AI. It offers users the opportunity to streamline data-driven decision-making, saving time and resources while enabling insightful analysis without specialized expertise
- Key beneficiaries of this use case include the Department of Culture and Tourism, and various government entities

1.2

Pain Points

- Time consuming manual reporting
- Human error and inconsistencies

1.3

Expected Impact

- Improved Productivity by reducing number of manhours
- Cost Saving
- Improved Quality of Insight
- Improved Decision Accuracy

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI/ Gen AI Scientists
- 2 Data Engineers
- 1 Full Stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5

Risks

- Inaccurate or biased AI-generated insights due to model hallucination or wrong prompts leading to incorrect decision-making
- Dependencies as it requires deep collaboration from Internal stakeholders

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Historical data	To be sourced	TBD	TBD	TBD	TBD	No PPI

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLMs, Knowledge Graphs, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No Yes

Computational Power

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4

Key Considerations

- Infrastructure Needs for Gen AI Model: Robust, scalable AI infrastructure
- Data Privacy Considerations: Ensure regulatory compliance
- Ethical Considerations: Govern with clear ethics
- Visualization tools and its connectivity with DB and analytics system



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Medium	High	Low	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	High	High	Medium	Medium

1. Business Overview

Description

- The purpose of this use case is to transform art and culture appreciation through utilizing GenAI to create digital personas of artists, enabling museum visitors to interact via a chatbot for in-depth discussions about artworks, creative processes and artists' backgrounds, all initiated through QR codes.
- Key beneficiaries of this use case include the Department of Culture and Tourism, the culture sector in general, and Abu Dhabi Museums

1.2

Pain Points

- Lack of visitor engagement
- Lack of experience personalization
- Language barriers of tourists
- Limitations in providing in-depth information

1.3

Expected Impact

- Higher Educational Impact
- Enhanced visitor engagement
- Higher Visitor Satisfaction
- Improved Visitor Engagement
- Higher Visitor Numbers
- Data driven insights

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI/ Gen AI Scientists
- 2 Data Engineers
- 1 Full Stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5

Risks

- Inaccurate or biased AI-generated insights due to model hallucination or wrong prompts may lead to incorrect decision-making

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Historical data	To be sourced	TBD	TBD	TBD	TBD	No PPI

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLMs, Knowledge Graphs, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Infrastructure Needs for Gen AI Model: Scalable AI-ready infrastructure
- Data Privacy Considerations: Strict compliance management
- Ethical Considerations: Robust ethical oversight



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	High	High	Medium

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Medium	Medium	AI Model Complexity	High	Low

1. Business Overview

Description

- The purpose of this use case is to deploy an AI based model to forecast hotel operational metrics, including guest count, number of nights, occupancy rates and revenue, to enhance operational and financial planning.
- Key beneficiaries of this use case include tourism operators and hotelier stakeholders, as well as the strategy team in DCT

1.2

Pain Points

- Time and effort required for effective inventory management
- Time and effort required for effective staffing optimization

1.3

Expected Impact

- Improved Guest Satisfaction
- Increased revenue
- Improved Occupancy
- Streamlined operations

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Insufficient availability of data for all source markets
- Volatile Macro Indicators like Wars/Conflicts, Pandemics etc

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Historic KPIs	36 months	TBD	TBD	TBD	TBD	TBD
External data like Mobility	36 months					
Flight searches & connectivity	36 months					

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

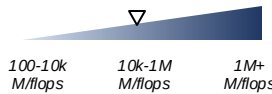
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for CPU Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

NoYes



2.4

Key Considerations

- KPIs and Historical Data: Accurate KPI collection
- External Data Usage: Enhance models with external data
- Visualization tools and its connectivity with DB and analytics system
- Ease of integration with internal systems



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	Medium	High	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	High	Medium	Medium

1. Business Overview

Description

- The purpose of this use case is to deploy an AI-driven model to help Emirati and Arab publishers classify literature based on readability levels, ensuring accurate targeting of readers across various age groups.
- Key beneficiaries of this use case include DCT culture sector, writers, publishers, and the wider reader community

1.2

Pain Points

- Difficulty in accurate targeting of readers
- Reduced reader experience

1.3

Expected Impact

- Reader Satisfaction
- Engagement Metrics
- Arabic reading proficiency
- No. of Arabic Books Read
- Improved Educational Impact
- Increased Reader Engagement

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Inaccurate or biased AI-generated insights due to model hallucination or wrong prompts may lead to incorrect decision-making

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Words & Texts	TBD	>10M words and >20,000 texts	TBD	Emirati writers & publishers	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

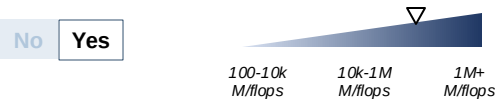
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Infrastructure Needs for Gen AI Model: Advanced, scalable AI infrastructure
- Data Size, Quality, and Integrity: Ensure comprehensive, high-quality datasets
- Ethical Considerations: Uphold ethical standards in AI application

Furthermore, an additional 15 use cases have been defined by DCT

DCT



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Identified AI Use Cases

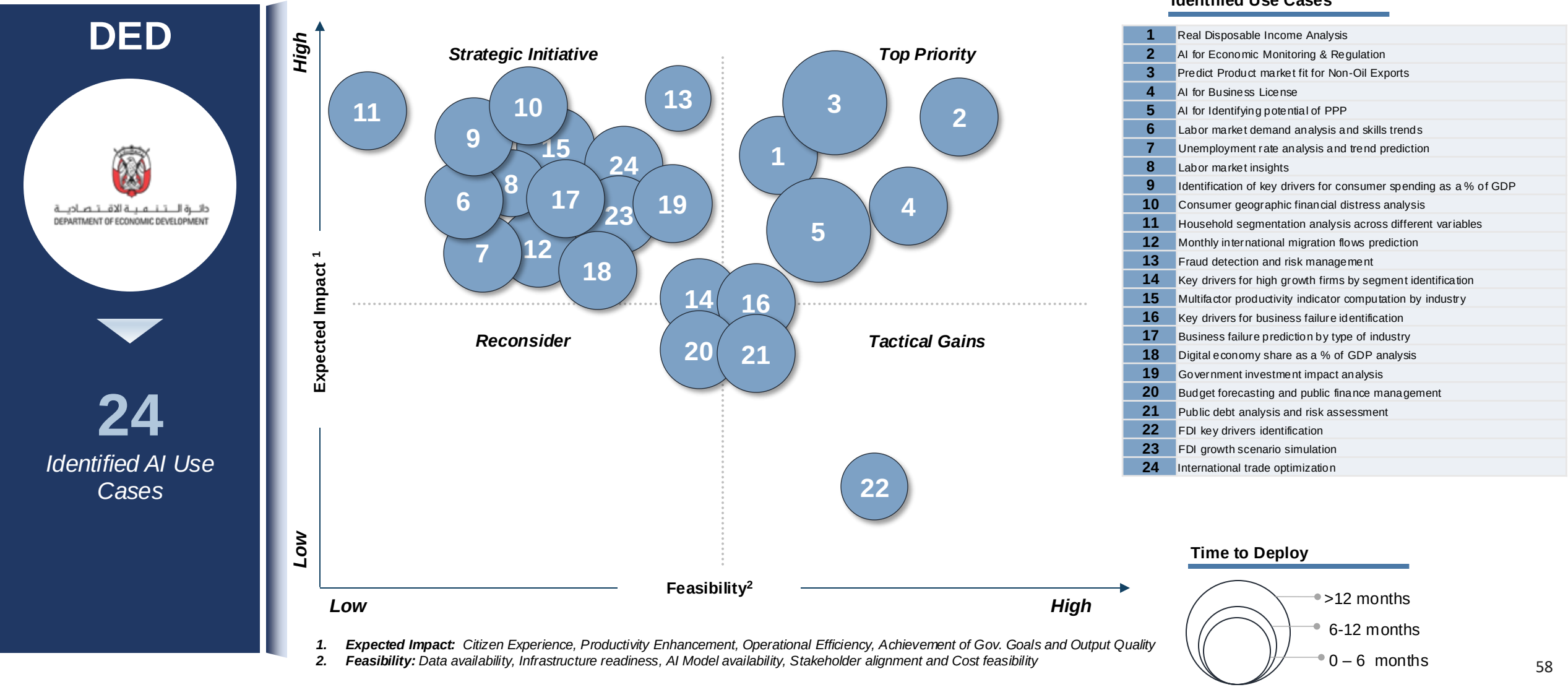
DCT Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
6	Pre-Tourism Planning	AI algorithms can analyze data and tourist travel patterns to optimize travel routes, reduce ecological impact, and promote sustainable tourism practices	●	●	0-6 Months
7	Personalized Advisor	AI-driven recommendation engines can analyze user preferences and behavior to offer personalized travel suggestions, enhancing the overall tourism experience	●	●	0-6 Months
8	Social Media Insights	AI tools can analyze social media trends and sentiments to identify popular destinations, create targeted marketing campaigns, and attract more tourists	●	●	0-6 Months
9	Virtual Tour Experiences	AI and virtual reality technologies can create immersive virtual tour experiences of cultural sites, allowing potential tourists to explore destinations remotely and encourage visitation	●	●	6-12 Months
10	Chatbot Assistance	AI-powered chatbots can provide real-time assistance to tourists, offering information about attractions, local services, and cultural events to enhance their travel experience	●	●	0-6 Months
11	Cross Sell/ Upsell Modeling	Leverage hotel guests' database for personalization campaigns and cross-selling adjacent services	●	●	6-12 Months
12	Dynamic Pricing	Use dynamic pricing algorithms to optimize packages and hotel accommodation process	●	●	0-6 Months
13	Augmented Reality Guides	AI-driven augmented reality apps can overlay digital information, historical facts, and interactive guides onto real-world scenes, enriching tourists' understanding and engagement with cultural sites	●	●	>12 Months
14	Predictive Maintenance	AI analytics can predict maintenance needs for tourist facilities and infrastructure, ensuring a seamless and enjoyable experience for visitors by minimizing downtime and disruptions	●	●	6-12 Months
15	Fraud Detection for Secondary Ticket Sales (concerts)	Use AI to identify sale/ purchase of concert tickets in secondary markets	●	●	6-12 Months
16	Art Restoration	AI-powered image analysis and restoration algorithms can assist in restoring damaged artworks, manuscripts, and artifacts, preserving their original aesthetic and historical value	●	●	0-6 Months
17	Chatbot Assistance	AI-powered chatbots can provide real-time assistance to tourists, offering information about attractions, local services, and cultural events to enhance their travel experience	●	●	0-6 Months
18	Cultural Heritage Monitoring	AI-based image recognition and monitoring systems can detect and prevent unauthorized activities such as vandalism or illegal excavation at cultural heritage sites	●	●	0-6 Months
19	Digital Archives	AI can help digitize and preserve cultural artifacts, documents, and historical records, ensuring their long-term accessibility and conservation	●	●	0-6 Months
20	24/7 Traveler Support	Deploy AI-driven chatbots to provide real-time assistance to citizens, available anytime, anywhere.	●	●	>12 Months



DED – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



DED has prioritized 5 use cases from their identified long list





Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					~5-20M AED		6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Medium	Low	Low	High	High				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
Low	Medium	Low	Medium	Medium	

1. Business Overview

1.1 Description

- The purpose of the use case is to leverage AI for real-time income tracking using various data sources (e.g., bank transactions) for a view on disposable income across different demographics and using predictive AI to forecast how disposable income might change based on economic indicators, policy changes, and other factors.
- Key beneficiaries of this use case include ESO

1.2 Pain Points

- Without AI-driven analysis, economic development agencies lack precise insights into market trends and investment opportunities, making it challenging to attract potential investors effectively.

1.3 Expected Impact

- Gain real-time and accurate view of financial well-being of the population
- Enable effective economic policy making and taking relevant measures to support economic growth

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineering
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 Tester (AI)

Engineering tester and Data Science tester)

1.5 Risks

- Ethical considerations around how income data is collected, used, and shared
- AI models heavily rely on the accuracy and quality of input data. Inaccurate or incomplete data can lead to biased or misleading analysis, affecting the reliability of income trends and insights derived from the analysis

2. Data, AI and Infrastructure Overview

2.1 Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3 Cloud Services and Providers

- Technology Stack: Databricks, Azure cognitive services, Azure ML studio, blob storage, SQL service, AKS
- Cloud Infrastructure: AWS/ Microsoft/ Google

GPU Enabled Infra

No

Yes

Computational Power

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4 Key Considerations

- MoU to be signed with related entities
- Visualization tool connectivity and accessibility with ML pipeline
- Sufficient ML build/operational infrastructure.

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Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility					~5M AED	6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility			
High	Low	Medium	High	High	Medium	Medium	Low	High	High			

1. Business Overview

1.1

Description

- The purpose of the use case is to leverage business licensing data and historical trends to generate industry-insights and recommend the types of businesses to encourage or discourage based on market potential and job creation
- Key beneficiaries of this use case include ADBC/ADRA, ESO

1.2

Pain Points

- Complex and changing regulatory environments renders it difficult to stay compliant with evolving regulations, leading to uncertainty, increased costs, and potential legal risks.

1.3

Expected Impact

- Offer fast, data-driven insights into economic activity to inform DED's market regulation

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineering
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 Tester (AI)

Engineering tester and Data Science tester)

1.5

Risks

- AI algorithms may lack transparency, making it difficult to understand how decisions are made and factors influencing regulations
- Insufficient data protection measures, potentially expose sensitive financial information to unauthorized access or breaches, compromising privacy and security.

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: Databricks, Azure cognitive services, Azure ML studio, blob storage, SQL service, AKS
- Cloud Infrastructure: AWS/ Microsoft/ Google

GPU Enabled Infra

No

Yes

Computational Power

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- The licensing data accessibility
- External datasets for augmentation like population density, infrastructure (buildings, Parking lots, Public transportation), electricity and water consumption would be beneficial
- Ease of integration
- Adequate data for solutions

CONFIDENTIAL



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~5-20M AED	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	Medium	High	Low

Data Availability

Low

Infrastructure Readiness

Low

2. Feasibility

AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Low	Medium

1. Business Overview

Description

- The purpose of the use case is to simulate scenarios, test policy changes, and ultimately make data-driven decisions for resource allocation, infrastructure development, and trade agreements, fostering economic diversification and growth beyond oil
- Key beneficiaries of this use case include ESO

1.2

Pain Points

- Without AI-driven analysis, economic development agencies lack precise insights into market trends and investment opportunities, making it challenging to attract potential investors effectively.

1.3

Expected Impact

- Empowers data-driven decisions for economic diversification and growth beyond oil

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineering
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 Tester (AI Engineering tester and Data Science tester)

1.5

Risks

- Real-world systems complexity and uncertainty can lead to oversimplification and inaccurate predictions. Data availability and quality pose challenges, requiring rigorous validation against historical data and expert knowledge

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

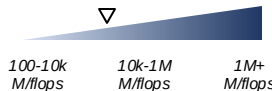
Cloud Services and Providers

- Technology Stack: Databricks, Azure cognitive services, Azure ML studio, blob storage, SQL service, AKS
- Cloud Infrastructure: AWS/ Microsoft/ Google CPU compute and storage

GPU Enabled Infra

No Yes

Computational Power



2.4

Key Considerations

- Adequate data for solution
- Visualization tool connectivity and accessibility with ML pipeline
- Sufficient ML build/operational infrastructure considering limited data volume



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~5M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	Low

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Medium	Medium	AI Model Complexity	High	High
		Low		

1. Business Overview

Description

- The purpose of the use case is to verify documents, identify potential issues, streamline inspections/fines process, collect financial statements and suggest necessary clarifications using AI.
- Key beneficiaries of this use case include ADRA/ADBC

1.2

Pain Points

- Current business registration and licensing procedures are often manual and time-consuming, leading to delays and frustration for entrepreneurs seeking to establish or expand their operations.

1.3

Expected Impact

- Reduction in time taken to issue business license
- Streamlining backend processes and increase in accuracy of licensing documents

1.4

Expected Squad Required

- | | |
|--|---|
| <ul style="list-style-type: none">1 Scrum Master1 Business Analyst2 AI Scientists2 Data Engineering0.5 UX/UI Designer0.5 Solution Architect1 Tester (AI) | Engineering tester and Data Science tester) |
|--|---|

1.5

Risks

- This use case requires multiples AI modules to be deployed to ease the workflow. The success of each AI model at different stage of application for license issuance/Renewal/cancellation etc would require the AI model to be highly accurate with its predictions.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- Technology Stack: Databricks, Azure cognitive services, Azure ML studio, blob storage, SQL service, AKS
- Cloud Infrastructure: AWS/ Microsoft/ Google

GPU Enabled Infra

Computational Power

No Yes

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4

Key Considerations

- The business workflow is available from ADBC Accela system
- Adequate data for solutions
- Ease of integration
- Sufficient GenAI build/operational infrastructure
- Gen AI model and infrastructure availability



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~5-20M AED	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	Medium	Low	High	Low

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Low	Low	AI Model Complexity Medium	Low	Medium

1. Business Overview

Description

- The purpose of the use case is to use data analytics and machine learning to assess viability and feasibility of PPP model for projects based on risk assessment, and collaboration potential between public and private sectors for infrastructure development
- Key beneficiaries of this use case include ESO

1.2 Pain Points

- Without AI-driven analysis, economic development agencies lack precise insights into market trends and investment opportunities, making it challenging to attract potential investors effectively.

1.3 Expected Impact

- Ease of assessment of government projects with highest potential and feasibility to be launched as PPP programs

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineering
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 Tester (AI Engineering tester and Data Science tester)

1.5 Risks

- AI models used to assess PPP potential may be trained on historical data that reflects biases or disparities
- AI systems used for PPP analysis may be complex and not easily interpretable by stakeholders, leading to concerns about transparency and accountability.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

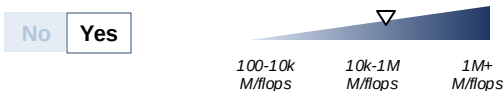
AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- Technology Stack: Databricks, Azure cognitive services, blob storage, SQL service, AKS
- Cloud Infrastructure: AWS/ Microsoft/ Google with AI Studio

GPU Enabled Infra

Computational Power



2.4 Key Considerations

- Adequate data for solutions
- Ease of integration
- Sufficient ML build/operational infrastructure

Furthermore, an additional of 19 use cases have been defined by DED

DED



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Identified AI Use Cases

DED Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
6	Labor market demand analysis and skills trends	Analyze online job posting data across public and private sector, identifying key labor trends over time, and better understanding the current demand across key job industries and occupations	High	Medium	6-12 Months
7	Unemployment rate analysis and trend prediction	Analyze unemployment rate across citizens and non-citizens with nowcasting by leveraging online employment and job searches to inform timely decision making	High	Medium	6-12 Months
8	Labor market insights	AI-driven analysis of employment data to identify emerging job sectors, skills in demand, and wage trends	High	Medium	0-6 Months
9	Identification of key drivers for consumer spending as a % of GDP	Identify key drivers for consumer spending fluctuations by population segments, spend type, and spending category leveraging administrative and payments transaction data	High	Medium	6-12 Months
10	Consumer geographic financial distress analysis	Track the consumer financial distress geographically leveraging existing administrative data on individuals as well as alternative, and proxy data sources to better understand the impact of various economic events on consumers financial distress and geographic disparities	High	Medium	6-12 Months
11	Household segmentation analysis across different variables	Enrich the current household segmentation by dividing Abu Dhabi households into different categories based on variables such as household composition, income level and nationality, leveraging existing data to support policy making decisions with evidence of impact on population segments	High	Low	6-12 Months
12	Monthly international migration flows prediction	Improve migration flows reporting with predictions leveraging online migration and travel related searches to provide timely statistics to policy makers and support decision making	High	Medium	6-12 Months
13	Fraud detection and risk management	Leverage AI systems to detect financial fraud and assess credit risks	High	Medium	0-6 Months
14	Key drivers for high growth firms by segment identification	Determine key drivers of high growth firms by segments leveraging administrative data sources, web scraped data, and proxy data to understand key trends driving business growth	Medium	Medium	6-12 Months

Furthermore, an additional of 19 use cases have been defined by DED

DED



24

Identified AI Use Cases

DED Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
15	Multifactor productivity indicator computation by industry	Compute the multifactor productivity indicator by comparing the amount of goods and services produced to the amount of inputs required to better understand the production efficiency by industry in Abu Dhabi and track the overall economic performance over time thereby informing policy decision making	●	●	6-12 Months
16	Key drivers for business failure identification	Determine key drivers of firms' risk of bankruptcy leveraging administrative and survey data sources to develop a risk score model by industry, enabling decision making	●	●	6-12 Months
17	Business failure prediction by type of industry	Forecast the fluctuations in statistically significant key drivers for business failures based on other policy changes, global economic decisions and black swan events to identify industries at risk of bankruptcy within a certain time frame	●	●	6-12 Months
18	Digital economy share as a % of GDP analysis	Provide a better understanding on how digital companies use the internet, their size and nature of their business by combining online private sector data, and existing data in business registers to understand the share of the digital economy to overall GDP	●	●	6-12 Months
19	Government investment impact analysis	Help investigate the government investment direct and indirect impact on a certain indicators by analyzing external circumstances, collecting administrative data sources to better understand the effect and value of the investments and how they stimulate economic growth	●	●	6-12 Months
20	Budget forecasting and public finance management	Use AI models to predict revenue and expenditure patterns for effective fiscal planning	●	●	6-12 Months
21	Public debt analysis and risk assessment	Deploy AI tools to assess the impact of public debt on economic stability and growth	●	●	6-12 Months
22	FDI key drivers identification	Analyze key drivers of FDI fluctuations for the oil versus non-oil sector by leveraging existing administrative, survey and alternative data sources to better understand the key economic variables impacting FDI growth in Abu Dhabi thereby informing policy decision making	●	●	0-6 Months
23	FDI growth scenario simulation	Simulate the impact of the change in key economic drivers on FDI growth in the oil versus non-oil sector to enable policy makers to understand how to better diversify from the oil sector and proactively make decisions that would ensure economic growth	●	●	6-12 Months
24	International trade optimization	AI-based trade analytics to identify market opportunities and optimize supply chain logistics	●	●	6-12 Months



DGE – Deep Dive



Digital Care – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility

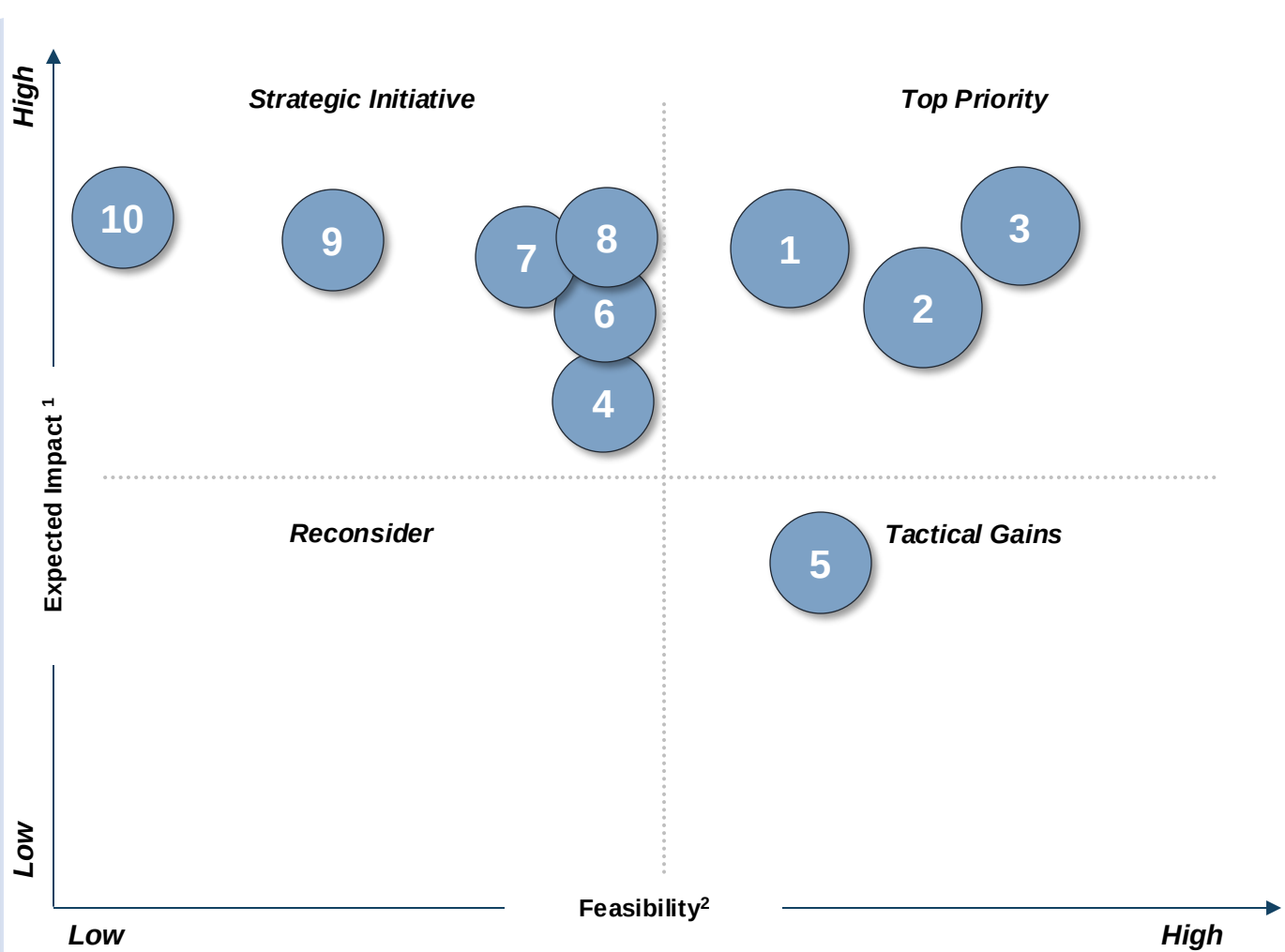
Digital Care



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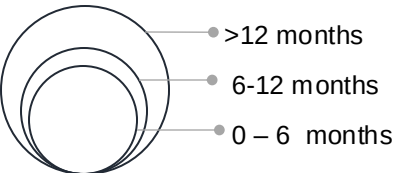
Identified AI Use Cases



Identified Use Cases

1	AI in project documentation and communication
2	Digital Assistant
3	AI License Manager
4	Code Generation/ Conversion
5	Requirements Analysis
6	Test Case Generation
7	SLDC Assistant
8	User-Facing Co-Pilot
9	GIS Data Accessibility
10	Public Records Management (Gen AI)

Time to Deploy



1. **Expected Impact:** Citizen Experience, Productivity Enhancement, Operational Efficiency, Achievement of Gov. Goals and Output Quality
2. **Feasibility:** Data availability, Infrastructure readiness, AI Model availability, Stakeholder alignment and Cost feasibility

DGE Digital Care has prioritized 3 use cases from their identified long list

1



AI in Project Documentation and Communication

- The purpose of the use case is to utilize GenAI to automate the summaries of meetings, generate comprehensive status reports, streamline progress updates and provide templates for accurate and consistent project documentation, ensuring alignment among teams and stakeholders, the reason behind that is to try to address inefficiencies in project management and improve outcomes
- This use case will benefit mainly project managers

Expected Impact

- Improve accuracy and consistency in project documentation
- Improve alignment among teams and stakeholders
- Improve productivity and project outcomes
- Enhance strategic planning and decision-making

2



Digital Assistant

- The purpose of the use case is to deploy a GenAI-enabled digital assistant to swiftly review knowledge management systems, analyze past tickets, and identify relevant components to assist in handling incoming requests, the reason behind that is to serve as a co-pilot for operators
- This use case will benefit mainly operators, end-users, directors, system administrators, business owners or executives

Expected Impact

- Streamline operational workflows and reduce the time required to handle incoming requests
- Help operators make more informed decisions, leading to fewer errors and higher-quality outcomes

3



AI License Manager

- The purpose of the use case is to leverage artificial intelligence algorithms to analyze software usage patterns, identify underutilized or unused licenses, and optimize license allocation across ADGEs, the reason behind that is to enable organizations to reduce costs and support overheads associated with unused or underutilized licenses
- This use case will benefit business owners, end-users and system administrators

Expected Impact

- Help organizations reduce unnecessary software spending and support costs
- Reduce the risk of legal and financial penalties associated with non-compliance
- Streamline license management processes, and save time and resources required



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					5-20M AED		6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Low	High	High	High	Medium				

2. Feasibility								
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility				
Low	Medium	High	Low	Medium				

1. Business Overview

1.1

Description

- The purpose of the use case is to utilize GenAI to automate the summaries of meetings, generate comprehensive status reports, streamline progress updates and provide templates for accurate and consistent project documentation, ensuring alignment among teams and stakeholders, the reason behind that is to try to address inefficiencies in project management and improve outcomes
- This use case will benefit mainly project managers

1.2

Pain Points

- In addition to being time-consuming and error-prone, manual documentation often lacks consistency among different teams

1.3

Expected Impact

- Improve accuracy and consistency in project documentation
- Improve alignment among teams and stakeholders
- Improve productivity and project outcomes
- Enhance strategic planning and decision-making

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineers
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Data quality issues that could impair model accuracy
- Ethical challenges related to AI predictions
- Technical challenges, such as the integration of legacy systems, data security vulnerabilities, and regulatory compliance requirements
- Resistance to change may adversely affect the adoption of new technologies

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM solutions , compatible with arabic language as some documentation can be written in Arabic
- Cloud Services: dependent on the solution architecture

GPU Enabled Infra

Computational Power

NoYes

100-10k M/flops10k-1M M/flops1M+ M/flops

2.4

Key Considerations

- Adequate data for solutions
- Gen AI build/operational infrastructure
- Arabic Language Handling: Ensures that AI solutions and are well-integrated for handling Arabic language documentation
- Data Integration: Focuses on seamless integration of data from various entities to ensure accurate documentation



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	High	High	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Medium	High	Medium

1. Business Overview

Description

- The purpose of the use case is to deploy a GenAI-enabled digital assistant to swiftly review knowledge management systems, analyze past tickets, and identify relevant components to assist in handling incoming requests, the reason behind that is to serve as a co-pilot for operators
- This use case will benefit mainly operators, end-users, directors, system administrators, business owners or executives

1.2

Pain Points

- Operators tasked with handling incoming requests often face challenges in efficiently accessing and synthesizing relevant information from knowledge management systems, past tickets, and components databases. This can lead to delays in response times, redundant efforts, and suboptimal resolution of issues

1.3

Expected Impact

- Streamline operational workflows and reduce the time required to handle incoming requests
- Help operators make more informed decisions, leading to fewer errors and higher-quality outcomes

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineers
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- AI accuracy
- Data quality and availability
- Data privacy and security
- User acceptance and adoption

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Internal data sources	Easy access if data is well-structured and up-to-date	Not yet defined	Not yet defined	Not yet defined	High	High
External data sources		Not yet defined	Not yet defined	Not yet defined	Varies	High

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM solutions , compatible with arabic language as some documentation can be written in Arabic
- Cloud Services: dependent on the solution architecture

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Adequate data for solutions
- Gen AI build/operational infrastructure
- Dependency on Internal Data: Critical data reliance



Prioritization Outcomes

1. Expected Impact				
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	Medium	Medium	High	Medium

2. Feasibility				
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Medium	Low	Medium

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	6-12 Months	8 FTEs

1. Business Overview

1.1 Description

- The purpose of the use case is to leverage artificial intelligence algorithms to analyze software usage patterns, identify underutilized or unused licenses, and optimize license allocation across ADGEs, the reason behind that is to enable organizations to reduce costs and support overheads associated with unused or underutilized licenses
- This use case will benefit business owners, end-users and system administrators

1.2 Pain Points

- The inability to accurately identify software products, licenses, versions, and usage across distributed computing environments, results in the allocation of underutilized or unused licenses, waste of resources and hindered cost optimization efforts

1.3 Expected Impact

- Help organizations reduce unnecessary software spending and support costs
- Reduce the risk of legal and financial penalties associated with non-compliance
- Streamline license management processes, and save time and resources required

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineers
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- AI accuracy
- Data quality and availability
- Data privacy and security
- User acceptance and adoption

2. Data, AI and Infrastructure Overview

2.1 Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3 Cloud Services and Providers

- Technology Stack: ML Studio, Visualization tools
- Cloud Services: Data storage, memory and compute based on the architecture

GPU Enabled Infra

No Yes

Computational Power

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4 Key Considerations

- Internal License historical data availability
- Infrastructure for ML train and inference
- Visualization tools and its connectivity with DB and analytics system

Furthermore, an additional 7 use cases have been defined by DGE DIGITAL CARE

Digital Care



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Identified AI Use Cases

DGE Digital Care Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
4	Code Generation/ Conversion	Gen AI to enable conversion of code from one platform or language to another as well as validate the conversion.	High	Medium	0-6 Months
5	Requirements Analysis	Gen AI Co-Pilot to evaluate tickets and tasks for validity, accurateness, and completeness before submission to tools such as Jira, etc.	Medium	High	0-6 Months
6	Test Case Generation	Automatically generate appropriate test cases leveraging Gen AI, business requirements, and other components	High	Medium	0-6 Months
7	SLDC Assistant	Ensure that all applicable SLDC requirements are followed for software & system development components, provide insights and suggestions for improvement. Auto document components, etc.	High	Medium	0-6 Months
8	User-Facing Co-Pilot	Using GenAI to build better knowledge management system and user experience for users prior to submitting tickets to help desk platforms	High	Medium	0-6 Months
9	GIS Data Accessibility	Enable non-GIS experts to query GIS data seamlessly with the help of conversational AI	High	Low	0-6 Months
10	Public Records Management (Gen AI)	Use AI to automate the categorization, indexing, and retrieval of public records, improving access and efficiency	Medium	Low	0-6 Months



GovAcademy – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility

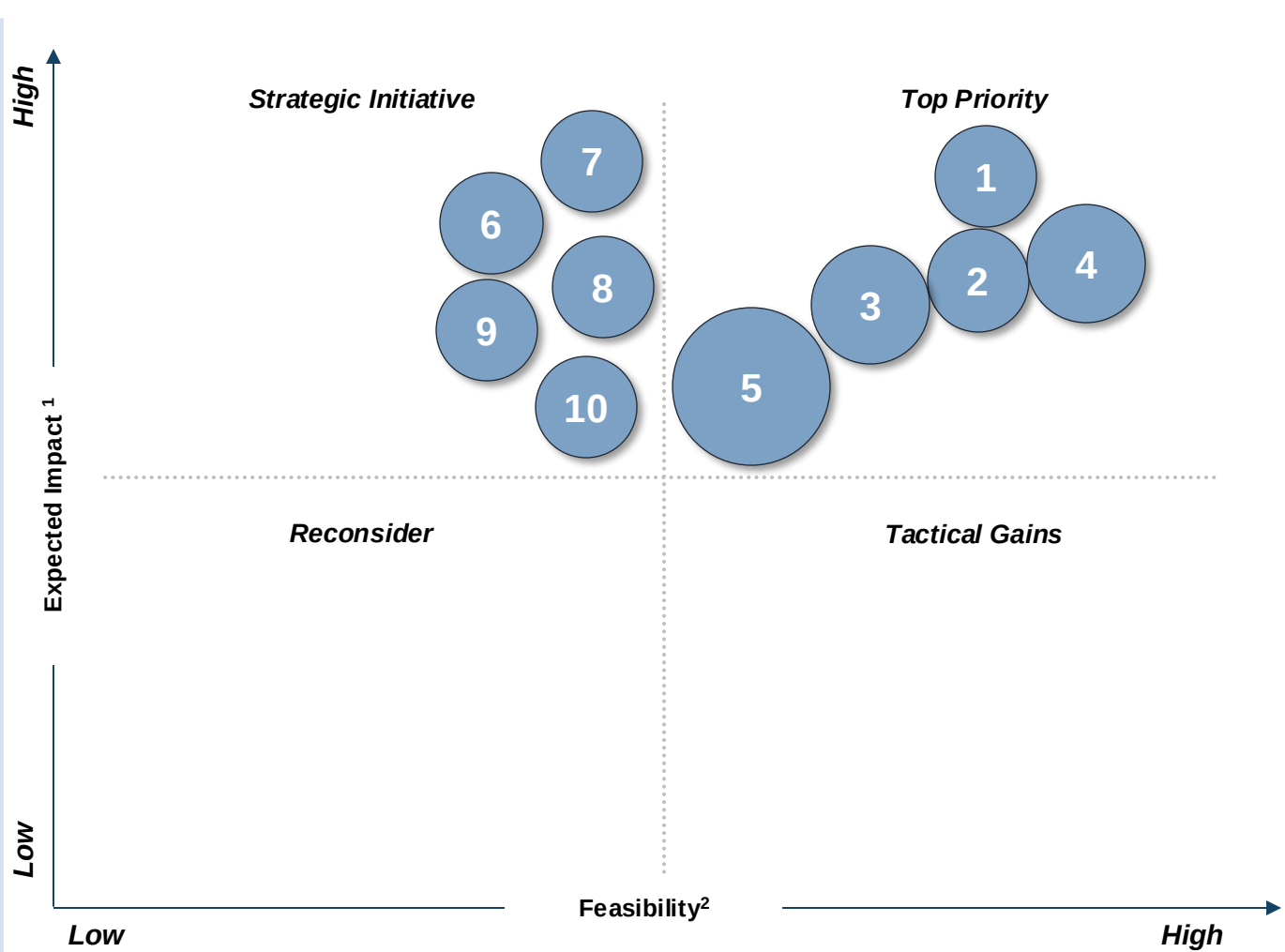
GovAcademy



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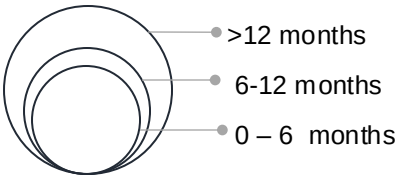
Identified AI Use Cases



Identified Use Cases

1	Individual Development Plan
2	Training Advisor
3	NLP-based Advisor
4	Content Creation
5	AR/ VR Driven Programs
6	Personalized/Adaptive and Curated Learning
7	Advanced Library Curation and Recommendation Engine
8	AI Oral Skills Coach for Users
9	AI-Powered Tutors
10	AI Powered Skill Gap Analysis

Time to Deploy



1. **Expected Impact:** Citizen Experience, Productivity Enhancement, Operational Efficiency, Achievement of Gov. Goals and Output Quality
2. **Feasibility:** Data availability, Infrastructure readiness, AI Model availability, Stakeholder alignment and Cost feasibility

DGE GovAcademy, has prioritized 5 use cases from their identified long list

1



Individual Development Plan

- Enhances individual development by mapping job profiles and facilitating professional growth and career advancement

Expected Impact

- Improved employee development and career progression
- Improved accuracy of mapping job profiles to potential candidate

2



Training Advisor

- Provides guidance and support to employees in identifying, accessing, and navigating training and development opportunities

Expected Impact

- Curated training for employees to achieve upskilling objectives
- Improved employee training outcome
- Customized plan for employee based on their need

3



NLP-based Advisor

- Conversational platform that enables employees to engage in interactive learning experiences through natural language interaction

Expected Impact

- Enhanced employee experience for learning
- Tailored training to employees' requirements

4



Content Creation

- Leverages GenAI to create new training content or enhance existing training content

Expected Impact

- Accelerated method for generating training content
- Increase in employee productivity
- Improved creative content generation

5



AR/ VR Driven Programs

- Leverages augmented reality and virtual reality technologies to provide immersive learning experiences that blend digital content with the real world

Expected Impact

- Improved employee experience for learning
- Generated insights from user experience during the training
- Enhanced employee engagement



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
<5M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	High	Medium	Low

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	Medium	Low	Medium	High

1. Business Overview

Description

- The purpose of the use case is to improve individual development by mapping job profiles and facilitating professional growth and career advancement
- Key beneficiaries to this use case include Government Employees, Talent Managers, and Line Managers

1.2

Pain Points

- Existing challenges in aligning employee skills with evolving job market demands and career opportunities

1.3

Expected Impact

- Improved employee development and career progression
- Increased accuracy in mapping job profiles to potential candidates

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Use case reliance on confidential data
- Data privacy and security

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
ADRP data	Yes	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Yes

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

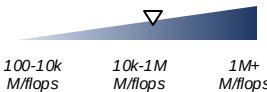
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Adequate data for solutions
- Sufficient ML build/operational infrastructure
- Visualization tool connectivity and accessibility with ML pipeline
- Ease of integration
- Regulatory Updates Readiness: Critical for compliance management



Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility							
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	<5M AED	0-6 Months	6 FTEs
Low	High	High	Low	Low	High	High	Low	Low	High			

1.1

1. Business Overview

Description

- The purpose of the use case is to provide guidance and support to employees in identifying, accessing, and navigating training and development opportunities
- Key beneficiaries to this use case include Government Employees

1.2

Pain Points

- Existing challenges in navigating and identifying beneficial trainings

1.3

Expected Impact

- Curated training for employees to achieve upskilling objectives
- Improved employee training outcome
- Customized plan for employee based on their need

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Use case reliance on confidential data
- Accuracy risks may affect output accuracy and use case effectiveness

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
3rd party data	Yes	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Yes

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No

Yes

Computational Power

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Adequate data for solutions
- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- Visualization tools and its connectivity with DB and analytics system



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	High	Low	Low

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Medium	Medium	AI Model Complexity	Low	Medium
		Medium		

1. Business Overview

Description

- The purpose of the use case is to have a conversational platform that enables employees to engage in interactive learning experiences through natural language interaction
- Key beneficiaries to this use case include Government Employees and Executives

1.2

Pain Points

- Inaccuracy in current interactive learning
- Delay in matching training to employees' needs

1.3

Expected Impact

- Enhanced employee experience for learning
- Tailored training to employees' requirements

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Use case reliance on confidential data
- Accuracy risks may affect output accuracy and use case effectiveness
- Data privacy and security

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
ADRP data	Low	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Yes
GovAcademy data	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

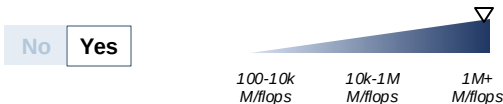
2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Adequate data for solutions
- Sufficient AI build/operational infrastructure
- Gen AI model and infrastructure availability
- Ease of integration
- Regular Model Updates: Maintains accuracy, relevance



Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility					5-20M AED	6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility			
Low	High	High	Low	Low	High	High	High	Low	Medium			

1.1

1. Business Overview

Description

- The purpose of the use case is to leverage GenAI and create new training content or enhance existing training content (e.g., development of educational materials, resources, and activities)
- Key beneficiaries to this use case include Talent Managers and GovAcademy team members

1.2

Pain Points

- Repetitive and outdated training content

1.3

Expected Impact

- Accelerated method for generating training content
- Increase in employee productivity
- Improved creative content generation

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Accuracy risks
- Use case effectiveness

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Online data	Yes	Not yet defined	Not yet defined	Not yet defined	Not yet defined	No

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No

Yes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Adequate data for solutions
- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- Ease of integration



Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility							
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	>20M AED	>12 Months	11 FTEs
Low	Low	Medium	Low	Low	Low	Low	High	Low	Low			

1. Business Overview

Description

- The purpose of the use case is to leverage augmented reality and virtual reality technologies to provide immersive learning experiences that blend digital content with the real world
- Key beneficiaries of this use case include Government Employees

1.2 Pain Points

- Lack of hands-on experience
- Employee accessibility issues
- Limited collaborative opportunities

1.3 Expected Impact

- Improved employee experience for learning
- Generated insights from user experience during the training
- Enhanced employee engagement

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineer
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5 Risks

- Accuracy risks may affect output accuracy and use case effectiveness
- Technical constraints

2. Data, AI and Infrastructure Overview

Data Requirements

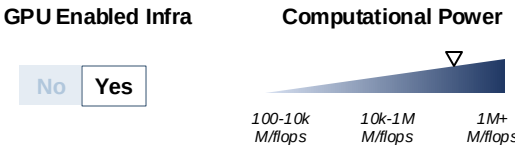
Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined

2.2 AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - AR/VR - Generative AI

2.3 Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage



2.4 Key Considerations

- Adequate data, specifically for AR/VR for the solution
- Sufficient CV build/operational infrastructure
- Ease of integration

Furthermore, an additional 5 use cases have been defined by DGE GovAcademy

GovAcademy



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DEPARTMENT OF GOVERNMENT ENABLEMENT

10

Identified AI Use Cases

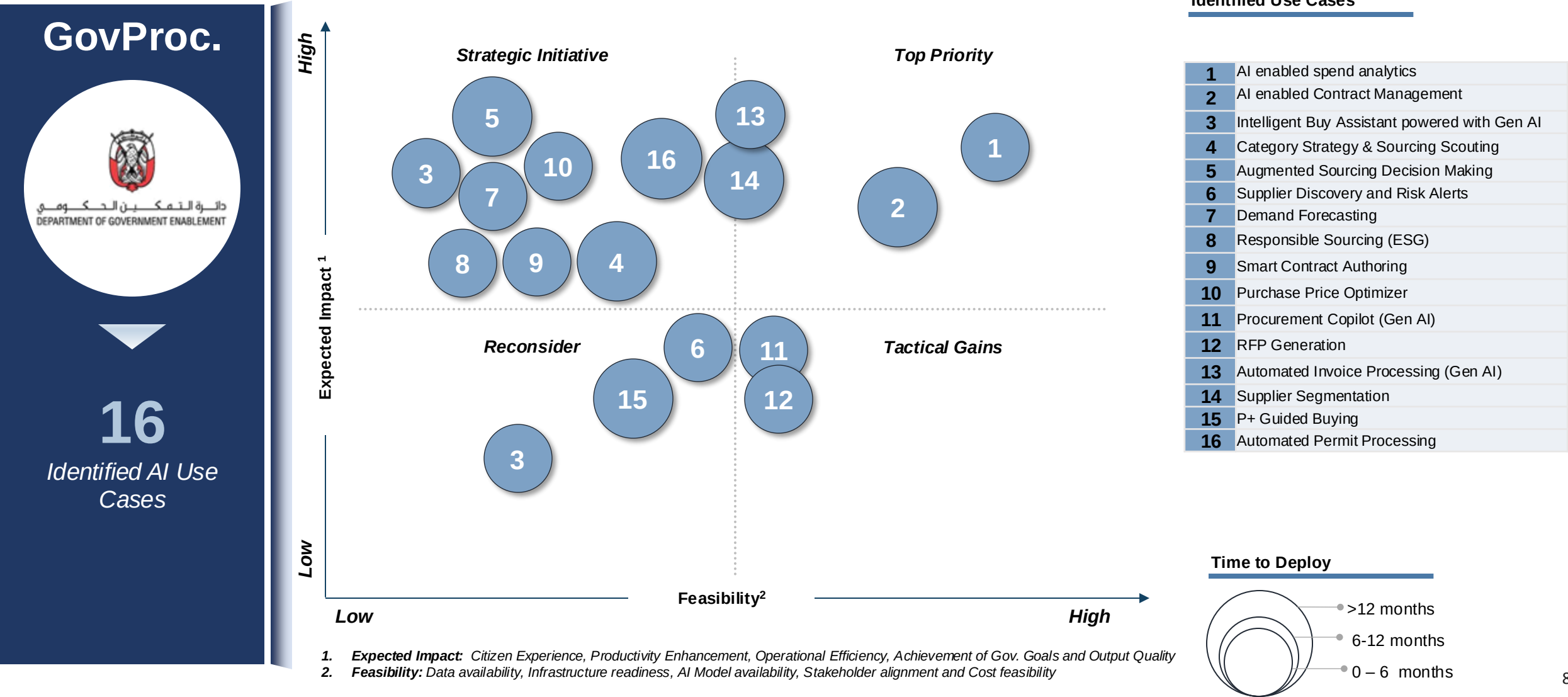
GovAcademy Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
6	Personalized/Adaptive and Curated Learning	Personalizes the learning journey by leveraging AI-enabled adaptive learning models to customize the training material	High	Medium	0-6 Months
7	Advanced Library Curation and Recommendation Engine	Recommends certain articles or books that are available in the entity's library to users based on their interests, courses, needs	High	Medium	0-6 Months
8	AI Oral Skills Coach for Users	Improves the user's oral presentation skills using NLP and speech recognition capabilities, by indicating the weak areas and suggesting improvements	High	Medium	0-6 Months
9	AI-Powered Tutors	Assists learners in their learning journey with a 1-to-1 AI based tutor	Medium	Medium	0-6 Months
10	AI Powered Skill Gap Analysis	Provides strategic recommendations to empower businesses and educational institutions to address critical skill gaps effectively	Medium	Medium	0-6 Months



GovProcurement – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



DGE GovProcurement has prioritized 2 use cases from their identified long list

1



AI-Enabled Spend Analytics

- *Alleviates the problem of complex and time-consuming activity of categorizing voluminous spend data into the defined category taxonomy (UNSPSC)*

Expected Impact

- Increased visibility into government procurement spend with high accuracy, quick turnaround and minimal manual intervention, leading to enhanced decision-making in procurement to drive cost reductions and increase efficiency
- Applicable to any processes that require data cleansing and classification (e.g., finance, budgeting, tax)

2



AI-Enabled Contract Management

- *Streamlines the manual and error prone process of contract authoring, review, analysis, and management, and automates various aspects of the contract lifecycle reducing risks and improving contract compliance*

Expected Impact

- Reduced spend leakage in 8-12%
- Decreased governance costs in 60%
- Reduced post-signature disputes in 80%
- Reduced cost of contract process optimization in 80%
- Leveraged AI enabled Contract Management tool for all ADGEs to manage their contacts
- Applicable to all types of contracts including sales contracts, manpower contracts etc.



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~5M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	High	High	Low

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	High	Medium	Medium	High

1. Business Overview

Description

- The Purpose of the use case is to alleviate the problem of complex and time-consuming activity of categorizing voluminous spend data into the defined category taxonomy (UNSPSC)
- Key beneficiaries of this use case include Procurement Practitioners, Entity Leadership

1.2 Pain Points

- Reducing procurement cycle times
- Driving analytics for cost optimizations
- Tracking value realization
- Reducing reaction time to market changes/supplier failures

1.3 Expected Impact

- Visibility on procurement spend with high accuracy, quick turnaround, minimal manual intervention
- Enhanced decision-making to drive cost reductions and increase efficiency
- Applicable to any processes that require data cleansing and classification

1.4 Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5 Risks

- Availability, completeness and accuracy of data
- Data security

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Contract PO Invoice data	Ready	Large Datasets	ERP to cloud		Simple APIs	Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

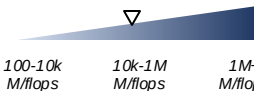
Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Adequate data for solutions
- Sufficient ML build/operational infrastructure
- Ease of integration
- Visualization tool connectivity and accessibility with ML pipeline



Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility					~5M AED	6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility			
Low	High	Medium	High	Low	Medium	Low	High	Low	High			

1. Business Overview

1.1

Description

- The Purpose of the use case is to streamline the manual and error prone process of contract authoring, review, analysis, and management, and automates various aspects of the contract lifecycle reducing risks and improving contract compliance
- Key beneficiaries of this use case include Procurement Practitioners, End-users, Entity Leadership

1.2

Pain Points

- Reducing procurement cycle times
- Driving analytics for cost optimizations
- Tracking value realization
- Reducing reaction time to market changes/supplier failures

1.3

Expected Impact

- Reduced spend leakage in 8-12%
- Decreased governance costs in 60%
- Reduced post-signature disputes in 80%
- Reduced cost of contract process optimization in 80%
- Available for all ADGEs to manage their contacts
- Applicable to all types of contracts

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Availability of contract data on tool/adoption
- Data security

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

NoYes

100-10k M/flops10k-1M M/flops1M+ M/flops

2.4

Key Considerations

- Adequate data for solutions
- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- Ease of integration

Furthermore, an additional of 14 use cases have been defined by DGE GovProcurement

GovProc.



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DEPARTMENT OF GOVERNMENT ENABLEMENT

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Identified AI Use Cases

DGE Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
3	Intelligent Buy Assistant powered with Gen AI	Assists users in making informed purchasing decisions by leveraging AI algorithms to improve category performance at lower effort and better buying experience	High	Medium	0-6 Months
4	Category Strategy & Sourcing Scouting	GenAI powered companion that is available to prepare, recommend, challenge and document all decisions as part of sourcing and category management activities	Medium	Medium	6-12 Months
5	Augmented Sourcing Decision Making	Transform procurement ways of working for sourcing-, category & business partner manager to prepare, recommend, challenge and document decisions.	High	Medium	6-12 Months
6	Supplier Discovery and Risk Alerts	Supplier discovery and recommendation tool based on performance factors, supplier capabilities & offerings, and internal and external insights	Medium	Medium	0-6 Months
7	Demand Forecasting	Develop ML models to forecast demand based on historical procurement data, sales forecasts, and market trends.	High	Medium	0-6 Months
8	Responsible Sourcing (ESG)	AI-powered Analytics tools set to provide ESG insights, that leverages multiple tools and AI to describe, predict, simulate and prescribe solutions to improve around the key three ESG pillars: Trusted, Net Zero, Circular.	High	Medium	0-6 Months
9	Smart Contract Authoring	LLM backed contract generation and contract compliance tracking can greatly reduce the redundant work whilst increasing risk mitigation capability and overall efficiency.	High	Medium	0-6 Months
10	Purchase Price Optimizer	AI Assistant who leverages the analytic modelling capabilities to prescribe purchasing strategies combining price trends, operational costs, supply risks and environmental, social and governance impact.	High	Medium	0-6 Months
11	Procurement Copilot (Gen AI)	Leverages chatbot capabilities to automate standard transactional procurement tasks. This system understand and executes user commands to carry out actions like creating purchase orders, processing invoices, and updating vendor information efficiently.	Medium	Medium	0-6 Months

Low

Medium

High

89

Furthermore, an additional of 14 use cases have been defined by DGE GovProcurement

GovProc.



16

Identified AI Use Cases

DGE Long List of Non-prioritized Use Cases

Title	Description	AI Impact	Feasibility	Time to Deploy
12 RFP Generation	Implement General AI to automate the generation of RFP documents, ensuring they meet industry standards and organizational requirements while incorporating specific project details. This AI system leverages natural language generation to create clear, precise, and legally compliant RFPs based on input parameters.	●	●	0-6 Months
13 Automated Invoice Processing (Gen AI)	Use AI to automatically extract data from invoices and match them to purchase orders and receipts.	●	●	0-6 Months
14 Supplier Segmentation	Use ML models to categorize suppliers based on various criteria like spend, risk, and performance	●	●	6-12 Months
15 P+ Guided Buying	GenAI powered Guided Buying Digital assistant to analyze and compare multiple vendors using internal and external data sources. Supported by Interactive Chatbot for Procurement Managers to filter and compare vendors	●	●	6-12 Months
16 Automated Permit Processing	Automate the review and approval of permit applications using AI to ensure timely and transparent processing	●	●	6-12 Months



GovTalent – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility

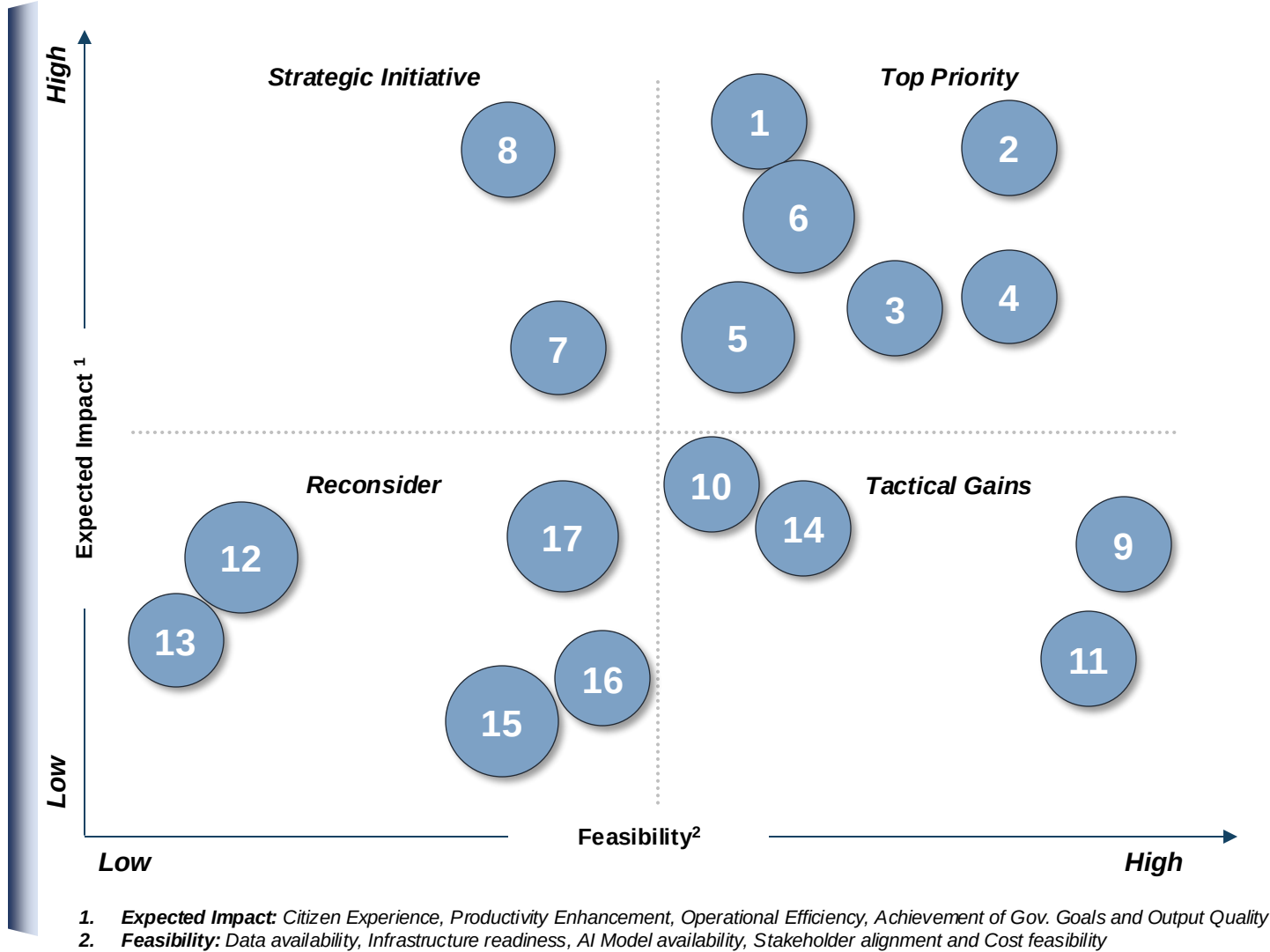
GovTalent



دائرة التمكين الحكومي
DEPARTMENT OF GOVERNMENT ENABLEMENT

17

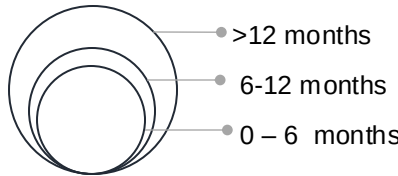
Identified AI Use Cases



Identified Use Cases

1	Labor Market Forecasting
2	Automated Job Matching
3	Interview Support
4	Crafting Job Descriptions for Posting
5	Professional Growth Pathway Creation
6	AI-Powered Skill Gap Analysis
7	Attrition Modeling
8	Talent Acquisition through Personalization
9	Job Offers Generation
10	Personalized Learning Paths Continuous Learning and Upskilling
11	Virtual Assistants/ Knowledge Bot
12	Talent Optimization
13	Employee Sentiment Analysis
14	Employee Turnover Prediction (Custom ML)
15	Automated Exit Surveys
16	Departure Checklist Generation
17	Human Capital Management

Time to Deploy



GovTalent has prioritized 6 use cases from their identified long list

1



Labor Market Forecasting

- The purpose of the use case is to utilize advanced predictive analytics to anticipate future job demands, the reason behind that is to try to enable the development of targeted hiring strategies and policy directives to align with emerging Abu Dhabi Government workforce (white color) needs
- This use case will benefit mainly Government agencies and departments, policy makers, as well as job seekers and workers within Abu Dhabi

Expected Impact

- Anticipate future talent needs
- Optimize workforce planning
- Effectively manage and develop talent

2



Automated Job Matching

- The purpose of the use case is to utilize automated algorithms to swiftly analyze resumes and pinpoint the best-fit candidates, the reason behind that is to try to significantly streamline the initial screening process and minimize time spent on candidate evaluation
- This use case will benefit mainly Government human resource departments and job applicants

Expected Impact

- Streamline recruitment workflows
- Accelerate candidate selection
- Enhance the efficiency of hiring processes

3



Interview Support

- The purpose of the use case is to leverage AI to automatically review CVs, the reason behind that is to assess suitability and generate relevant interview questions based on the candidate's background and job requirements
- This use case will benefit mainly Government human resource departments and job applicants

Expected Impact

- Improve allocation of time and effort
- Enhance candidate matching to job requirements
- Improve consistency and reduce human bias

GovTalent has prioritized 6 use cases from their identified long list

4



Crafting Job Descriptions for Posting

- The purpose of the use case is to utilize AI to generate job descriptions, the reason behind that is to optimize diversity and search visibility and ensure job descriptions attract a broad range of qualified candidates*
- This use case will benefit mainly Government human resource departments and job applicants*

Expected Impact

- Increase operational efficiency
- Improve candidate attraction
- Reduce bias and enhance standardization

5



Professional Growth Pathway Creation

- The purpose of the use case is to support employees and job seekers in developing a personalized plan, the reason behind that is to support their professional growth and skill development through a guided process based on set priorities, proven skills and other dimensions*
- This use case will benefit mainly Government human resource departments, employees, and job seekers*

Expected Impact

- Improve employee engagement, job satisfaction, and motivation
- Align employee growth with organizational goals and objectives
- Improve employee adaptability and career advancement

6



AI-Powered Skill Gap Analysis

- The purpose of the use case is to use AI to analyze current employee skills, the reason behind that is to compare them to future job requirements, and recommend necessary training programs*
- This use case will benefit mainly Government human resource departments and employees*

Expected Impact

- Improve strategic workforce planning
- Optimize training investments
- Improve employee performance and engagement
- Enhance talent development and succession planning



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	High	High	Medium

Data Availability	Infrastructure Readiness	2. Feasibility	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Medium	Low	Medium	Medium	Medium

1. Business Overview

Description

- The purpose of the use case is to utilize advanced predictive analytics to anticipate future job demands, the reason behind that is to try to enable the development of targeted hiring strategies and policy directives to align with emerging Abu Dhabi Government workforce (white color) needs
- This use case will benefit mainly Government agencies and departments, policy makers, as well as job seekers and workers within Abu Dhabi

1.2

Pain Points

- Prolonged hiring cycles and increased recruitment costs due to skill shortages and mismatches

1.3

Expected Impact

- Anticipate future talent needs
- Optimize workforce planning
- Effectively manage and develop talent

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Model Accuracy: Risks associated with the predictive accuracy of AI models, potentially leading to suboptimal decision-making and poor forecasts.
- Regulatory Compliance: Navigating the complex legal landscape regarding data use and AI applications in public administration.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- Infrastructure and tool to support advanced analytics like AI studio
- Cloud Services : For AI processing, memory, and storage

GPU Enabled Infra

Computational Power

No Yes

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4

Key Considerations

- Data Security and Privacy: Ensuring that all personal data complies with legal standards
- Scalability: Ability to handle increasing data volumes as the workforce and job market dynamics evolve
- Integration: Seamless integration with existing HR systems and databases for real-time data access and analysis
- Stakeholder Engagement: Continuous involvement of stakeholder entities to align the forecast with actual market needs



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Medium	High	High	High

Data Availability	Infrastructure Readiness	2. Feasibility	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	High	High	High	Medium

1. Business Overview

Description

- The purpose of the use case is to utilize automated algorithms to swiftly analyze resumes and pinpoint the best-fit candidates, the reason behind that is to try to significantly streamline the initial screening process and minimize time spent on candidate evaluation
- This use case will benefit mainly Government human resource departments and job applicants

1.2 Pain Points

- Manually screening large volumes of resumes is time-consuming and can lead to inconsistencies and oversight

1.3 Expected Impact

- Streamline recruitment workflows
- Accelerate candidate selection
- Enhance the efficiency of hiring processes

1.4 Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Data Privacy Concerns: Potential risks associated with handling sensitive applicant information, including breaches of confidentiality.
- Bias in AI Algorithms: Risk of AI models perpetuating existing biases in hiring practices, leading to unfair candidate selection.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Applicant resumes, Job descriptions	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- Infrastructure and tool to support advanced analytics like AI studio
- Cloud Services : For AI processing, memory, and storage

GPU Enabled Infra

Computational Power

No Yes

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4 Key Considerations

- Data Security and Privacy: Compliance with legal standards
- Scalability: Handling increasing data volumes
- Integration: Seamless integration with existing HR systems
- Stakeholder Engagement: Continuous stakeholder involvement



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	High	High	Medium

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Low	Medium	Medium

1. Business Overview

Description

- The purpose of the use case is to leverage AI to automatically review CVs, the reason behind that is to assess suitability and generate relevant interview questions based on the candidate's background and job requirements
- This use case will benefit mainly Government human resource departments and job applicants

1.2

Pain Points

- HR professionals need to sift through numerous applications and prepare effectively for interviews, which is a resource-intensive process

1.3

Expected Impact

- Improve allocation of time and effort
- Enhance candidate matching to job requirements
- Improve consistency and reduce human bias

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Resistance to Change: Potential resistance from HR personnel to adopt and trust AI-driven recruitment processes.
- Integration Issues: Challenges in seamlessly integrating the new system with existing HR platforms and databases.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Candidate CVs, Job descriptions, Historical interview data	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

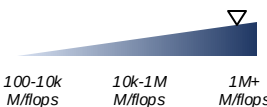
Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No Yes

Computational Power



2.4

Key Considerations

- User Training and Adoption: Effective training programs
- Ethical AI and Bias Mitigation: Ensuring fairness and transparency
- Performance and Reliability: High availability and cost efficiency



Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility					5-20M AED	0-6 Months	6 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility			
Low	High	Medium	Low	Medium	High	High	Medium	Medium	Medium			

1. Business Overview

1.1

Description

- The purpose of the use case is to utilize AI to generate job descriptions, the reason behind that is to optimize diversity and search visibility and ensure job descriptions attract a broad range of qualified candidates
- This use case will benefit mainly Government human resource departments and job applicants

1.2

Pain Points

- Writing effective and inclusive job descriptions can be time-consuming and may not consistently align with organizational standards or SEO (search engine optimization) best practices

1.3

Expected Impact

- Increase operational efficiency
- Improve candidate attraction
- Reduce bias and enhance standardization

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Quality Control: Risk of generating job descriptions that do not meet organizational standards or SEO best practices.

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Job descriptions, HR guidelines, SEO keyword data	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No

Yes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Data Security and Privacy: Compliance with legal standards
- Scalability: Handling increasing data volumes
- Integration: Seamless integration with existing HR systems
- Stakeholder Engagement: Continuous involvement of stakeholders

CONFIDENTIAL



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	Medium	Medium	Low	High

Data Availability	Infrastructure Readiness	2. Feasibility	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Medium	Medium	Medium	Medium

1. Business Overview

Description

- The purpose of the use case is to support employees and job seekers in developing a personalized plan, the reason behind that is to support their professional growth and skill development through a guided process based on set priorities, proven skills and other dimensions
- This use case will benefit mainly Government human resource departments, employees, and job seekers

1.2 Pain Points

- Many employees struggle with defining clear career goals and pathways for growth, leading to confusion and uncertainty about skill development priorities

1.3 Expected Impact

- Improve employee engagement, job satisfaction, and motivation
- Align employee growth with organizational goals and objectives
- Improve employee adaptability and career advancement

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineers
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Inaccurate Pathways: Potential for the AI to suggest career paths that do not align with employee aspirations or organizational needs.
- Technical Integration: Potential difficulties in integrating the system seamlessly with diverse HR software and databases.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

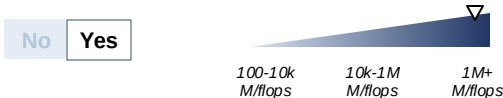
AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4 Key Considerations

- Data Security: Protecting employee data with strict security measures
- User Experience: Providing an intuitive and user-friendly interface for employees and HR managers
- Regulatory Compliance: Adhering to legal standards related to employee data and career development practices
- Scalability: Ability to expand the system as needed



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	Medium	High	High	High

Data Availability

Medium

Infrastructure Readiness

Medium

2. Feasibility

AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Medium

1. Business Overview

Description

- The purpose of the use case is to use AI to analyze current employee skills, the reason behind that is to compare them to future job requirements, and recommend necessary training programs
- This use case will benefit mainly Government human resource departments and employees

1.2

Pain Points

- Identifying and addressing skills gaps within the organization to better plan for future training and development

1.3

Expected Impact

- Improve strategic workforce planning
- Optimize training investments
- Improve employee performance and engagement
- Enhance talent development and succession planning

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineers
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Dependence on Technology: Over-reliance on AI for skill gap analysis might overlook qualitative insights that human judgment can provide.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Performance reviews, Training records, Job req.	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

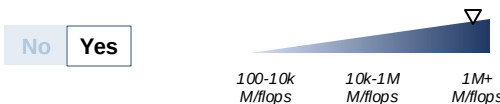
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Data Quality and Integrity: Ensure the accuracy and reliability of performance reviews, training records, and job requirements data
- Employee Confidentiality: Protect the privacy of individual employee performance and training data
- User Engagement: Encourage employee and manager engagement with the skill gap analysis tools to maximize adoption and effectiveness
- Cross-Department Collaboration: Facilitate collaboration across departments to gather comprehensive skill data

Furthermore, an additional of 11 use cases have been defined by DGE HRA

GovTalent



17

Identified AI Use Cases

DGE GovTalent Long List of Non-prioritized Use Cases				
Title	Description	AI Impact	Feasibility	Time to Deploy
7 Attrition Modeling	Leveraging AI to model attrition and workforce turnover and help plan for upcoming hiring needs and strategy	High	Medium	2-3 months
8 Talent Acquisition through Personalization	Create hyper-personalized content to attract the right talent for workforce (white color jobs)	High	Medium	2-3 months
9 Job Offers Generation	Customize job offer package to ensure compliance with legal and organizational guidelines	Medium	High	2-3 months
10 Personalized Learning Paths Continuous Learning and Upskilling	Customize onboarding materials based on a new hire's role, experience, and background, ensuring a more relevant and effective onboarding experience	Medium	High	2-3 months
11 Virtual Assistants/ Knowledge Bot	Guide new employees through the onboarding process by answering FAQs and providing information in real-time	Medium	High	2-3 months
12 Talent Optimization	Visualize and assess how existing roles can evolve in the future and how to free capacity in the organization by automatically assigning learnings to roles to upskill existing employees	Medium	Low	6-12 months
13 Employee Sentiment Analysis	Analyze employee feedback from various communication channels to gauge sentiment and identify common themes or concerns	Medium	Low	3-9 months
14 Employee Turnover Prediction (Custom ML)	Use machine learning models to predict which employees are likely to leave the company based on factors such as job satisfaction, salary, work environment, and performance reviews.	Medium	High	3-9 months
15 Automated Exit Surveys	Gather feedback automatically from departing employees to identify patterns and areas for improvement, streamlining the exit survey process	Low	Medium	6-12 months
16 Departure Checklist Generation	Generate dashboards and automatically update based on what the departing employee has completed and what is outstanding to complete to ensure covering the end-to-end offboarding process milestones	Low	Medium	3-6 months
17 Human Capital Management	Use AI to analyze skills gaps, predict employee turnover, and strategize proactive human resources management	Medium	Medium	Not received yet



TAMM – Deep Dive

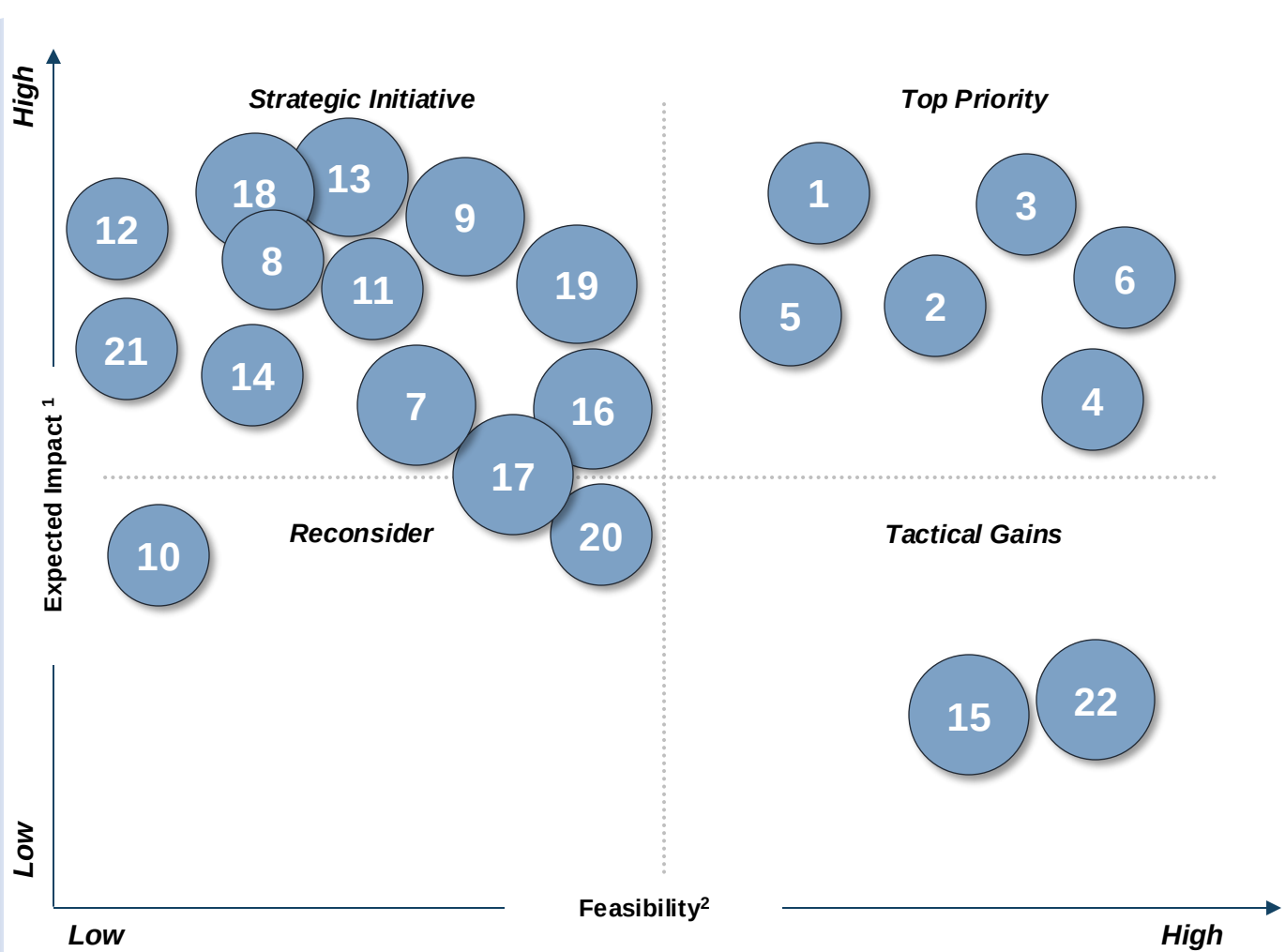
Summary and prioritization of identified use cases, across impact and feasibility

TAMM



22

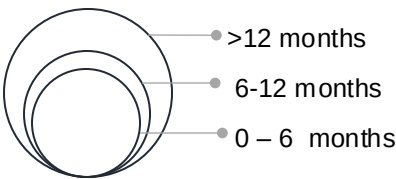
Identified AI Use Cases



Identified Use Cases

1	Contact Center Voice Bot
2	Automated Case Triage
3	Knowledge base content
4	Tailor Knowledge and Training Solutions for Individual Frontline Agents
5	Identify the Effectiveness of Knowledge Articles by Linking it to Outcome / Customer Satisfaction
6	Voice-to-Text Transcription
7	Document Verification and Processing (Custom ML)
8	AI-Powered Language Translation Services (Gen AI)
9	Intelligent Response Suggestions
10	Predictive Ticket Resolution
11	Converting emails to cases
12	Automated Follow-Up/ Escalation
13	Personalized Service Recommendations (Gen AI)
14	AI-Powered Government Services Assistant
15	SME Journey Mapping
16	Misinformation Detection
17	Multilingual Digital Customer Service
18	Service Demand Forecasting (Custom ML)
19	Intelligent Observability (Gen AI)
20	AI-Driven Content Creation (Gen AI)
21	Public Services Copilot (Gen AI)
22	Assess the sentiment and quality of the interaction in real time

Time to Deploy



1. **Expected Impact:** Citizen Experience, Productivity Enhancement, Operational Efficiency, Achievement of Gov. Goals and Output Quality
2. **Feasibility:** Data availability, Infrastructure readiness, AI Model availability, Stakeholder alignment and Cost feasibility

DGE Tamm, has prioritized 6 use cases from their identified long list

1



Contact Center Voice Bot

- The voice bot leverages speech to text, NLP, text to speech to act as first line support for the CC incoming calls by seamlessly integrating with existing tools, managing incoming calls, and enhancing service offerings. It accurately interprets customer queries and recognizes their intents, streamlining call routing and information retrieval and provides tailored responses to individual preferences. It mainly benefits end-users and agents

Expected Impact

- Reduce number of calls to agents by 25pp
- Reduce waiting time and handling time by 5pp
- Increase CSAT towards agents by 3pp

2



Automated Case Triage

- The purpose of the use case is to deploy automated case triaging, the reason behind that is to analyze incoming support tickets and automatically classify and prioritize them based on priority, type of incident, topic and channel the incident to the right stakeholders
- This use case will benefit mainly contact center agents

Expected Impact

- Reduced need for staff interventions
- Expedited case handling
- Improved operational efficiency

3



Knowledge Base Content

- The purpose of the use case is to turn customer enquiries and incidents resolution into clear, comprehensive and easy-to-read knowledge articles by sorting, labeling and transforming them in English and Arabic, classifying and categorizing them as per the existing KA forms and leveraging different intelligent and search tools to present them to both customers and agents
- It mainly benefits contact centers & service centers

Expected Impact

- Increase "knowledge satisfaction" by 10pp
- Improvement of knowledge accuracy by 10pp
- Reduce number of cases escalated to the technical team by 10pp
- Increase in the volumes of information and incidents FCR by 10pp

DGE TAMM, has prioritized 6 use cases from their identified long list

4

Tailor Knowledge and Training Solutions for Individual Frontline Agents

- The purpose of this use case is to deploy AI technologies to deliver personalized knowledge and training solutions to frontline agents in customer service or support roles, the reason behind that is to enhance the performance, efficiency, and effectiveness of individual agents by providing them with targeted learning resources and on-demand support*
- The use cases will mainly benefit frontline agents*

Expected Impact

- Improve agent performance, efficiency and productivity
- Enhance customer experience
- Offer a tailored learning experience for frontliners
- Improve consistency in service delivery

5

Identify the Effectiveness of Knowledge Articles by Linking it to Outcome / Customer Satisfaction

- The purpose of this use case is to evaluate the effectiveness of the defined and availed knowledge articles for the purpose of improving customer satisfaction, knowledge sharing effectiveness and useability, by measuring their impact on outcomes*
- The key beneficiaries of this use case are primarily the contact center agents and beneficiaries*

Expected Impact

- Improved quality of knowledge articles
- Reduced average handling time of customer inquiries
- Increased first contact resolution
- Reduced repeated interactions
- Improved customer satisfaction

6

Voice-to-Text Transcription

- The purpose of this use case is to transcribe voice interactions between customers and call center agents into text in real-time, enabling faster and more accurate documentation of support cases.*
- This use case primarily benefits contact center agents and end-users by streamlining the support process*

Expected Impact

- Speed up case creation - staff efficiency
- Enhance the accuracy of data entered
- Improve the quality of analytics
- Better inform the agent about the sentiment
- enable flagging possible call issues to the team leads/managers in real time



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					TBD		0-6 Months	6 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Medium	High	High	Medium	High				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
Medium	Medium	Medium	Medium	Medium	

1.1

1. Business Overview

Description

- The voice bot leverages speech to text, NLP, text to speech to act as first line support for the CC incoming calls by seamlessly integrating with existing tools, managing incoming calls, and enhancing service offerings
- It accurately interprets customer queries and recognizes their intents, streamlining call routing and information retrieval and provides tailored responses to individual preferences
- It mainly benefits end-users and agents

1.2

Pain Points

- Agents are answering around 684K yearly calls as of the type of "inquires ", impacting the overall average waiting time as well as handling time

1.3

Expected Impact

- Reduce number of calls to agents by 25pp
- Reduce waiting time and handling time by 5pp
- Increase CSAT towards agents by 3pp

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Accuracy STT, LLM and TTS engines (or multimodal LLM) that supports Arabic where 90% of the calls are on Emirati dialects
- Availability of up-to-date required knowledge articles from ADGE side

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Call recordings	Yes	~684K of type inquiry	Genesys	Not yet defined	Not yet defined	No
Knowledge articles	Yes	~1500	CRM	Not yet defined	Not yet defined	No

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, rein for cement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No

Yes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Gen AI model and infrastructure availability
- Adequate data for the solutions
- Secure personal data handling



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					TBD		0-6 Months	6 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Medium	High	High	Medium	High				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
High	High	Low	High	High	

1.1

1. Business Overview

Description

- The purpose of the use case is to deploy automated case triaging, the reason behind that is to analyze incoming support tickets and automatically classify and prioritize them based on priority, type of incident, topic and channel the incident to the right stakeholders
- This use case will benefit mainly contact center agents in optimizing the incident management process and improve response time and end-user satisfaction

1.2

Pain Points

- TAMM has deployed Case Officers who review each open case and decide on the next steps needed, which is time consuming

1.3

Expected Impact

- Reduced need for staff interventions
- Expedited case handling
- Improved operational efficiency

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Incomplete or incorrect data
- Data Availability

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, rein for cement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

NoYes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Adequate data for the solutions
- Ease of integration
- Scalable infrastructure to handle large amounts of tickets



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	High	Medium	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	High	Low	High	High

1. Business Overview

1.1

Description

- The purpose of the use case is to turn customer enquiries and incidents resolution into clear, comprehensive and easy-to-read knowledge articles by sorting, labeling and transforming them in English and Arabic, classifying and categorizing them as per the existing KA forms and leveraging different intelligent and search tools to present them to both customers and agents
- It mainly benefits contact centers & service centers

1.2

Pain Points

- Ability to convert solved problems into knowledge articles to address customers with similar needs (questions, issues, etc., especially for digital services)
- Adherence of ADGEs to write and update KA and their own internal knowledge for writing articles (high governance effort)

1.3

Expected Impact

- Increase “knowledge satisfaction” by 10pp
- Improvement of knowledge accuracy by 10pp
- Reduce number of cases escalated to the technical team by 10pp
- Increase in the volumes of information and incidents FCR by 10pp

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Accuracy of existing Speech to Text model
- Different data source types, content structure and language diversity
- Accuracy of agents' responses and coverage of services based on similarity threshold

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Transcript from interactions	Yes	170,000 calls (5 months)	Genesys	Not yet defined	Not yet defined	Yes
Knowledge articles	Yes	~1500	MSD-CRM	Not yet defined	Not yet defined	No
Training Modules	Yes	14	PPT	Not yet defined	Not yet defined	No
Agent-level training completion	Yes	1	Excel	Not yet defined	Not yet defined	No
Cases	Yes	800K p/year	CRM	Not yet defined	Not yet defined	No
ServiceNow Incidents	Yes	38K p/year	ServiceNow	Not yet defined	Not yet defined	No

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

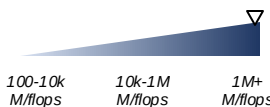
Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No Yes

Computational Power



2.4

Key Considerations

- Include feedback loop to incorporate the customer/agent feedback
- Gen AI model and infrastructure availability
- Consumption/integration of the model (batch scoring)
- Ease of integration



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					TBD		0-6 Months	6 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Medium	High	High	Medium	High				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
High	High	Low	High	High	

1. Business Overview

1.1

Description

- The purpose of this use case is to deploy AI technologies to deliver personalized knowledge and training solutions to frontline agents in customer service or support roles, the reason behind that is to enhance the performance, efficiency, and effectiveness of individual agents by providing them with targeted learning resources and on-demand support
- The use case is aimed towards frontline agents to help them better leverage existing knowledge (explicit and tacit) across the organization and optimize their productivity

1.2

Pain Points

- Inability to provide specific feedback and training to the agent based on the quality and accuracy of the data shared with end-users

1.3

Expected Impact

- Improve agent performance, efficiency and productivity
- Enhance customer experience
- Offer a tailored learning experience for frontliners
- Improve consistency in service delivery

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Incomplete or incorrect data
- Data Availability

2. Data, AI and Infrastructure Overview

2.1

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, rein for cement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: LLM, Graph DB, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No

Yes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- Ease of integration
- Adequate data for solutions
- Secure personal data handling
- Implement mechanisms for feedback and continuous improvement based on agent interaction with the system

CONFIDENTIAL

Identify the effectiveness of knowledge articles by linking it to outcome / customer satisfaction



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					TBD		0-6 Months	6 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
Medium	High	High	Medium	High				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
High	High	Low	High	High	

1. Business Overview

1.1 Description

- The purpose of this use case is to evaluate the effectiveness of the defined and availed knowledge articles for the purpose of improving customer satisfaction, knowledge sharing effectiveness and useability, by measuring their impact on outcomes
- The key beneficiaries of this use case are primarily the contact center agents and beneficiaries

1.2 Pain Points

- Inability to assess the accuracy and effectiveness of the information shared with the customer
- Low usage of knowledge articles and limited impact

1.3 Expected Impact

- Improved quality of knowledge articles
- Reduced average handling time of customer inquiries
- Increased first contact resolution
- Reduced repeated interactions
- Improved customer satisfaction

1.4 Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Data privacy concerns linked to the processing of personal customer data
- Outdated knowledge articles in the data leading to decreasing customer satisfaction

2. Data, AI and Infrastructure Overview

2.1 Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3 Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

NoYes

Computational Power

100-10k M/flops10k-1M M/flops1M+ M/flops

2.4 Key Considerations

- Ease of integration
- The end product could vary based on data quality and preference
- Secure personal data handling
- Responsible AI should be considered

CONFIDENTIAL



Prioritization Outcomes

Preliminary Budget

Expected Duration

Required Team Size

1. Expected Impact				
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	High	Medium	High

2. Feasibility				
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Medium	Medium	Medium

TBD

0-6 Months

6 FTEs

1. Business Overview

Description

- The purpose of this use case is to transcribe voice interactions between customers and call center agents into text in real-time, enabling faster and more accurate documentation of support cases.
- This use case primarily benefits contact center agents and end-users by streamlining the support process

1.2

Pain Points

- Tamm agents spend time capturing and documenting the case at hand, and sometimes the captured data is not accurate.

1.3

Expected Impact

- Speed up case creation - staff efficiency
- Enhance the accuracy of data entered
- Improve the quality of analytics
- Better inform the agent about the sentiment
- enable flagging possible call issues to the team leads/managers in real time

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Accuracy STT, LLM and TTS engines (or multimodal LLM) that supports Arabic where 90% of the calls are on Emirati dialects
- Availability of up-to-date required knowledge articles from ADGE side

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Calls recordings	Yes	684K of type inquiry	Genesys	DGE Tamm	Not yet defined	No
Knowledge Articles	Yes	1500	CRM	DGE Tamm	Not yet defined	No

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

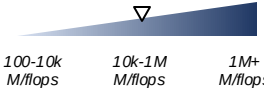
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Secure personal data handling
- Ease of integration
- Integration of the IVR system
- Display location of the speech-to-text output
- A planned revamp of the IVR system aims to address these key considerations effectively

Furthermore, an additional 16 use cases have been defined by TAMM

TAMM Long List of Non-prioritized Use Cases					
	Title	Description	AI Impact	Feasibility	Time to Deploy
7	Document Verification and Processing (Custom ML)	Use ML models to automate document verification and data extraction			6-12 Months
8	AI-Powered Language Translation Services (Gen AI)	Utilize General AI to provide real-time, automated translation of government services, documents, and communications. This solution leverages advanced neural machine translation technologies to ensure translations are not only accurate but also contextually appropriate across multiple languages			0-6 Months
9	Intelligent Response Suggestions	Provide call center agents with real-time suggestions for responding to customer inquiries based on historical interactions, knowledge base articles, and best practices			6-12 Months
10	Predictive Ticket Resolution	Analyze historical support data to predict the resolution time and likelihood of successfully resolving a support ticket based on various factors			0-6 Months
11	Converting emails to cases	Monitor the conversation or email box, identify the type of case and create it automatically, then assign it to the relevant team or queue			0-6 Months
12	Automated Follow-Up/ Escalation	Automate follow-up communications with customers after support interactions, such as status updates and trigger escalations for unresolved issues based on predefined criteria			0-6 Months
13	Personalized Service Recommendations (Gen AI)	Implement General AI to analyze user profiles and behavior, enriched with a Customer 360 view, to recommend relevant government services. This solution combines data from multiple sources, including user activity history, service usage patterns, demographic information, and interactions across different channels, to provide personalized recommendations			6-12 Months
14	AI-Powered Government Services Assistant	Provide convenient and secure access to government services through conversational AI			6-12 Months
15	SME Journey Mapping	Automate support for SMEs through AI-driven journey mapping, offering tailored guidance based on their developmental stage			0-6 Months
16	Misinformation Detection	Detect and address misinformation and rumors effectively with AI-driven algorithms			6-12 Months
17	Multilingual Digital Customer Service	Offer customer support through digital humans, equipped with a UAE-specific persona, in multiple languages			6-12 Months

TAMM



22

Identified AI Use Cases

Furthermore, an additional 16 use cases have been defined by TAMM

TAMM



22

Identified AI Use Cases

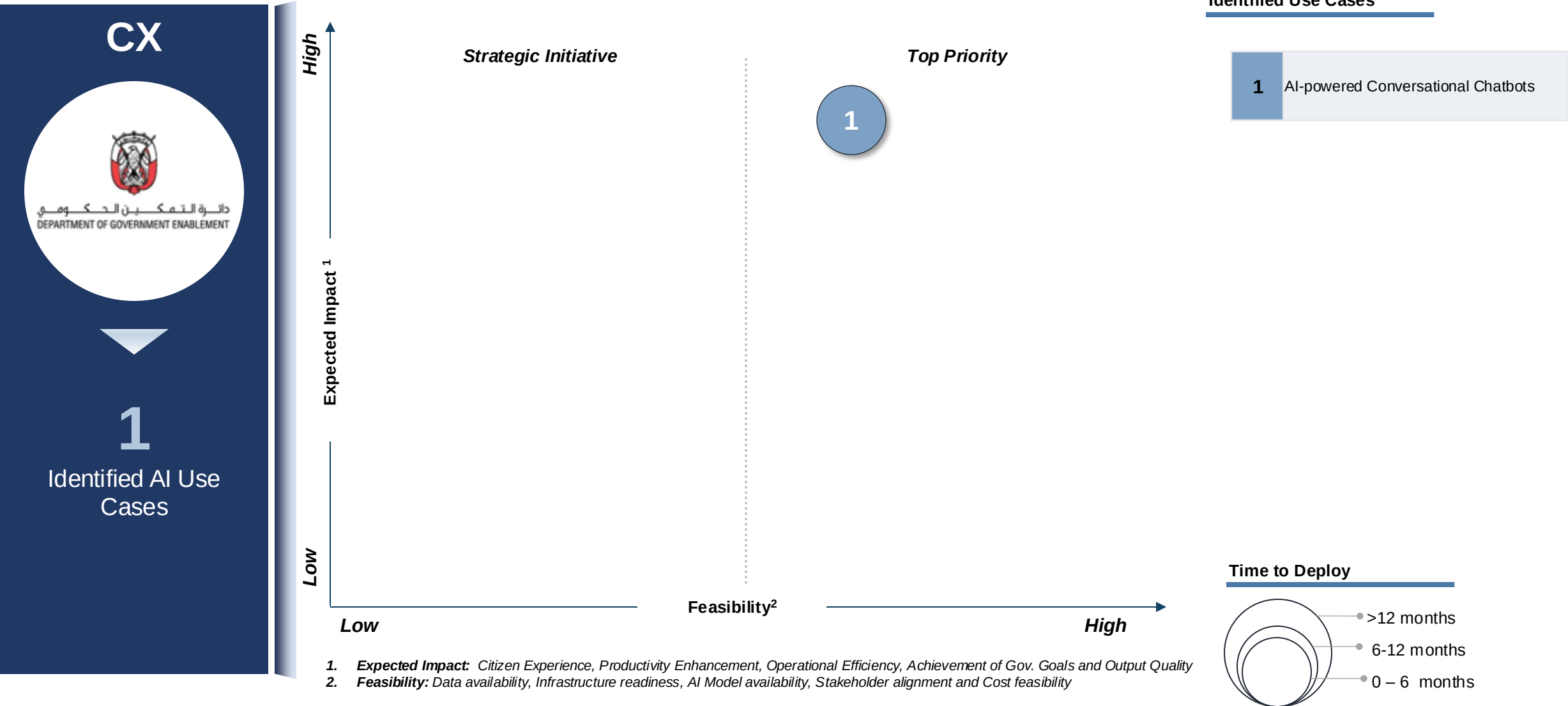
TAMM Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
18	Service Demand Forecasting (Custom ML)	Implement custom Machine Learning models to forecast service demand accurately based on usage patterns, demographic data, economic indicators, and external factors such as events or policy changes	●	●	6-12 Months
19	Intelligent Observability (Gen AI)	Implement General AI to provide intelligent observability across TAMM's digital infrastructure. This solution leverages AI to analyze data from various sources, predict potential system failures, identify performance bottlenecks, and detect security threats before they impact users. By using machine learning algorithms, the system can learn from past incidents and optimize its response strategies	●	●	6-12 Months
20	AI-Driven Content Creation (Gen AI)	Implement General AI to automate the creation of dynamic content for public communications, official notifications, and service updates. This AI system uses natural language generation (NLG) techniques to produce well-structured, contextually appropriate content that adapts to various service needs	●	●	0-6 Months
21	Public Services Copilot (Gen AI)	Deploy a General AI-driven Public Services Copilot that automates and simplifies interactions such as form submissions and appointment scheduling. This copilot uses pre-trained models for natural language understanding to efficiently process citizen inquiries and execute necessary actions by interfacing with backend systems	●	●	0-6 Months
22	Assess the Sentiment and Quality of the Interaction in Real Time	Implement real-time sentiment analysis during customer interactions across various channels (voice, video, text) using AI. This capability is aimed at enhancing the responsiveness and effectiveness of customer service.	●	●	0-6 Months



CX – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



DGE CX, has prioritized 1 use case from their identified long list

1



AI-powered Conversational Chatbots

- *The purpose of the AI-powered Conversational Chatbots is to integrate extensive customer experience service catalogue encompassing detailed information such as service names, descriptions, application procedures, associated costs, and required documentation into AI chatbots, to provide users with accurate guidance tailored to their specific service needs*
- *It mainly benefits end-users and agents*

Expected Impact

- Reduce number of calls to agents
- Reduce waiting time and handling time
- Enhance user experience
- Improve operational efficiency
- Improve consistency and accuracy in information provided to end-users



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
TBD	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Medium	Medium	High	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	Medium	Medium	Medium	Medium

1. Business Overview

Description

- The purpose of the AI-powered Conversational Chatbots is to integrate extensive customer experience service catalogue encompassing detailed information such as service names, descriptions, application procedures, associated costs, and required documentation into AI chatbots, to provide users with accurate guidance tailored to their specific service needs
- It mainly benefits end-users and agents

1.2

Pain Points

- Agents face a large volume of inquiries regarding various services, impacting overall response times and consistency in information delivered

1.3

Expected Impact

- Reduce number of calls to agents
- Reduce waiting time and handling time
- Enhance user experience
- Improve operational efficiency
- Improve consistency and accuracy in information provided to end-users

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- TBD

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- TBD

GPU Enabled Infra

Computational Power

No Yes

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4

Key Considerations

- Gen AI model and infrastructure availability
- Adequate data for the solutions
- Secure personal data handling

Furthermore, an additional X use cases have been defined by CX

CX



دائرة التمكين الحكومي
DEPARTMENT OF GOVERNMENT ENABLEMENT



1

Identified AI Use Cases

CX Long List of Non-prioritized Use Cases

Title	Description	AI Impact	Feasibility	Time to Deploy
TBD	TBD	TBD	TBD	TBD

118



DMT – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility

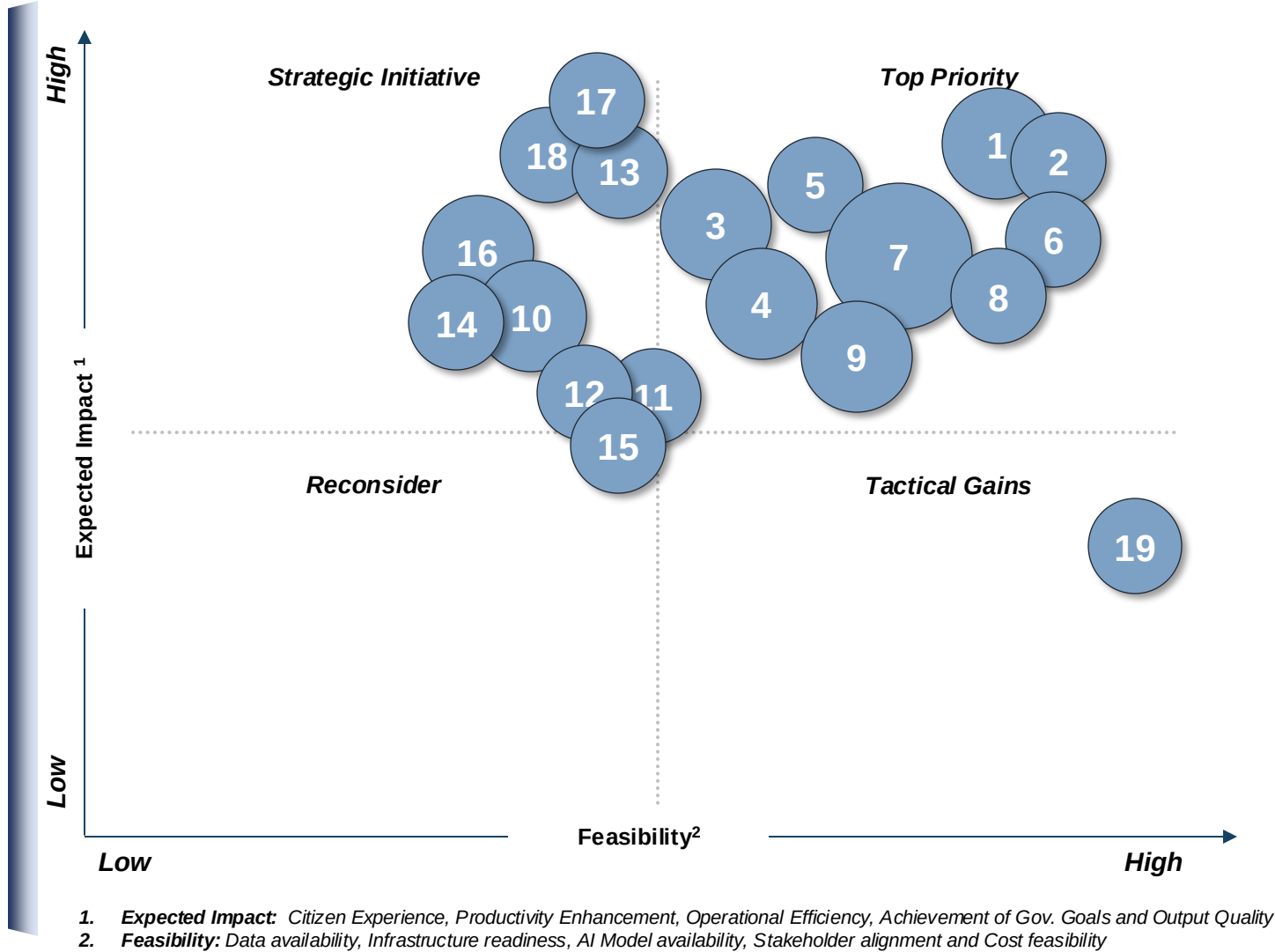
DMT



دائرة البلديات والنقل
DEPARTMENT OF MUNICIPALITIES
AND TRANSPORT

19

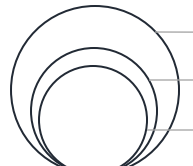
Identified AI Use Cases



Identified Use Cases

1	Capture City Distortions through AI
2	Sentiment Analysis based on Services Comments
3	Facial Recognition & Driver / Passenger Behavior Analysis
4	AI to Enhance Transportation Planning and Traffic Management
5	Real Estate Market Prediction
6	Construction Approval, Development and Monitoring
7	Smart City Platform
8	Traffic Management
9	Public Parking Management
10	Digital Twin for Smart City
11	Visual Pollution Detection
12	Traffic Modeling
13	Geospatial Intelligence for Mobility Planning
14	Climate Command Center
15	GenAI Flood Warning
16	Public Safety Monitoring
17	Automated Permit Processing
18	Citizen Chatbot
19	Address Normalization

Time to Deploy



- >12 months
- 6-12 months
- 0 – 6 months

DMT has prioritized 9 use cases from their identified long list

1



Capture City Distortions through AI

- Abu Dhabi's infrastructure expansion and focus on green spaces complicate manual inspection processes due to time-consuming tasks and challenging environmental conditions. Innovative approaches are needed to efficiently monitor and maintain these areas.

Expected Impact

- Enhance the efficiency of operations
- Lower substantially the costs and time required for detecting violations

2



Sentiment Analysis based on Services Comments

- DMT offers various public services and traditionally gathers feedback through surveys and complaints, which can be slow and limited in scope, capturing feedback after experiences occur.

Expected Impact

- Enhance service management efficiency and consistency
- Allow staff to concentrate on key responsibilities
- Improve timely interactions with the audience.

3



Facial Recognition & Driver / Passenger Behavior Analysis

- Current CCTV systems on public buses and taxis lack real-time analysis capabilities for safety and behavior monitoring. AI can analyze footage to detect driver fatigue, incidents, automate fare payments, and enhance safety for vulnerable passengers.

Expected Impact

- Increased safety
- Faster response times
- Improved efficiency
- Scalable & reusable across mode

4



AI to Enhance Transportation Planning and Traffic Management

- Traditional transportation planning and traffic management methods rely on historical data and limited real-time information. AI can analyze large datasets for predictive traffic flow, congestion prevention, and optimized infrastructure planning.

Expected Impact

- Enhance the accuracy of transportation and traffic modelling outcomes
- Reduced traffic congestion
- Shorter travel times
- Improved fuel efficiency
- Data-driven infrastructure development

DMT has prioritized 9 use cases from their identified long list

5



Real Estate Market Prediction

- Use data analysis and machine learning to forecast trends, pricing changes, and demand fluctuations in the real estate market

Expected Impact

- Improve investment and pricing strategies

6



Construction Approval, Development & Monitoring

- Employ AI-based monitoring systems for quality control and project oversight

Expected Impact

- Enhance construction monitoring
- Automate permit approvals
- Reduce inspector exposure to hazardous conditions
- Enable quicker information gathering

7



Smart City Platform

- Use data and sensors to optimize city operations, improve services like transportation and energy management, and enhance quality of life for residents through real-time data analysis and decision-making

Expected Impact

- Unifies subsystems under a single platform
- Streamlines management processes
- Enhances ease of use and operational efficiency

8



Traffic Management

- Leverage data analytics and real-time monitoring to optimize traffic flow, reduce congestion, and enhance road safety.
- Improve the efficiency of transportation networks and the overall commuting experience

Expected Impact

- Reduced traffic congestion
- Improved road safety
- Enhanced public satisfaction with transportation services
- Increased economic productivity

9



Public Parking Management

- Optimize the management of public parking spaces by adopting real-time monitoring of parking availability, implementing dynamic pricing strategies, optimizing enforcement efforts, and providing convenient parking guidance to drivers

Expected Impact

- Better utilization of parking spaces
- Reduced traffic congestion
- Improved traffic flow
- Easier parking access and guidance



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~10M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	Medium	High	High	Medium

1. Business Overview

Description

- The purpose of the use case is to utilize AI-driven methodologies to monitor and manage the ongoing expansion of urban infrastructure and green spaces effectively. This involves employing innovative techniques to identify, analyze, and address distortions or disruptions in the urban landscape, ensuring optimal utilization of resources and preservation of environmental sustainability
- Key beneficiaries of this use case include Citizens & Residents, Public Cleanliness & Appearance Division, ADDC, Etisalat, Du, Tadweer, AWQAF

1.2 Pain Points

- Time and human resources required to segment and prepare the training datasets for those customs AI models

1.3 Expected Impact

- Enhance the efficiency of operations
- Lower substantially the costs and time required for detecting violations

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5 Risks

- AI accuracy risks
- Privacy risks
- Data Quality
- Integration with multiple systems could be challenging

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Accessible	High

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

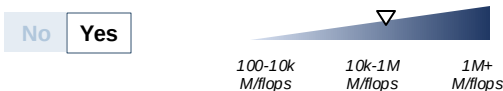
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Adequate data for solutions
- Technology partners required for implementation:
- Secure personal data handling



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~11M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	Medium	High	High	Medium

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
High	Medium	AI Model Complexity	High	Medium

1. Business Overview

Description

- The purpose of the use case is to use text data to understand public opinions, identify whether feedback is positive, negative, or neutral. The use case also helps decision-makers improve services based on public sentiment
- Key beneficiaries of this use case include Citizens, DMT and Affiliates

1.2

Pain Points

- Planning the development of the city, improving traffic congestion, simplifying parking way finding, driving revenue through violation detection via LLM with vision

1.3

Expected Impact

- Enhance service management efficiency and consistency
- Allow staff to concentrate on key responsibilities
- Improve timely interactions with the audience.

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- AI accuracy risks
- Privacy risks
- Bias

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Accessible	High

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes

100-10k M/flops 10k-1M M/flops 1M+ M/flops

2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- The implementation of the use case requires collaboration and alignment with media team
- Secure personal data handling

Facial Recognition & Driver / Passenger Behavior Analysis



DMT

Status:

Not started

3

Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility					~44M AED	6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility			
High	High	High	Medium	Medium	Low	Low	High	Medium	Low			



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~27M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	Medium	Medium	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	Low	High	Low	Low

1. Business Overview

Description

- The purpose of this use case is to analyze vast datasets for predictive traffic flow, congestion prevention, and optimized transportation modelling and infrastructure planning
- Key beneficiaries of this use case include Commuters, ITC Planning and Operations Sector and City Planners

1.2

Pain Points

- Planning the development of the city, improving traffic congestion, simplifying parking way finding, driving revenue through violation detection via LLM with vision

1.3

Expected Impact

- Enhance the accuracy of transportation and traffic modelling outcomes
- Reduced traffic congestion
- Shorter travel times
- Improved fuel efficiency
- Data-driven infrastructure development

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Data quality and integration with fast IoT devices and sources
- AI Accuracy

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Some data is available with ITC, others require input	Not yet defined	Not yet defined	Not yet defined	Medium	Not yet defined

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

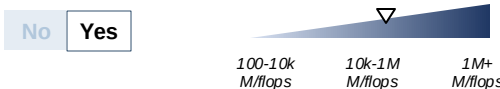
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Adequate data for solutions
- Ease of integration
- Ensuring SLA with external stakeholders to ensure availability and reliability of data sources



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~8M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	Medium	High	Medium

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
High	Medium	AI Model Complexity	Medium	Medium
		Medium		

1. Business Overview

Description

- The purpose of the use case is to leverage data analysis and machine learning to forecast trends, pricing changes, and demand fluctuations in the real estate market
- Key beneficiaries of this use case include Government and Policy Makers, Real Estate Investors, Property Developers, Real Estate Agents and Brokers, Homebuyers and Renters, and Financial Institutions

1.2

Pain Points

- Planning the development of the city, improving traffic congestion, simplifying parking way finding, driving revenue through violation detection via LLM with vision

1.3

Expected Impact

- Improving investment and pricing strategies

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Accuracy and Volatility of the Real Estate Market and Policy Makers

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Accessible	High

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

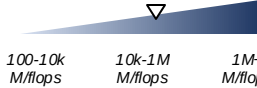
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Adequate data for solutions.
- Technology partners required for implementation:
- Visualization tool connectivity and accessibility with ML pipeline
- Ease of integration



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~15M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	Medium	High	High	High

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Medium	High	AI Model Complexity	Medium	Medium
		Medium		

1. Business Overview

Description

- The purpose of this use case is to employ AI-based monitoring systems for quality control and project oversight. Leveraging AI's diagnostic capabilities, they identify root causes of deviations, enabling proactive measures to optimize project outcomes and mitigate risks effectively
- Key beneficiaries for this use case include Citizens, Abu Dhabi Leadership, DMT Leadership and ADREC

1.2

Pain Points

- Challenges in project monitoring, compliance with regulations, and resource management
- Traditional methods of site monitoring and inspection can be slow, labor-intensive, and prone to human error, leading to delays, cost overruns, and potential safety hazards

1.3

Expected Impact

- Enhance construction monitoring
- Automate permit approvals
- Reduce inspector exposure to hazardous conditions
- Enable quicker information gathering

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Vision model accuracy
- Privacy risks
- Data quality
- Integration with multiple systems
- Regulatory compliance
- Operational risks

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Complex	High

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

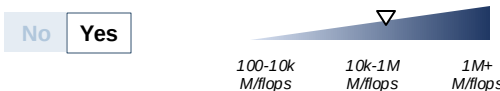
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Technology partners required for implementation:
- Secure personal data handling
- Ease of integration



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~57M AED	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Low	Medium	High	High

Data Availability	Infrastructure Readiness	2. Feasibility	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	High	High	Low	

1. Business Overview

Description

- The purpose of this use case is to use data and sensors to optimize city operations, improve services like transportation and energy management, and enhance quality of life for residents through real-time data analysis and decision-making
- Key beneficiaries for this use case include Citizens, Abu Dhabi Leadership, DMT Leadership, Planning & Infrastructure, ITC, ADREC, AD Police, ADDC, DCT, Etisalat, and Du

1.2 Pain Points

- The lack of a centralized management system for urban subsystems like traffic control, urban development plans, street lighting, wastewater networks and sensor networks results in fragmented and inefficient operations

1.3 Expected Impact

- Unifies subsystems under a single platform
- Streamlines management processes
- Enhances ease of use and operational efficiency

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineers
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5 Risks

- Integration with the existing tools
- Absence of MDM
- Data quality issues
- Absence of reference architecture for whole Government

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Complex	High

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation Augmentation - Advisory - Interaction	- Dashboard Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

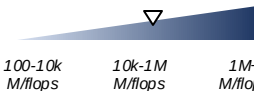
Cloud Services and Providers

- Technology Stack: Cloud, IoT, Streaming, ML
- Cloud Services: On-premise or cloud with GPU

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Technology partners required for implementation:
- Ethical considerations need to be accounted for when planning and implementing the use case
- Comply with related data laws



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5-20M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Medium	High	Medium	High

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Medium	Medium	AI Model Complexity	Medium	Medium
		Medium		

1. Business Overview

Description

- The purpose of this use case is to leverage data analytics and real-time monitoring to optimize traffic flow, reduce congestion, and enhance road safety. It seeks to integrate various data sources such as traffic cameras, vehicle sensors, and weather forecasts to improve the efficiency of transportation networks and the overall commuting experience
- Key beneficiaries for this use case include the whole of Abu Dhabi society such as residents and commuters and public safety operators

1.2

Pain Points

- Persistent traffic congestion during peak hours
- Inefficient utilization of existing road infrastructure
- Difficulty in identifying and responding to traffic incidents and emergencies
- Lack of real-time insights into traffic patterns and trends for informed decision-making

1.3

Expected Impact

- Reduced traffic congestion
- Improved road safety
- Enhanced public satisfaction with transportation services
- Increased economic productivity

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Data privacy and security
- Integration into multiple systems
- Reliance on accurate and up-to-date data for effective traffic management
- Potential resistance to change

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation Augmentation - Advisory - Interaction	- Dashboard Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive Diagnostic Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition ML (supervised, unsupervised, reinforcement learning) - Generative AI

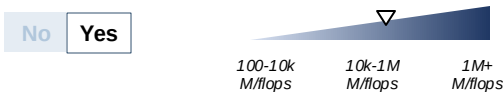
2.3

Cloud Services and Providers

- Technology Stack: Cloud, IoT, GIS, Sensor data
- Cloud Services: Azure/ AWS / Google Cloud with AI studio

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Ethical considerations need to be accounted for when planning and implementing the use case
- Secure personal data handling
- Adequate data for solutions.



Prioritization Outcomes					Preliminary Budget		Expected Duration	Required Team Size
1. Expected Impact					~20M AED		6-12 Months	8 FTEs
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality				
High	Medium	Medium	Medium	High				

2. Feasibility					
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	
Medium	Low	High	Medium	Low	

1.1

1. Business Overview

Description

- The purpose of this use case is to optimize the management of public parking spaces. This includes real-time monitoring of parking availability, implementing dynamic pricing strategies, optimizing enforcement efforts, and providing convenient parking guidance to drivers
- Key beneficiaries for this use case include Abu Dhabi residents and tourists

1.2

Pain Points

- Limited parking availability leading to traffic congestion
- Manual monitoring and enforcement processes
- Ineffective pricing strategies leading to underutilization of premium parking spaces

1.3

Expected Impact

- Better utilization of parking spaces
- Reduced traffic congestion
- Improved traffic flow
- Easier parking access and guidance

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5

Risks

- Data Privacy and Security
- Integration of multiple systems
- Compliance with local regulations and laws

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined	Not yet defined

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: Cloud, IoT, GIS, Sensor data
- Cloud Services: Azure/ AWS / Google Cloud with AI studio

GPU Enabled Infra

Computational Power

NoYes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Sufficient ML build/operational infrastructure
- Ethical considerations need to be accounted for when planning and implementing the use case
- Secure personal data handling
- Adequate data for solutions

Furthermore, an additional of 10 use cases have been defined by DMT

DMT



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Identified AI Use Cases

DMT Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
10	Digital Twin for Smart City	Digital Twin to analyze traffic patterns, pedestrian movement, and other components. Gen AI model to help optimize transport options and help with city planning initiatives	●	●	6-12 months
11	Visual Pollution Detection	Utilizes AI to assess and recommend scalable solutions for visual pollution detection	●	●	0-6 months
12	Traffic Modeling	Examine the flow of traffic and movement for patterns and anomalies	●	●	0-6 months
13	Geospatial Intelligence for Mobility Planning	Assist in making digital twins, visualizations, computational analysis and simulations using geospatial data	●	●	0-6 months
14	Climate Command Center	Copilot to Assist in Analyzing Options for Climate Risk Mitigation. Provides Task Based automations to reduce time, complexity and risk to respond to climate change	●	●	0-6 months
15	GenAI Flood Warning	Early warning for flood and other disaster components, leverage GenAI to help operators identify most critical areas of concern and next best actions	●	●	0-6 months
16	Public Safety Monitoring	Use ML to analyze surveillance footage in real-time to detect unusual behaviors or potential threats and alert authorities	●	●	6-12 months
17	Automated Permit Processing	Automate the review and approval of permit applications using AI to ensure timely and transparent processing.	●	●	0-6 months
18	Citizen Chatbot	Deploy an AI-powered chatbot to handle common inquiries and provide instant responses to citizens	●	●	0-6 months
19	Address Normalization	Utilize generative AI to transform noisy, human-inputted addresses into clean data compatible with navigation systems.	●	●	0-3 months



DoE – Deep Dive

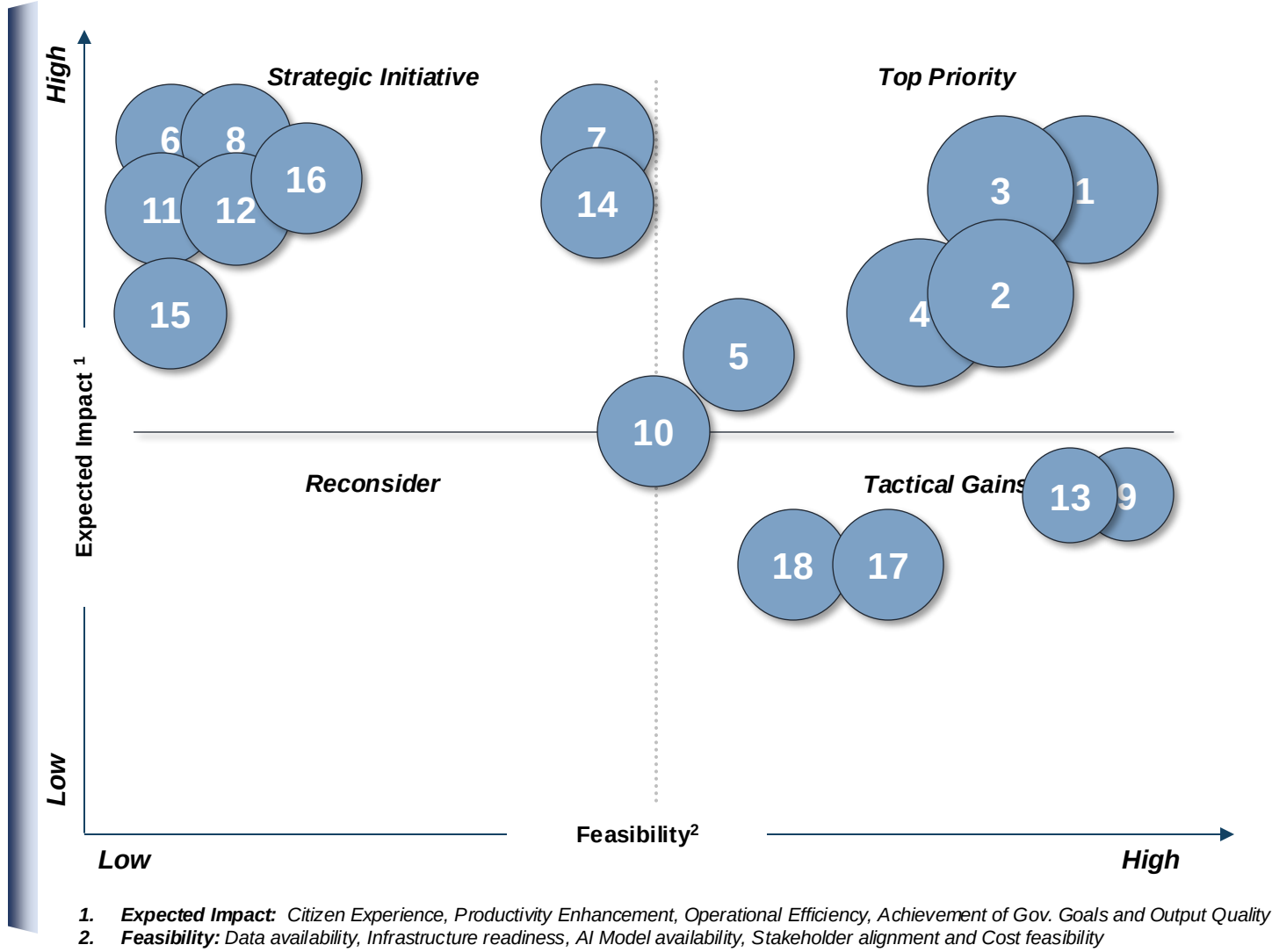
Summary and prioritization of identified use cases, across impact and feasibility

DOE

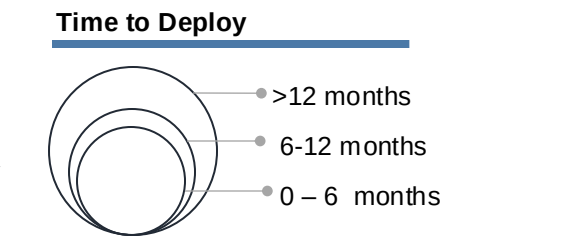


18

Identified AI Use Cases



Identified Use Cases	
1	Demand Side Management
2	Smart Grid Optimization
3	Renewable Energy Forecasting
4	Carbon Capture Utilization and Storage
5	Infrastructure Predictive Maintenance
6	Dynamic Capital Allocation
7	Regulatory Compliance Monitoring
8	Multi-variate Margin Optimization
9	Supply Planning Optimization
10	Failure Prediction & System Retrofit
11	Site Inspection Optimization
12	Equipment Efficiency Optimization
13	Demand Response Management
14	Renewable Asset Output Optimization
15	Systems Performance Optimization
16	Decarbonization Scenario Planning
17	Environmental Assessment & Action
18	Environmental Monitoring (Custom ML)



DOE has prioritized 5 use cases from their identified long list

1



Demand Side Management

- Predicts and optimizes water and electricity consumption patterns

Expected Impact

- Improved anticipation of future demand
- Optimized operations
- Improved resource management

2



Smart Grid Optimization

- Facilitates the management of electricity flow across the grid, enhancing efficiency by predicting demand patterns and integrating renewable energy sources smoothly

Expected Impact

- Improved energy efficiency, grid stability, reliability, security, and flexibility, while also supporting the integration of renewable energy

3



Renewable Energy Forecasting

- Forecast the output from renewable energy sources such as solar and wind

Expected Impact

- Optimized energy production
- Improved grid integration
- Reduced cost
- Enhanced market participation
- Increased reliability and resilience
- Improved support for renewable energy planning and policy development, and technological innovation

4



Carbon Capture Utilization and Storage

- enhancing the efficiency and effectiveness of carbon capture, utilization, and storage technologies

Expected Impact

- Improved capture efficiency
- Reduced costs
- Optimized storage selection
- Enhanced monitoring
- Efficient utilization of carbon and climate change mitigation
- Improved innovation

5



Infrastructure Predictive Maintenance

- Monitors infrastructure and equipment performance based on pre-defined measures, leveraging AI to identify anomalies and predict required improvements and maintenance efforts to avoid failure or downtime

Expected Impact

- Reduced downtime and costs
- Optimized maintenance scheduling
- Enhanced equipment safety
- Improved asset visibility and monitoring
- Enhanced energy efficiency
- Improved remote monitoring and diagnostics
- Improved innovation and competitiveness

Demand Side Management

Forecasting Water and Electricity Consumption



Department of Energy –
DoE

Status: Not started

1

Prioritization Outcomes

1. Expected Impact				
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	High

2. Feasibility				
Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	High	High	Medium

Preliminary Budget	Expected Duration	Required Team Size
5M AED	>12 Months	11 FTEs

1. Business Overview

1.1 Description

- The purpose of the use case is to forecast water and electricity consumption more accurately by utilizing predictive analytics and AI. This will in turn improve resource management and efficiency by predicting the demand fluctuations based on various factors like weather, time of day, and consumer behavior
- Key beneficiaries of this use case are utility providers, consumers, and regulatory bodies

1.2 Pain Points

- Limited ability to detect and mitigate wastage or unauthorized usage effectively
- Inability to handle peak demand efficiently, resulting in potential service interruptions

1.3 Expected Impact

- More accurate forecasts will lead to better resource planning and reduced wastage
- Ability to manage peak loads effectively, ensuring consistent service delivery
- Improved customer satisfaction through more accurate billing and fewer service disruptions

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineer
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5 Risks

- Incomplete or incorrect data
- Data Availability
- Potential inaccuracies in predictions due to variable factors like unexpected weather conditions or economic shifts
- Performance Bottlenecks

2. Data, AI and Infrastructure Overview

2.1 Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Back-end Systems of Energy Sector Companies	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3 Cloud Services and Providers

- AWS or Azure (based on DoE current cloud service provider) for scalable storage and computational power
- Data Science Environment and tools
- Visualization tools

GPU Enabled Infra

Computational Power

No

Yes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4 Key Considerations

- Provision for scalable processing power to handle complex predictive modeling and real-time data processing
- Visualization tool connectivity and accessibility with ML pipeline
- Regularly update models to adapt to new data and changing conditions



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~20M AED	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	High

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	High	High	Low

1. Business Overview

Description

- The purpose of the use case is to facilitate the management of electricity flow across the grid, enhancing efficiency by predicting demand patterns and integrating renewable energy sources smoothly
- Key beneficiaries of this use case include grid operators predominantly as it will help them improve load balancing and reducing energy wastage, and consumers who can benefit from lower energy prices

1.2 Pain Points

- Difficulty in predicting and managing fluctuating demand
- Inefficiencies in energy distribution can lead to higher operational costs and energy waste.

1.3 Expected Impact

- Improving energy efficiency, grid stability, reliability, security, and flexibility.
- Supporting the integration of renewable energy,

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineer
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5 Risks

- Incomplete or incorrect data
- Data Availability
- Performance Bottlenecks

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Back-end Systems of Energy Sector Companies	TBD	TBD	TBD	TBD	TBD	TBD

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

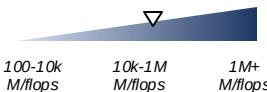
Cloud Services and Providers

- AWS or Azure (based on DoE current cloud service provider) for scalable storage and computational power
- Data Science Environment and tools
- Visualization tools

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Provision for scalable processing power to handle complex predictive modeling and real-time data processing
- Ease of integration
- Visualization tool connectivity and accessibility with ML pipeline



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5M AED	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	High

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	High	High	Medium

1. Business Overview

Description

- The purpose of the use case is to leverage AI to forecast demand for renewable energy sources such as solar and wind. Accurate forecasting will enable better integration of renewable energy into the power grid, optimize energy production, and enhance market participation
- Key beneficiaries of this use case include utilities providers, energy producers and grid operators

1.2

Pain Points

- Difficulty in optimizing energy production to meet demand fluctuations
- Challenges in maximizing economic benefits due to inefficient forecasting and energy pricing

1.3

Expected Impact

- Improved Grid Integration
- More accurate energy production forecasts allow for better alignment with demand patterns
- Efficient energy management and reduced need for expensive energy storage or backup solutions

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineer
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5

Risks

- Incomplete or incorrect data
- Data Availability
- Risk of misestimating energy yields, potentially leading to energy deficits or oversupply
- Market Volatility
- Performance Bottlenecks

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Back-end Systems of Energy Sector Companies	TBD	TBD	TBD	TBD	TBD	TBD

2.2

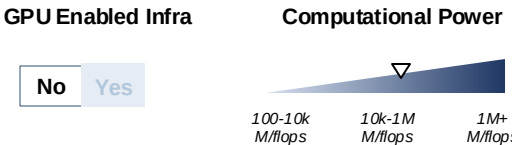
AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

Cloud Services and Providers

- AWS or Azure (based on DoE current cloud service provider) for scalable storage and computational power
- Data Science environment and tools
- IoT and Sensor Data Integration



2.4

Key Considerations

- Provision for scalable processing power to handle complex predictive modeling and real-time data processing
- Ease of integration
- Visualization tool connectivity and accessibility with ML pipeline



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~20M AED	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	High

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	High	High	Low

1. Business Overview

Description

- The purpose of this use case is to enhance the efficiency and effectiveness of carbon capture, utilization, and storage technologies. This will directly support DoE in reducing their carbon footprint effectively and economically, thereby aiding in meeting global climate targets and promoting environmental sustainability
- Key beneficiaries of this use case include DoE and consumers

1.2 Pain Points

- Inefficiencies in current carbon capture technologies leading to high operational costs
- Challenges in identifying optimal sites for carbon storage and in monitoring these sites effectively
- Risks associated with carbon leakage

1.3 Expected Impact

- Optimized Capture Efficiency
- Cost reduction in capture and storage operations
- Enhanced Storage Site Selection
- Improved Monitoring and Leakage Detection

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineer
- 1 Full stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5 Risks

- Incomplete or incorrect data
- Data Availability
- Performance Bottlenecks

2. Data, AI and Infrastructure Overview

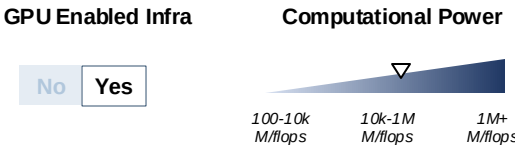
Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Back-end Systems of Energy Sector Companies	TBD	TBD	TBD	TBD	TBD	TBD

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- AWS or Azure (based on DoE current cloud service provider) for scalable storage and computational power
- Data Science environment and tools
- IoT and Sensor Data Integration



2.4 Key Considerations

- Provision for scalable processing power to handle complex predictive modeling and real-time data processing
- Ease of integration
- Regularly update models to adapt to new data and changing conditions
- Visualization tool connectivity and accessibility with ML pipeline



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
5M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	High

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	High	High	Low

1. Business Overview

Description

- The purpose of this use case is to leverage AI to monitor infrastructure and equipment performance through predefined measures. The goal is to identify anomalies, predict maintenance needs, and schedule improvements proactively to minimize downtime and prevent failures.
- Key beneficiaries of this use case include infrastructure managers, utilities providers and consumers

1.2 Pain Points

- Unplanned downtime and failures leading to high repair costs and operational disruptions
- Inefficiencies in routine maintenance schedules that do not align with actual equipment condition

1.3 Expected Impact

- Proactive maintenance allows for scheduling repairs before failures occur
- Cost reduction in capture and storage operations
- Optimizing maintenance schedules reduces unnecessary checks and focuses resources where they are needed most

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Testers

1.5 Risks

- Inadequate data can lead to inaccurate predictions, affecting maintenance effectiveness
- Data Availability
- Performance Bottlenecks
- Resistance to Change

2. Data, AI and Infrastructure Overview

Data Requirements

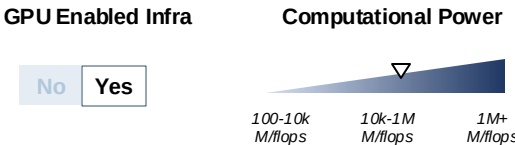
Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Back-end Systems of Energy Sector Companies	TBD	TBD	TBD	TBD	TBD	TBD

2.2 AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- AWS or Azure (based on DoE current cloud service provider) for scalable storage and computational power
- Data Science environment and tools
- IoT and Sensor Data Integration



2.4 Key Considerations

- Provision for scalable processing power to handle complex predictive modeling and real-time data processing
- Ease of integration
- Regularly update models to adapt to new data and changing conditions
- Visualization tool connectivity and accessibility with ML pipeline
- Implement effective change management strategies

Furthermore, an additional of 13 use cases have been defined by DOE

DOE



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Identified AI Use Cases

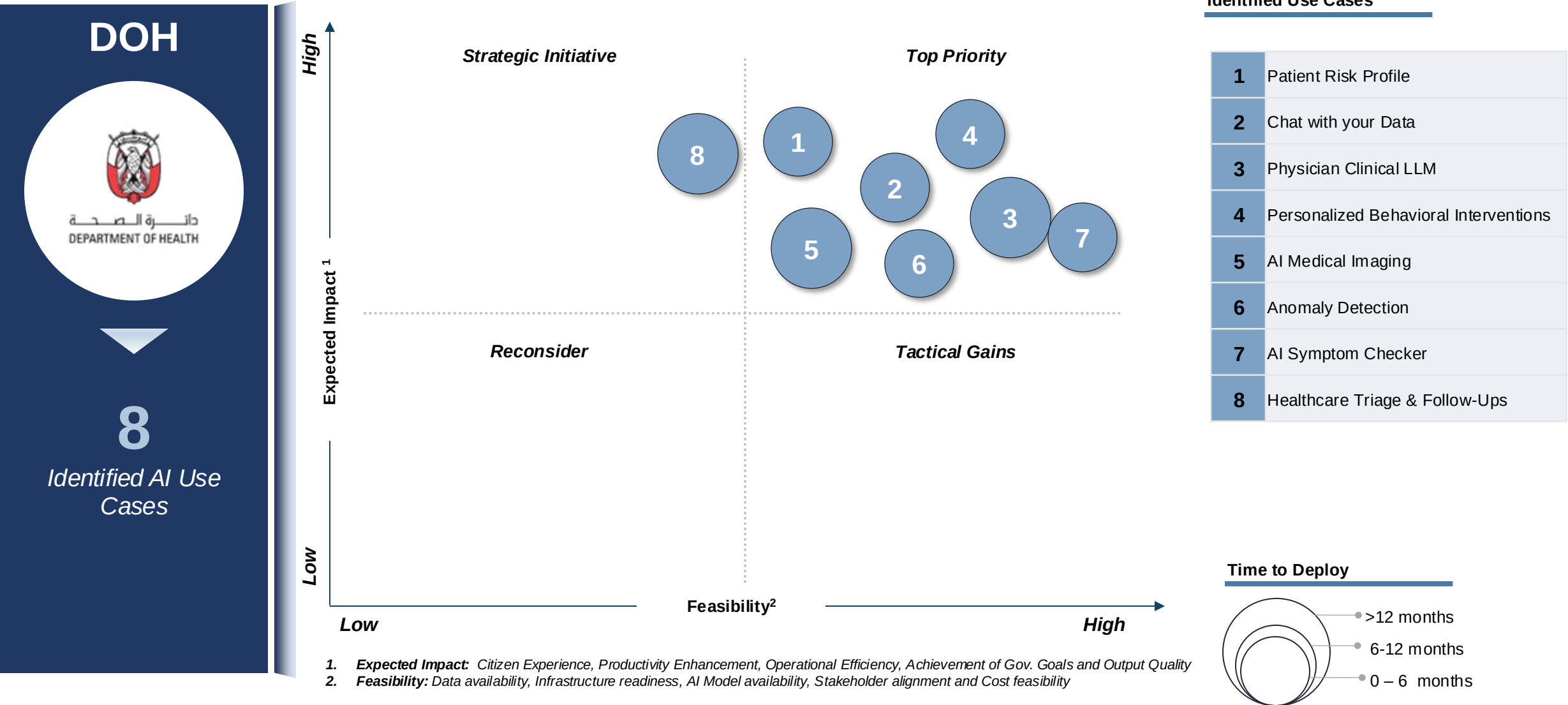
DOE Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
6	Dynamic Capital Allocation	Optimize CAPEX allocation across the operations value chain based on net returns, sector-specific demand curves and other key operational metrics	High	Low	6-12 Months
7	Regulatory Compliance Monitoring	Monitor compliance submissions for expenses and operating costs for anomalies and irregularities, assisting with management of operations expenses using GenAI	High	Medium	6-12 Months
8	Multi-variate Margin Optimization	Balance dynamically between cost, production, margin and carbon based on real-time market sensing to optimize utilization and improve equipment maintenance	High	Low	6-12 Months
9	Supply Planning Optimization	Optimize overall supply by using AI to forecast energy load using real-time demand data, deriving the required supply from the different sources such as generators and the required gas reserves to achieve this supply	Medium	High	0-6 Months
10	Failure Prediction & System Retrofit	Apply a predictive AI model that utilizes downhole sensor data (current and historical), and principal component analysis model data (PCA) to predict failures, provide an early warning system, and allow operators to make better-informed decisions	Medium	Medium	6-12 Months
11	Site Inspection Optimization	Optimize the collection and management of site inspection documents, leveraging GenAI to generate result documents and outcome documents for inspectors and regulators	High	Low	6-12 Months
12	Equipment Efficiency Optimization	Improve efficiency of equipment using an AI model that utilizes performance metrics from the equipment (e.g. turbine blade cooling, fuel consumption, etc.) to provide operators with proactive measures, enhance planning for maintenance services and management of spare parts, and alert relevant stakeholders	High	Low	6-12 Months
13	Demand Response Management	Analyze consumption patterns to optimize energy usage during peak times, shifting non-essential consumption to off-peak periods	Medium	High	0-6 Months
14	Renewable Asset Output Optimization	Predict the best locations for renewable assets like wind turbines and solar panels and optimize their output based on weather conditions	High	Medium	6-12 Months
15	Systems Performance Optimization	Leverage AI to monitor the performance of renewable energy systems in real-time and recommend real-time actions to maximize efficiency and reduce waste	High	Low	6-12 Months
16	Decarbonization Scenario Planning	Predict future emissions based on current trends and anticipated changes in policy, economics, and technology and simulate various future scenarios based on changes in energy production methods, carbon pricing, and technology adoption rates	High	Low	6-12 Months
17	Environmental Assessment & Action	Employ AI to perform environmental impact assessments and devise mitigation strategies	Medium	Medium	6-12 Months
18	Environmental Monitoring (Custom ML)	Implement ML models to analyze environmental data from sensors and satellites to predict pollution levels and identify sources	Medium	Medium	6-12 Months



DOH – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility



DOH has prioritized 7 use cases from their identified long list

1



Patient Risk Profile

- Identify the probability a patient has of developing a disease (14 diseases e.g., diabetes, hypertension) and provide the risk level to physicians for more accurate and timely diagnosis

Expected Impact

- Increased accuracy of diagnosis by assisting physicians in identifying potential risks
- Improved health outcomes due to early identification of risks, diagnosis and treatment
- Time savings for physicians

2



Chat with your Data

- Provide real time answers to questions on all available healthcare data (e.g., disease registries, etc.) making data easier and quicker to access and interpret

Expected Impact

- Time savings for decision makers
- Improved regulator decision making due to more intuitive, quicker and easier to interpret access to insights

3



Physician Clinical LLM

- Provide physicians with a search tool within Malaffi to quickly access active patient information, for example anomalies in a patient's latest blood test results.

Expected Impact

- More accurate diagnosis by assisting physicians in identifying anomalies
- Improved health outcomes due to improved diagnosis leading to precise treatment plans
- Time savings for physicians

4



Personalized Behavioral Interventions

- Behavioral nudges for the population, leveraging AI, to increase their physical activity levels with the aim to reduce overall obesity rates.

Expected Impact

- Healthcare cost savings for obesity through prevention and early management
- Improved health outcomes due to personalization for patients
- Data collected from PBI can provide further insights into population health trends, behavior patterns and effectiveness of interventions

DOH has prioritized 7 use cases from their identified long list

5



AI Medical Imaging

- Early and more accurate detection of diseases by analyzing X-ray and CT scans across a range of diseases (e.g., lung diseases)

Expected Impact

- Improved patient outcomes due to more accurate and/or early detection
- Healthcare cost savings for diseases due to accurate/early detection
- Optimized use of radiologists

6



Anomaly Detection

- Proactively detects and flags anomalous physician and provider transactions (claims, sick leaves and prescription of select drugs) initiating ongoing investigations for improvements in outcomes and experiences

Expected Impact

- Cost savings by preventing fraudulent claims from being paid out
- Improved patient safety due to prevention of fraudulent prescriptions
- Improved compliance and reducing risk of regulatory violations

7



AI Symptom Checker

- AI-based symptom diagnostic tool based on symptom-to-disease correlation. Enables patients to determine potential diagnoses and risk levels for their potential conditions.

Expected Impact

- Improved access to healthcare
- Reduced healthcare costs through self-assessing for minor ailments
- Reduced healthcare costs due to early detection of diseases



Prioritization Outcomes

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	Medium	Medium	High	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	High	High	Medium	High

Preliminary Budget

~2.8M AED

Expected Duration

0-6 Months

Required Team Size

6 FTEs

1. Business Overview

Description

- The purpose of this use case is to identify the probability a patient has of developing a disease (14 diseases e.g., diabetes, hypertension) and provide the risk level to physicians for more accurate and timely diagnosis, for overall improved patient outcomes
- Key beneficiaries of this use case are mainly the physicians and the healthcare providers. Other indirect beneficiaries include the patients

1.2

Pain Points

- Access to Data and AI capabilities is a continuous area of improvement
- Effective top tech talent attraction
- Strict cybersecurity requirements

1.3

Expected Impact

- Increased accuracy of diagnosis by assisting physicians in identifying potential risks
- Improved health outcomes due to early identification of risks, diagnosis and treatment
- Time savings for physicians

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Privacy and security concerns with patient data
- Potential inaccuracies in risk stratification
- Bias in risk identification due to incomplete training data
- Resistance from physicians to use risk stratification

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet identified	Available	TBD	TBD	TBD	TBD	Highly confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

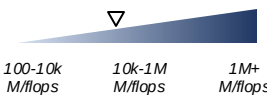
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Secure personal data handling
- Visualization tool connectivity and accessibility with ML pipeline
- Sufficient ML build/operational infrastructure
- Adequate data for solutions
- Implement effective change management strategies to facilitate the transition from old practices to new systems



Prioritization Outcomes

Preliminary Budget

~0.8M AED

Expected Duration

0-6 Months

Required Team Size

6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	High	Low	High	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	High	High	Medium	High

1. Business Overview

Description

- The purpose of this use case is to provide real time answers to questions on all available healthcare data (e.g., disease registries, etc.) making data easier and quicker to access and interpret, and allowing easier access to insights and indicators
- Key beneficiaries of this use case are mainly the patients, healthcare providers and regulators

Pain Points

- Difficulty accessing sensitive data
- Lengthy analytical processes
- Delayed decision making

Expected Impact

- Time savings for decision makers
- Improved regulator decision making due to more intuitive, quicker and easier to interpret access to insights

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

Risks

- Potential privacy and security risks related to patient personal data
- Language barriers and ambiguous questions may lead to inaccurate insights
- Staff may resist leveraging insights via the chatting tool and continue decision making based on existing practices

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet identified	Easy to obtain	TBD	TBD	TBD	TBD	Highly confidential

AI Requirements

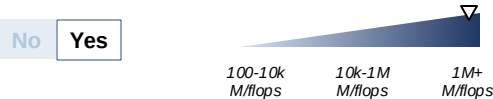
AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Statistics - ML (supervised, unsupervised, reinforcement learning) - Generative AI

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



Key Considerations

- Secure personal data handling
- Ease of integration
- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- Adequate data for solutions
- Implement effective change management strategies to facilitate the transition from old practices to new systems



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~1.8M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	Medium	Medium	Medium	Medium

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Medium	High	High	Medium	High

1. Business Overview

Description

- The purpose of this use case is to provide physicians with a search tool within "Malaffi" to quickly access active patient information, for example anomalies in a patient's latest blood test results
- Key beneficiaries of this use case are the physicians. Indirect beneficiaries include the patients and healthcare providers

1.2 Pain Points

- Long information retrieval times
- Risk of missed information
- Room to improve efficiency in coordinating healthcare delivery

1.3 Expected Impact

- More accurate diagnoses by assisting physicians in identifying anomalies
- Improved health outcomes due to improved diagnoses leading to precise treatment plans
- Time savings for physicians

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Potential privacy and security risks related to patient personal data
- Low adoption / potential push back from physicians who may not want to use the search tool

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet identified	Easy to obtain	TBD	TBD	TBD	TBD	Highly confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

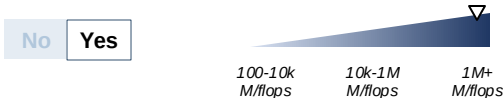
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Secure personal data handling
- Ease of integration
- Sufficient ML build/operational infrastructure
- Gen AI model and infrastructure availability
- Adequate data for solutions
- Implement effective change management strategies to facilitate the transition from old practices to new systems



Prioritization Outcomes										Preliminary Budget	Expected Duration	Required Team Size
1. Expected Impact					2. Feasibility							
Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality	Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility	~0.4M AED	0-6 Months	6 FTEs
High	Low	Low	High	High	High	High	Medium	Medium	High			

1.1

1. Business Overview

Description

- The purpose of this use case is to provide behavioral nudges for the population, leveraging AI, to increase their physical activity levels with the aim to reduce overall obesity rates
- Key beneficiaries of this use case are primarily Abu Dhabi population, indirect beneficiaries are the wider Abu Dhabi healthcare system

1.2

Pain Points

- Lack of motivation and engagement among the population
- Lack of awareness of behavioral factors
- Lack of personalized approaches to obesity prevention and tackling
- Decreased community well-being

1.3

Expected Impact

- Healthcare cost savings for obesity through prevention and early management
- Improved health outcomes due to personalization for patients
- Data collected from PBI can provide further insights into population health trends, behavior patterns and effectiveness of interventions

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Potential privacy and security risks related to patient personal information
- Patients could become over-reliant on interventions, or have increased stress/anxiety by constant monitoring

2.1

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet identified	Easy to obtain	TBD	TBD	TBD	TBD	Highly confidential

2.2

AI Requirements

AI Capabilities

- Automation
- Augmentation
- Advisory
- Interaction

End Product

- Dashboard
- Product/ app.
- Embedded Model
- Other:

AI Methodology

- Descriptive
- Prescriptive
- Diagnostic
- Predictive
- Cognitive

AI Technology

- Computer Vision
- NLP
- Speech Recognition
- ML (supervised, unsupervised, reinforcement learning)
- Generative AI

2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

NoYes

100-10k M/flops

10k-1M M/flops

1M+ M/flops

2.4

Key Considerations

- Secure personal data handling
- Ease of integration
- Sufficient ML build/operational infrastructure
- Adequate data for solutions



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~1.8M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	High	High	Medium

Data Availability

Low

Infrastructure Readiness

High

2. Feasibility

AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	Medium	High

1. Business Overview

Description

- The purpose of this use case is to significantly improve early detection and accuracy of diagnosing diseases through the analysis of X-ray and CT scans across a range of diseases (e.g., lung disease). This capability not only speeds up the diagnostic process but also enhances the precision of the results, potentially leading to early interventions and better patient outcomes
- Key beneficiaries of this use case are radiologists and patients

1.2

Pain Points

- Attracting top tech talent and access to expertise
- Stringent cybersecurity requirements
- Workload on radiologists
- Delayed diagnoses

1.3

Expected Impact

- Improved patient outcomes due to more accurate and/or early detection
- Healthcare cost savings for diseases
- Optimized use of radiologists and reduced fatigue
- Standardized diagnostics

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientists
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Data privacy and security
- Integration with existing systems
- Regulatory and compliance challenges
- Flawed predictions due to data issues
- Models tuned to past data may not perform well with new data
- Difficulty in understanding Complex AI models

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet identified	Available	TBD	TBD	TBD	TBD	Highly confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

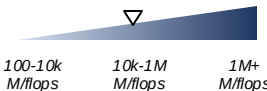
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No Yes

Computational Power



2.4

Key Considerations

- Collaboration with providers to adopt AI based imaging technology
- Ease of integration
- Secure personal data handling
- Implement effective change management strategies to facilitate the transition from old practices to new systems



Prioritization Outcomes

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Low	Medium	High	Medium	Low

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	High	Low	Medium	High

Preliminary Budget

~0.01M AED

Expected Duration

0-6 Months

Required Team Size

6 FTEs

1. Business Overview

Description

- The purpose of this use case is to proactively detect and flag anomalous physician and provider transactions (claims, sick leaves and prescription of select drugs) initiating ongoing investigations for improvements in outcomes and experiences
- Key beneficiaries of this use case are DOH sectors

1.2

Pain Points

- Difficulties in compliance monitoring
- Difficulties in detecting fraudulent transactions
- Reduced operational efficiencies

1.3

Expected Impact

- Cost savings by preventing fraudulent claims from being paid out
- Improved patient safety due to prevention of fraudulent prescriptions
- Improved compliance and reducing risk of regulatory violations

1.4

Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Potential privacy and security risks related to patient personal data
- Risk that legitimate transactions are flagged as anomalous
- Staff may resist leveraging tool and continue managing anomalies based on existing practices

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet identified	Available	TBD	TBD	TBD	TBD	Highly confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3

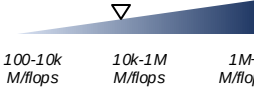
Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power

No Yes



2.4

Key Considerations

- Secure personal data handling
- Ease of integration
- Implement effective change management strategies to facilitate the transition from old practices to new systems



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~0.01M AED	0-6 Months	6 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	Low	Low	Medium	Medium

2. Feasibility

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
High	High	Low	High	High

1. Business Overview

Description

- The purpose of this use case is to provide an AI-based symptom diagnostic tool based on symptom-to-disease correlation. Such a tool enables patients to determine potential diagnoses and risk levels for their potential conditions.
- Key beneficiaries of this use case are primarily the patients

1.2 Pain Points

- Continuous need for accurate and timely diagnosis
- Need for faster decision making

1.3 Expected Impact

- Improved access to healthcare
- Reduced healthcare costs through self-assessing for minor ailments
- Reduced healthcare costs due to early detection of diseases

1.4 Expected Squad Required

- 0.5 Scrum Master
- 0.5 Business Analyst
- 2 AI Scientists
- 1 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Potential privacy and security risks related to patients' personal information
- Bias in interventions identified if training data is not comprehensive and representative of the population
- Patients could become over-reliant on symptom checker and may not engage with healthcare providers when needed

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
Not yet identified	Available	TBD	TBD	TBD	TBD	Highly confidential

2.2 AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

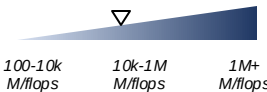
2.3 Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, G42, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

No Yes

Computational Power



2.4 Key Considerations

- Adequate data for solutions
- High confidentiality as it deals with personal patients' data
- Visualization tool connectivity and accessibility with ML pipeline

Furthermore, an additional use case has been identified by DOH

DOH



8

Identified AI Use Cases

DOH Long List of Non-prioritized Use Cases

Title	Description	AI Impact	Feasibility	Time to Deploy
8 Healthcare Triage & Follow-Ups	Implement AI to triage patients, summarize medical interactions, and suggest pertinent follow-up questions for healthcare providers.			6-12 months



SCAD – Deep Dive

Summary and prioritization of identified use cases, across impact and feasibility

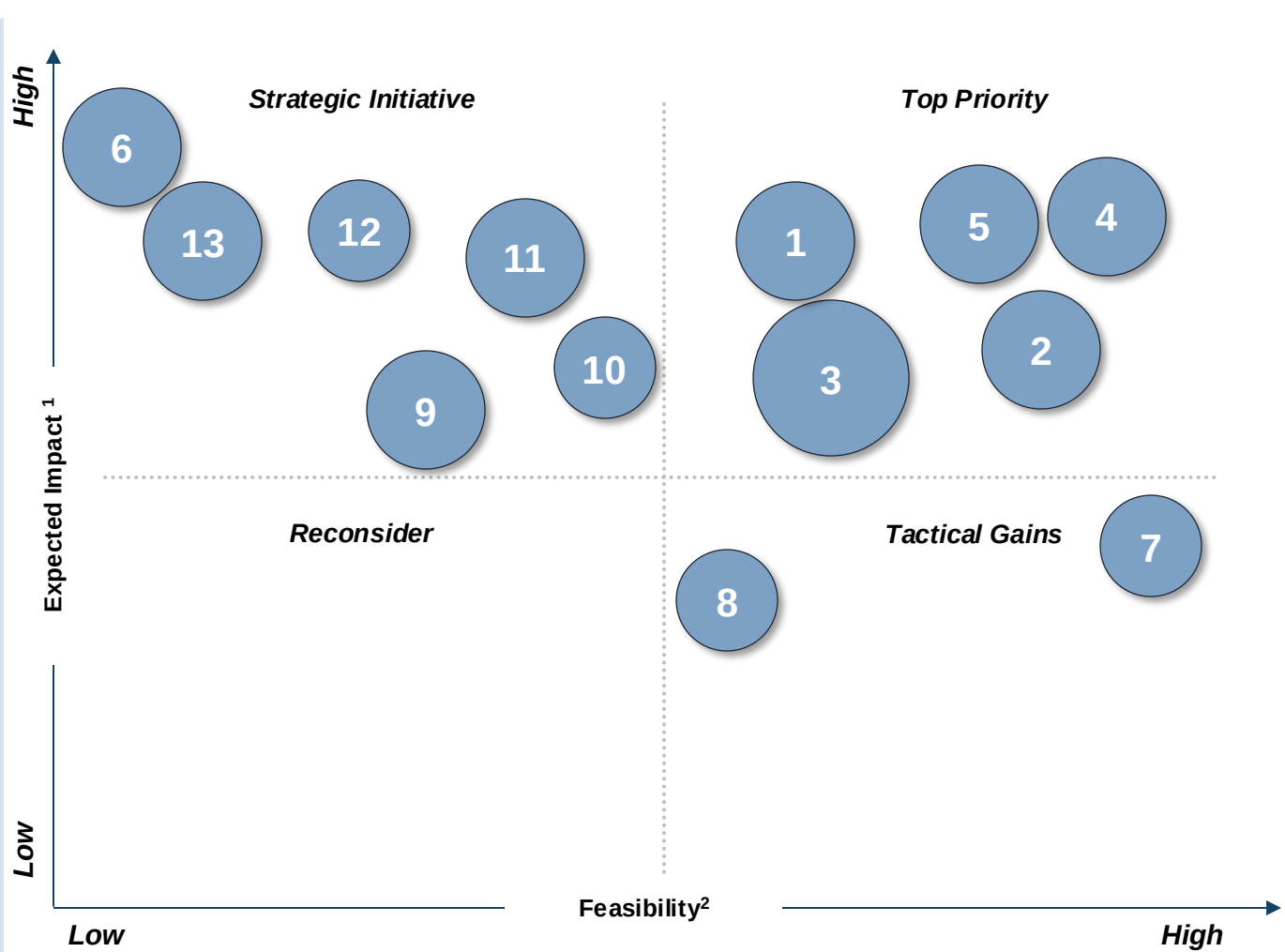
SCAD



مركز الإحصاء
STATISTICS CENTRE

13

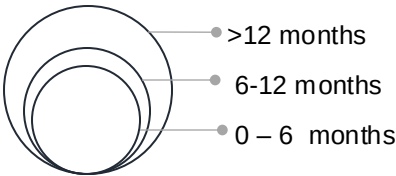
Identified AI Use Cases



Identified Use Cases

1	Forecasting GDP up to 5 years
2	Population and Migration Forecast up to 5 years
3	Domestic Agriculture
4	Environment Pollution
5	Livability – GenAI
6	Unstructured Data Analysis
7	Adaptive Survey Design
8	Survey Response Rate Simulation
9	Sampling Errors Prediction
10	Outlier Correction
11	Statistical Forecasting
12	ML-powered Encryption
13	Risk Detection and Mitigation

Time to Deploy



1. **Expected Impact:** Citizen Experience, Productivity Enhancement, Operational Efficiency, Achievement of Gov. Goals and Output Quality
2. **Feasibility:** Data availability, Infrastructure readiness, AI Model availability, Stakeholder alignment and Cost feasibility

SCAD has prioritized 5 use cases from their identified long list

1



Forecasting GDP up to 5 years

- *Forecasts sector-level GDP accurately*
- *Provides timely insights for economic planning, resource allocation, and policy formulation*

Expected Impact

- Improved forecasting accuracy and better decision-making by economists, policymakers, and businesses

2



Population and Migration Forecast up to 5 years

- *Predicts future population demographics across different age group*
- *Provides insights into changing demographic trends*
- *Enables decision-making by organizations and policymakers*

Expected Impact

- Improved trade forecasting, enhanced decision-making by trade analysts, policymakers, and businesses involved in international trade

3



Domestic Agriculture

- *Addresses the challenge of forecasting domestic agricultural production*
- *Provides insights into crop yields, production volumes, and agricultural trends*

Expected Impact

- Enhanced agricultural planning, improved resource allocation, and increased food security by farmers, policymakers, and food supply chains

4



Environment Pollution

- *Tackles the challenge of predicting pollution levels in specific areas*
- *Provides insights into air, water, or soil pollution levels*
- *Enables key stakeholders to take proactive measures to mitigate pollution and safeguard public health*

Expected Impact

- Improved pollution management, enhanced public health protection, and preservation of natural ecosystems by environmental agencies, policymakers, and local communities

5



Livability GenAI

- *Addresses challenges in assessing and enhancing livability factors in urban environments*
- *Provides insights into various aspects of livability*
- *Informs decision-making by urban planners, policymakers, and community stakeholders*

Expected Impact

- Enhanced urban livability, improved quality of life for residents, and sustainable urban development by urban planners, policymakers, and community stakeholders



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~1.5M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	Medium	High	Medium

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Medium	Low	AI Model Complexity	Medium	High
		Medium		

1. Business Overview

Description

- The Purpose of the use case is to forecast GDP for the next 5 years across various sectors in Abu Dhabi. This initiative aims to predict economic trends, assist in resource allocation, and support decision-making processes that influence the region's development and economic stability.
- Key beneficiaries of this use case include government policymakers, economic planners, and strategic departments involved in crafting and adjusting visa regulations, trade policies, and business licensing.

1.2

Pain Points

- Inaccuracy in current GDP forecasting methods.
- Delays in data processing affecting timely decision-making.
- Inadequate tools for sector-specific economic forecasting.

1.3

Expected Impact

- Enhance economic forecasting efficiency.
- Support governmental economic goals.
- Improve the quality of economic planning outputs.
- Address medium data availability challenges.
- Upgrade necessary infrastructure for AI deployment.

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Inaccurate forecasts due to model limitations or unforeseen variables.
- Poor generalization to new data.
- Difficulties in understanding AI decision-making processes.
- Limited or biased data impacting model performance.
- Not enough data for GDP calculation

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	Medium	Large Datasets	Data Warehouse	ADEO, ADGE	Requires Integration	Open and Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

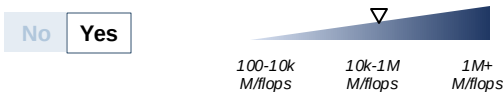
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Stakeholder Alignment: Ensure collaborative efforts across departments
- GDP Calculation Methodology: Accurate and reliable economic modeling
- AI Model Complexity: Handle complex neural networks



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~1.5M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	High	Medium	High	Medium

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	High	High	High

1. Business Overview

Description

- The Purpose of the use case is to forecast population dynamics and migration patterns over the next 5 years, aiming to understand demographic shifts and their implications on urban planning, workforce development, and social services.
- Key beneficiaries of this use case include demographic analysts, urban planners, and policy makers who will use these forecasts to shape housing policies, education systems, and healthcare services to better accommodate changing population needs.

1.2

Pain Points

- Challenges in predicting demographic shifts affecting policy and resource planning.
- Lack of accurate, timely data on migration patterns affecting economic and social planning.
- Difficulty in adapting to rapid demographic changes due to outdated forecasting models.

1.3

Expected Impact

- Improve efficiency in demographic analysis.
- Support urban and social service planning goals.
- Enhance the quality of demographic insights.
- Address challenges in data access and integration.
- Develop infrastructure to support advanced AI models.

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5

Risks

- Potential for inaccurate predictions due to data quality issues or model limitations.
- Model may not generalize well to new, unseen data.
- Complex models may be difficult to understand or explain.
- High dependency on the availability and quality of demographic data.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	Hard	Large Datasets	Data Warehouse	ADEO, ADGE	High complexity integration required	Open and Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

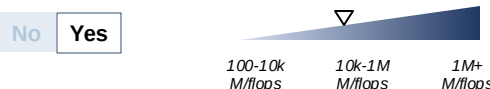
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Stakeholder Alignment: Collaborative approach involving stakeholder entities and the development team
- AI Model Complexity: Utilization of advanced neural networks



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~1.5M AED	>12 Months	11 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
Medium	Medium	Medium	Medium	Medium

Data Availability	Infrastructure Readiness	AI Model Complexity	Stakeholder Alignment	Cost Feasibility
Low	Low	Medium	Medium	High

1. Business Overview

Description

- The Purpose of the use case is to predict agricultural production levels, helping to enhance food security and optimize resource management across domestic markets. This forecasting aims to support agricultural planning and sustainable farming practices.
- Key beneficiaries of this use case include agricultural departments, farm management organizations, and food supply chain stakeholders who rely on accurate production forecasts to make informed decisions regarding crop planting, distribution logistics, and market pricing.

1.2

Pain Points

- Inaccuracies in current agricultural forecasting impacting planning and resource allocation.
- Difficulty in adapting quickly to changing agricultural conditions due to lack of real-time data.

1.3

Expected Impact

- Improve accuracy of agricultural forecasting.
- Enhance resource allocation and planning.
- Support government goals related to food security.
- Address medium challenges in data availability.
- Upgrade infrastructure to support AI technology.

1.4

Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 3 AI Scientists
- 2 Data Engineer
- 1 Full Stack Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 2 AI Testers

1.5

Risks

- Potential for inaccurate forecasts due to data or model limitations.
- Risk of models being too tailored to historical data, affecting their applicability to new conditions.
- Complex models might be difficult to interpret by end-users.

2. Data, AI and Infrastructure Overview

Data Requirements

Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	Medium	Large Datasets	Data Warehouse	ADEO, ADGE	Requires significant integration effort	Open and Confidential

2.2

AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

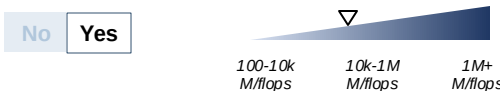
2.3

Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage

GPU Enabled Infra

Computational Power



2.4

Key Considerations

- Stakeholder Alignment: Close collaboration required with stakeholder entities
- AI Model Complexity: Utilizes neural networks
- Visualization tools and its connectivity with DB and analytics system

Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~1.7M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	High	High	High

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
High	High	AI Model Complexity	Medium	High

1. Business Overview

Description

- The Purpose of the use case is to utilize the Economic Complexity Index model leveraging AI and data analytics to monitor, analyze, and mitigate sources of pollution, enabling data-driven decision-making for environmental protection and sustainability initiatives.
- Key beneficiaries of this use case include environmental organizations, investment planners, and government policymakers, who utilize these insights to formulate strategies that promote innovation and environmental sustainability.

1.2 Pain Points

- Current methods may not adequately predict local pollution trends.
- Lack of real-time pollution data impedes effective response and planning.

1.3 Expected Impact

- Enhance pollution management capabilities.
- Improve public health protection and environmental preservation.
- Enable effective adaptation to different pollutants and regions.
- Tackle challenges in data availability and infrastructure readiness.

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Potential for inaccurate predictions due to data limitations or model complexity.
- Risk of models not generalizing well to new data scenarios.
- Complexity of AI models might make them hard to understand.
- Challenges due to the quality or timeliness of data collection.

2. Data, AI and Infrastructure Overview

Data Requirements

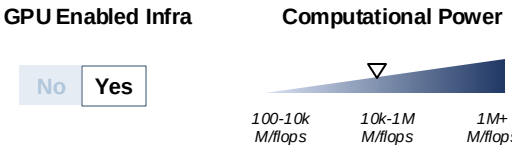
Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	Medium	Large Datasets	Data Warehouse	ADEO, ADGE	High Integration Complexity	Open and Confidential

2.2 AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other: Predictive Analytics Tools	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- Technology Stack: AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage



2.4 Key Considerations

- Stakeholder Alignment: Needs collaboration with stakeholder entities and internal teams
- AI Model Complexity: Involves complex machine learning model and may need neural networks



Prioritization Outcomes

Preliminary Budget	Expected Duration	Required Team Size
~1.8M AED	6-12 Months	8 FTEs

1. Expected Impact

Citizen Experience	Productivity Enhancement	Operational Efficiency	Achievement of Gov. Goals	Uplift Output Quality
High	High	Low	High	High

Data Availability	Infrastructure Readiness	2. Feasibility	Stakeholder Alignment	Cost Feasibility
Medium	Low	AI Model Complexity Medium	High	High

1. Business Overview

Description

- The Purpose of the use case is to evaluate and enhance urban livability factors including safety, accessibility, and environmental quality, facilitating smarter urban planning and improved living conditions.
- Key beneficiaries of this use case encompass urban planners, city council members, and community development organizations. These stakeholders use the insights provided to implement projects that improve public transportation, housing quality, and recreational facilities, contributing to a higher quality of urban life.

1.2 Pain Points

- Limited tools for assessing comprehensive livability factors in real-time.
- Challenges in making data-driven decisions for urban development due to data integration complexities.

1.3 Expected Impact

- Improve urban planning and development efficiency.
- Enhance quality of life for city residents.
- Support tailored urban interventions for specific livability challenges.
- Enable broad application across different urban contexts and projects.

1.4 Expected Squad Required

- 1 Scrum Master
- 1 Business Analyst
- 2 AI Scientist
- 2 Data Engineer
- 0.5 UX/UI Designer
- 0.5 Solution Architect
- 1 AI Tester

1.5 Risks

- Risks of inaccurate predictions affecting urban planning decisions.
- The model could be overly tailored to historical data, reducing its applicability.
- Complexity may obscure model decision processes.
- Dependence on comprehensive and timely data, which might be limited.

2. Data, AI and Infrastructure Overview

Data Requirements

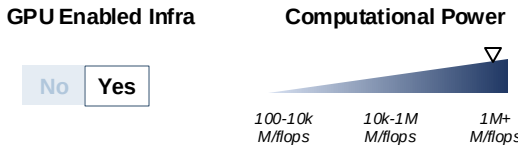
Dataset	Availability	Size	Source System	Source Entity	Accessibility	Confidentiality
TBD	Medium	Extensive Datasets	Data Warehouse	ADEO, ADGE	High Integration Complexity	Open and Confidential

2.2 AI Requirements

AI Capabilities	End Product	AI Methodology	AI Technology
- Automation - Augmentation - Advisory - Interaction	- Dashboard - Product/ app. - Embedded Model - Other:	- Descriptive - Prescriptive - Diagnostic - Predictive - Cognitive	- Computer Vision - NLP - Speech Recognition - ML (supervised, unsupervised, reinforcement learning) - Generative AI

2.3 Cloud Services and Providers

- Technology Stack: LLMs, Knowledge Graphs, AI development platform for model creation, data storage, and visualization tools
- Cloud Services: AWS, Google Cloud, or Azure, for Compute, Data processing tools, and scalable data storage



2.4 Key Considerations

- Stakeholder Alignment: Requires effective collaboration between stakeholders for successful implementation
- AI Model Complexity: Utilizes advanced neural networks
- Visualization tools and its connectivity with DB and analytics system
- LLM/ML Model: Falcon 40b

Furthermore, an additional 8 use cases have been defined by SCAD

SCAD



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Identified AI Use Cases

SCAD Long List of Non-prioritized Use Cases

	Title	Description	AI Impact	Feasibility	Time to Deploy
6	Unstructured Data Analysis	Leveraging the power of Natural Language Processing (NLP) to enrich collected and analyzed data sets and include unstructured data such as reports, surveys and administrative records	●	●	6-12 Months
7	Adaptive Survey Design	Improve the design of surveys by using AI to dynamically adjust questionnaires based on past responses	●	●	0-6 Months
8	Survey Response Rate Simulation	Optimize survey response rates by simulating expected responses based on past trends and key metrics to suggest adjustments to target population, sample sizes and questionnaire	●	●	0-6 Months
9	Sampling Errors Prediction	Analyze historical data on sampling patterns and outcomes to accurately predict sampling errors and identifying biases in sampling techniques, target populations or data collection methods	●	●	6-12 Months
10	Outlier Correction	Deploy Machine Learning to automate the detection of outliers and anomalies in collected datasets and accordingly suggest data corrections to improve data quality	●	●	0-6 Months
11	Statistical Forecasting	Complement statistical outputs with predictive models to provide statistical forecasting on different topics (e.g. economic trends, population growth, etc.) with higher accuracy and enable proactive decision-making	●	●	6-12 Months
12	ML-powered Encryption	Employ encryption techniques while ensuring that the statistical information remains useful but not personally identifiable by implementing Machine Learning algorithms, while complying with National regulations	●	●	0-6 Months
13	Risk Detection and Mitigation	Build a toolkit comprised of analytical algorithms that will detect and mitigate risks associated with datasets dissemination to automate the masking of data thereby ensuring the privacy of citizens while measuring and evaluating the loss in data quality and usefulness	●	●	6-12 Months

دائرة التمكين الحكومي
DEPARTMENT OF GOVERNMENT ENABLEMENT



Thank You

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